Snowflake 7TS Fl 29-31 - CW pump controls - Siemens call 10/30/2026 10:00

Siemens: Alan Parrish (Ops Mgr), Brendan Jackson (PM), Brian Greda, Rachael Werner. NW: Ruslan, Christino, Andrei. Tabs 01-08 refer to the files in this folder / bookmarks in the combined pack.

Goal of the call

• Agree the pump control scope: no separate panel, no $50k/floor. Pumps do duty/standby themselves; Siemens provides the points already in its proposal.

• Get a date for ATC submittal R1 (CRAC units on site mid-November, start-up after).

• Close the 9/28 items list (below).

Our position in four lines

• The design never had a separate panel: EOR specifies pumps with **integrated VFD and built-in pump system controller** (M-101 2.07.D items 5, 15, 17) and schedules them as **two pumps primary-standby** (M-401). No revision, no bulletin on this (tab 03).

• Sequence 3b asks for lead/lag *sequencing*: 'one pump running and the other as stand-by'. That is the Grundfos multipump 'Alternating operation': duty/standby, standby starts on alarm, no external controller (IOM 7.49 p.61, Dolphin guide p.3 - tabs 05, 06).

• Siemens had the approved pump submittal and the Grundfos manual on **6/29** (our email proposing multipump instead of a panel, resent 7/9). No objection; R0 came 7/22 still drawn with a panel and VFDs (tab 07).

• Siemens' own proposal p.5 already lists the field points (flow switches, leak detectors to BB BMS and autodialer, start/stop and status, DP transmitter, AC interlock). If a panel is still wanted: Alt 3 is **$2,800 per panel** (tab 01).

Answers to Alan (emails 10/28 and 10/29)

Alan says	Answer	Where
No panel, something in the VFDs does the rotation - so no BMS interaction for rotation?	Correct. Pumps are linked by GENIbus and run Alternating operation. No BMS logic for rotation.	05 p.61; 06 p.3
No alarm point in the documents / proposal; '100% standalone, no connection to the BMS at all'	'No connection to base building BMS' is about pump control. Alarms ARE in scope: leak detectors 'to alarm at BB BMS and Tenant AutoDialer', flow switches, start/stop + status (p.5); Excl. 6 'except Leak Detection'; seq 3c/3d local A/V alarm + landlord BMS - copied on their own 300C; their 300A already wires flow and leak alarms to DXRs.	01 p.5, p.8; 02 marks G, H, J, K; 03 M-102
Did the VFD vendor confirm the drives act as the lead/lag panel?	Yes - Dolphin 9/28: duty/standby guide, 'Standard startup/programming can be used', 'Contractor needs to run the interconnecting wiring', 'One pressure sensor is required per MLE motor'. Start-up scope confirmation requested.	06; 07
Pump panels should come with the mechanical package (Alan 10/27)	On this job the EOR specified the controller inside the pump; nothing else is scheduled. M-102 ATC spec: control panels/power are controls contractor's work.	03

Scope split we want to agree (per floor, x3)

Item	Who	Basis
Pumps, check valves, drip pans, piping, DP taps; install DP transmitters and flow switches (pipe devices)	NW	Siemens Excl. 18 'installation of pipe mounted devices'
GENIbus cable between the two pumps (A-Y-B, 16 ga shielded)	NW / EC (low voltage) - agree on call	Dolphin: 'contractor'
Pump multipump setup, sensor setup, I/O settings at start-up	Dolphin (confirming)	Barry 9/28
CRAC run (enable) contact -> DI1 of both pumps, jumper 2-6 removed	Siemens	p.5 'Hardwire interlock to AC unit'
Flow switches FS-1/FS-2: furnish and wire; contact to DXR (alarm) + to that pump's DI2 (External fault)	Siemens	p.5 'Flow Switches - furnish and wire'

Item	Who	Basis
Leak detectors LD (pump pans): to DXR, BB BMS, autodialer + contact to that pump's DI3	Siemens	p.5
DP transmitters: 2 per floor (one per pump) wired to AI1 4-20 mA	Siemens	Grundfos / Barry; IOM 'running pump sensor'
Pump status / alarm (relay 1 Alarm, OC1 or relay 2 Running) to DXR / autodialer	Siemens	p.5 'Start/stop and Status'
Pump isolation valves V-1/V-2, one relay per valve (leak on one pump closes only that pump)	Siemens	p.5; seq 3d
Local audio/visual alarm (horn/strobe) for loss of flow and leak	Siemens	seq 3c/3d on their 300C
1 h in 4 h run timer (3f)	Met if CRAC runs 24/7 (4a IDF); else DXR timer contact in parallel on DI1	seq 3f / 4a
Delete from R0: lead/lag panel box, VFD panels, VFD run/speed wiring, CS-1/CS-2 optional	Siemens R1	9/28 email

Questions to ask Siemens

• With no panel and no BMS rotation, what exactly is left in the '$50k per floor'? Itemize it.

• Do you agree the field points on proposal p.5 stay in base scope (flow switches, leak detectors, valves, status, interlock, DP)?

• How will you provide the local audio/visual alarm (3c/3d) and the 3f timer - both are in your own 300C sequence but not on any sheet?

• Flow switches: proposal says 'furnish and wire', R0 shows 'FBO' and not in BOM. Confirm you furnish.

• V-1/V-2: missing from the valve schedule; both on RE-1. Will R1 show one relay per valve?

• CRAC valves (100A): open with unit, close when off, 4 h daily flush (4h); 5-minute delay (4i/4j); separate condensate alarm.

• Which contact is the 'unit status' RE-x-1 - fan/enable or compressor? Must be enable/fan, or CRAC and pumps never start (flow switch 4k).

• Who powers the valve actuators and relays at start-up - DXR 24 V as on 300A?

• Date for ATC submittal R1, and who from Siemens attends start-up with Dolphin.

• If you still want a panel: Alt 3 at $2,800 per panel - what would it do that the pumps do not?

Keep in mind

• DP transmitters: 2 per floor needed (R0 has 1). Do not raise the extra cost - wait for Siemens to bring it up.

• Alternation: Grundfos recommends 'energy'; 'time' needs FM310/FM311 per current IOM p.63 (our pumps FM 300 per submittal I/O). Spec says 150 h - minor deviation; note for the EOR if asked.

• Do not use the pump DI 'Flow switch' function (that is low-flow stop, needs a diaphragm tank). Low-flow stop = Not active.

• Bypass HD-053 still open (Siemens RFI). Ask status.

• Fallback if Siemens insists on a panel: (1) Alt 3 at $2,800/panel ($8,400); (2) NW furnishes a Tekmar 132 per pair (~$800-1,100; tab 08) and Siemens installs, powers and wires it per p.5 'Install and power manufacturer provided Lead / Lag panel'. Mention only if needed.

Still open from our 9/28 email

• Flow switches in BOM

• V-1/V-2 in valve schedule

• One relay per pump valve (3d)

• 3c/3d/3f alarm and timer

• 300B typos (CWP-29 labels on fl. 31, LD-8 twice, DPTE-1)

• CRAC valve logic 4h; 4i/4j delay; condensate alarm

- DP range 0-15 vs 0-25 psi
- Sheet 100 relay table all WCCU-29-1

Tab 02 legend - marks on Siemens ATC R0

Mark	Sheet	What it shows
A	300	'LEAD LAG PUMP CONTROLLER' box = Siemens' own RIBU1C relays RE-1..5 (BOM 15) - their relays were to sit at the by-others panel
B	300	Note 2 'VFD mounted in ceiling near pump set' - the VFD is integral to the pump motor
C	300	Flow switches FS-1/FS-2 marked 'FBO'; proposal p.5 says furnish and wire
D	300	One DP transmitter per floor, 0/15 psi (BOM says 25 psi); Grundfos needs one per pump
E (red, 300A/300B)	300A/B	'CWP-xx-1&2 LEAD LAG PANEL' by others - not in BOM, no model, no power; terminal numbers 2/6, 4/6, 12/18, 10/20 are the Grundfos CRE motor terminals
F (blue)	300A/B	'VFD panels' - do not exist; 300B labels the 31st floor as CWP-29
G (green)	300A/B	Flow alarms RE-2-1 / RE-4-1 already wired to Siemens DXRs
H (green)	300A/B	Leak detectors LD-1..4 already wired to Siemens DXRs / autodialer
I	300C	BOM: no flow switches, no panel, no horn/strobe
J, K, L	300C	Their sequence copies M-102 3c (local A/V alarm), 3d (leak local + landlord BMS), 3f (timer)

Tab 07 - Barry (Dolphin) 9/28 and our 6/29 email to Siemens

Date / from	Text (verbatim)
9/28 8:09 Barry	Wiring diagram was included in the submittal. I am attaching it again. (generic 'control by BMS' diagram)
9/28 10:50 Barry	I am not sending you the generic drawing but that's how the pump was designed, specified, approved & released. Our startup tech will also start the pump based on this drawing only. Any advanced programming or pump setup will be by others. I will try to get answers to your comments below from the factory.
9/28 3:31 pm Barry	Duty/standby instructions are attached. -Standard startup/programming can be used. -Contractor needs to run the interconnecting wiring. -One pressure sensor is required per MLE motor.
Open with Barry	Draft 10/27 (not sent): confirm start-up does multipump + DP sensor + I/O settings on all 3 pairs, price if extra, notice to schedule; panel option price. Weak spot: '10:50' line says advanced programming by others - get the confirmation in writing.
6/29 Ruslan to Brendan + Brian	Since each floor operates as an independent system serving only one data room AC unit, we can use the native multi-pump functionality built into the Grundfos CRE drives. See attached Manual ... Section 7.49 ... By linking the two pump heads directly via their GENIbus communication terminals, they will automatically handle all duty rotation, staging, and standby redundancy on their own. With this setup and since pumps not going to base building BMS, i think lead/lag panel not necessary here. (Attached: Grundfos IOM and approved H-02.00.) Resent 7/9 'please confirm'. No reply on this point; R0 issued 7/22.

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Scope of Work

Siemens proposes the following:

Furnish and install the Siemens Building Management System (BMS) in accordance with the drawings and specifications by AMA Consulting Engineers.

- Mechanical plans dated 4/03/2026 - Issued for pricing
- Specification on plans
- Mechanical plans dated 4/22/2026 - Addendum #1

Detailed Scope of Work

The following list of equipment will be controlled and/or monitored by the BAS:

Qty	System Description
3	PXC BMS Primary Controllers
93	Fan Powered Boxes
3	Water Cooled AC units
6	CW Pumps
5	Fire Smoke Dampers
11	Pantry Sink / closet Leak Detectors
6	HW Heater Leak Detectors
3	Tenant Auto Dialers
	Alternate pricing - See below

The following equipment/systems will be controlled and/or monitored by the Siemens BMS as described. For counts per area refer to the above list of equipment per the breakdown:

General (building standards)

-
  - Use of plenum rated cabling for low voltage in open ceilings
  - Orange FLN / Comm Wire
  - Thermostat – Sensing only
  - 2/3 Static pressure resides in AC unit - run out Double gang poly and tip
  - Only tenant leak alarms to BMS. All other tenant equipment alarms to tenant Autodialer.
  - Leak Detector Liebert LT-410 only (building standard)
  - Laptop for ABT project on site. Use Laptop for all DXR projects. Contact Siemens CSM.
  - Field coordinate with Fire alarm contractor to provide new Fan Powered Box Shut Down contact at Mammoth AC unit DDC panel.

- **OPEN SHOP Electrical Subcontractor**

PXC DDC Controller

- Siemens to replace / upgrade existing primary DDC controller to enable communication to new FPB controllers via BACnet MSTP.
  - Re- connect existing BLN to new controller
  - Install new Ethernet cable to adjacent Base Building IT closet and safe-off for future connection (BB IT contractor to install future network switch).
  - Fire Alarm contractor to provide new shut down / status relays within 3ft of controllers
  - Re-install new differential pressure transmitter for existing AHU (removed during demo).

Water Cooled AC Units

- Water cooled AC units to be provided with manufacturer controls. Siemens to provide wiring and devices as follows:
  - Interlock wiring between CRAC and WCCU
  - Install and wire manufacturer's temperature/humidity room sensor
  - Hardwire interlock unit operation to pumps
  - Furnish (2) Isolation valves
  - Furnish and wire (2) Leak detectors (1-Condesing Unit / 1-to alarm at Base Building BMS and Tenant Autodialer. Leak detectors to close isolation valves upon alarm.

FP Boxes w/ Hot Water Reheat

- Siemens to provide factory mounted controls on FP boxes, and field FLN communication wiring. The following is a list of the factory mounted controls, monitored and controlled by Siemens:
  - (1) DXR BACnet MSTP equipment controller with damper actuator
  - (1) Fan - Start/stop and status
    - Current sensor p/n: ECMCS (Automation Components)
  - (1) Room Temperature Sensor – Sensing Only
  - (1) Discharge Air Temperature Sensor
  - (1) Hot Water Valve and Actuator
  - **Add Freeze protection program per Building standard.**
  - **Add Unoccupied mode program: Dampers full-open to prevent Mammoth unit shut down.**

CW Pumps

- Siemens to wire pumps and associated sensors to local lead / lag panel (provided by others).
  - (1) Install and power manufacturer provided Lead / Lag panel
  - (2) Isolation valves - furnish and wire
  - (2) Leak detectors - furnish and wire. To alarm at BB BMS and Tenant AutoDialer
  - (2) Flow Switches - furnish and wire
  - (2) Pump - Start/stop and Status
  - (1) Differential pressure transmitter - to maintain pump speed
  - (1) Hardwire interlock to AC unit operation

- **No connection to base building BMS**
- **Lead Lag panel provided by others**

Fire Smoke Dampers

- Siemens to furnish pneumatic labor to connect pneumatic tubing to new FSDs and existing Fire Alarm contacts. FSD dampers to operate in conjunction with existing FSDs serving Mammoth Unit

Leak Detectors - Pantry Sink and HW Heaters

- Siemens to furnish and install Leak detectors. Hardwire interlock to isolation valve (valve furnished, installed, and powered by others).
  - Leak detectors to alarm at BB BMS and Tenant AutoDialer

Autodialer

- Siemens to furnish and install tenant autodialer. Tenant to provide IP connection.

Alternate Price

1. ~~**Add $125 per Fan Powered Box -** (1) Field install controller in lieu of factory mounting.~~
2. ~~**Add $575 per CO2 room Sensor -** if required. Specification calls for CO2 monitoring in large areas, but plans do not show CO2 room sensors…. only show room temperature sensors.~~
3. ~~**Add $2,800 per Panel -** Furnish (1) pump Lead / Lag DDC panel with software. Stand-alone. No tie-in to BMS~~
4. ~~**Add $62,200 -** Conduit stub-ups and backboxes for thermostat installation (if required)~~
5. **Deduct $84,300** - Open shop labor for electrical labor **ACCEPTED and INCLUDED**

Exclusions

1. Conduit stub-ups and backboxes for thermostats
2. Conduit in open ceilings for low voltage & communication wiring (cable/wire to be plenum rated)
3. Instantaneous hot water heaters (i.e. serving eye wash stations)
4. PRV Stations, water meters, electric meters.
5. Furnishing Lead / lag panels for pumps
6. Base building BMS monitoring/control of (except Leak Detection):
    1. Pumps
    2. Water Cooled Computer room AC units
    3. Transfer Fan
    4. **Exhaust fan Dish Room**
7. Field installation of Fan Powered Box controls
8. Furnishing plumbing Isolation valves (i.e. HW heaters, sinks, etc..)
9. Air Purification System
10. CO2 sensors (none shown on plans)
11. Overtime
12. Per project aggregate general liability insurance
13. 120VAC Power wiring to all terminal box controllers
14. Furnishing, installing, or wiring of duct smoke detectors
15. Wiring for fire alarm shutdown of Fan Powered Box and/or CW AC equipment
16. Furnishing CW AC thermostats
17. FPB box control transformers
18. Installation of pipe mounted devices (i.e. control valves, flow meters, sensor wells)
19. Access doors
20. Temporary or standby labor and services

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Clarifications

1. Cost for Phasing and/or Escalation has not been included. Project completion estimated within twelve (12) calendar months from reception of Purchase order / contract.
2. Price based on a **Capital Improvement** project. Appropriate taxes will be added to final sale price if certificate is not provided.
3. Sample Certificate of Insurance must be provided with contract / purchase order. Required Insurance must be listed in commitment document.
4. **Estimated durations** are as follows from acceptance of PO / Contract:
    1. Purchase order / contract processing to activate Job number (internal) - 6 to 8 business days
    2. Submittal - 1st submission - 2 to 4 weeks from job activation
    3. On site installation to begin - upon approval of submittals
    4. Material shipments are experiencing unusually long lead times
5. The project cost includes the tie in of the associated BMS system to the existing Siemens BMS front end workstation
6. Graphic floorplan can only be generated from the Autocad files provided. Please provide the correct floorplan area(s) (full/partial) required to be displayed on the BMS.
7. Control wiring shall be run in conduit where exposed to outdoor conditions. Control wiring shall be run using plenum rated cable without conduit in all other locations
8. Miscellaneous wiring or power to 3rd party equipment has not been included, unless noted in Scope of Work section of this proposal.
9. If Per project Aggregate insurance is required: Add 1.1% to final sale (Per Aggregate Insurance would be noted in Cert. Of Insurance provided by GC / Building owner).

A: 'LEAD LAG PUMP CONTROLLER' = Siemens relays RE-1..5

CONDENSER WATER PUMPS
LOCATION & SERVES: SEE SCHEDULE

LEAD LAG PUMP CONTROLLER

- RIBU1C RE-1 (3)
- RIBU1C RE-3(3)
- RIBU1C RE-2(3)
- RIBU1C RE-4(3)
- RIBU1C RE-5(3)

DRAWING NOTES:

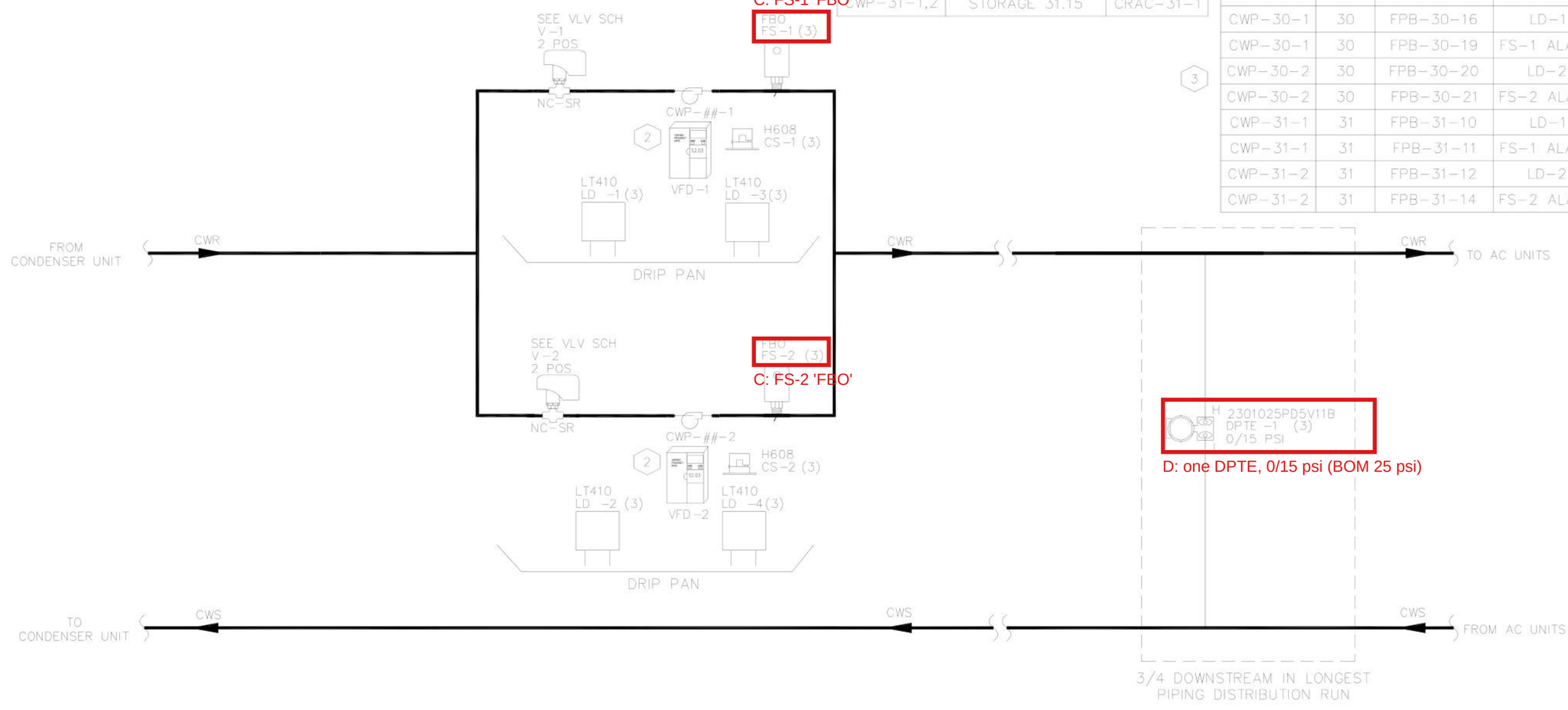
1. LEAK DETECTOR PROVIDED BY SIEMENS.
2. VFD MOUNTED IN CEILING NEAR PUMP SET.
3. REFER BELOW SCHEDULE FOR NEAREST DXR.

B: note 2 'VFD mounted in ceiling' - no separate VFD

CWP SCHEDULE		
UNIT TAG	LOCATION	SERVES
CWP-29-1,2	OPEN OFFICE 29.17	CRAC-29-1
CWP-30-1,2	DISH ROOM 30.13	CRAC-30-1
CWP-31-1,2	STORAGE 31.15	CRAC-31-1

TAG	FLOOR	NEAREST DXR	DEVICE
CWP-29-1	29	FPB-29-15	LD-1
CWP-29-1	29	FPB-29-16	FS-1 ALARM
CWP-29-2	29	FPB-29-17	LD-2
CWP-29-2	29	FPB-29-18	FS-2 ALARM
CWP-30-1	30	FPB-30-16	LD-1
CWP-30-1	30	FPB-30-19	FS-1 ALARM
CWP-30-2	30	FPB-30-20	LD-2
CWP-30-2	30	FPB-30-21	FS-2 ALARM
CWP-31-1	31	FPB-31-10	LD-1
CWP-31-1	31	FPB-31-11	FS-1 ALARM
CWP-31-2	31	FPB-31-12	LD-2
CWP-31-2	31	FPB-31-14	FS-2 ALARM

C: FS-1 'FBO'



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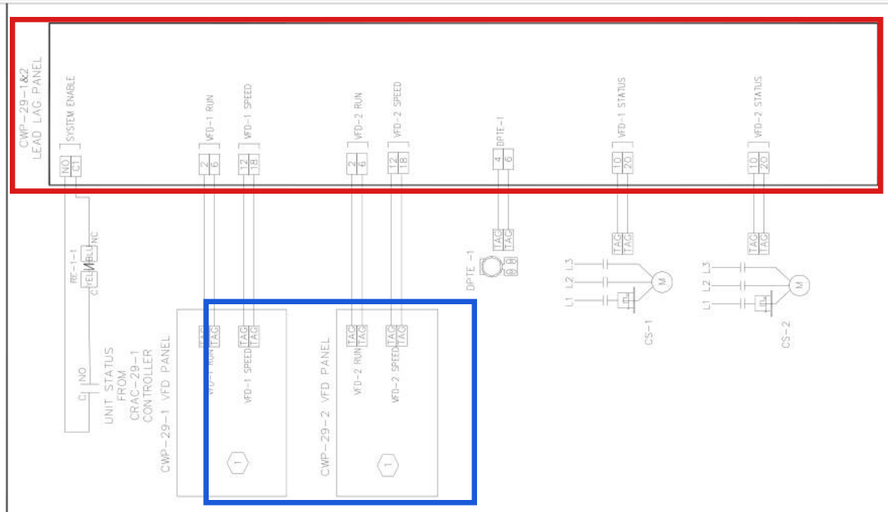
CONDENSER WATER PUMPS(MECH)

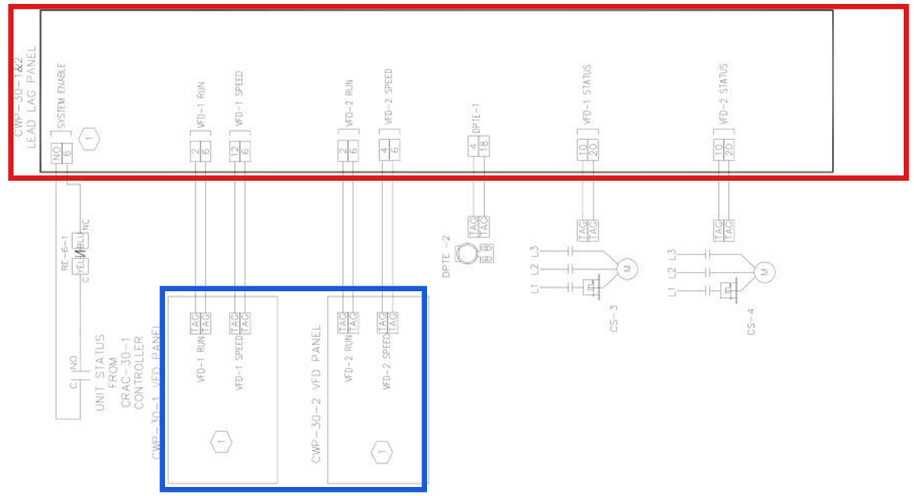
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VDF WIRING

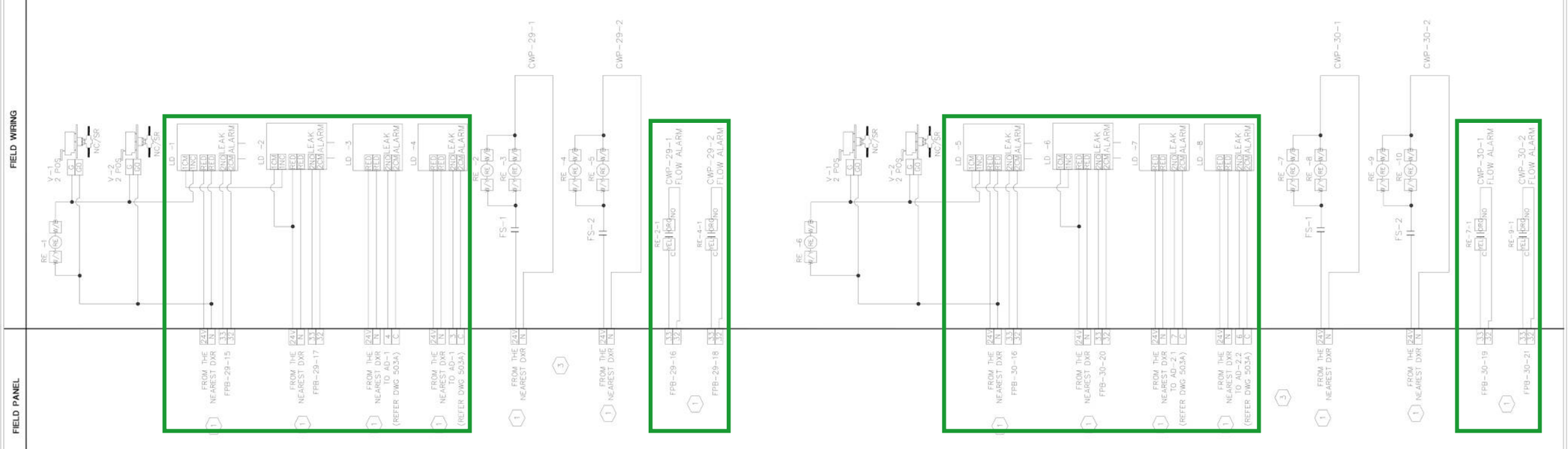




DRAWING NOTES:
1. TERMINATION DETAILS TO BE FIELD DETERMINED.
2. REFER DWG 300 FOR THE NEAREST DXR DETAILS.
3. ALL THE POINTS ARE CONNECTED TO DXR PROGRAM THRU SOFTWARE.

FIELD WIRING

FIELD PANEL



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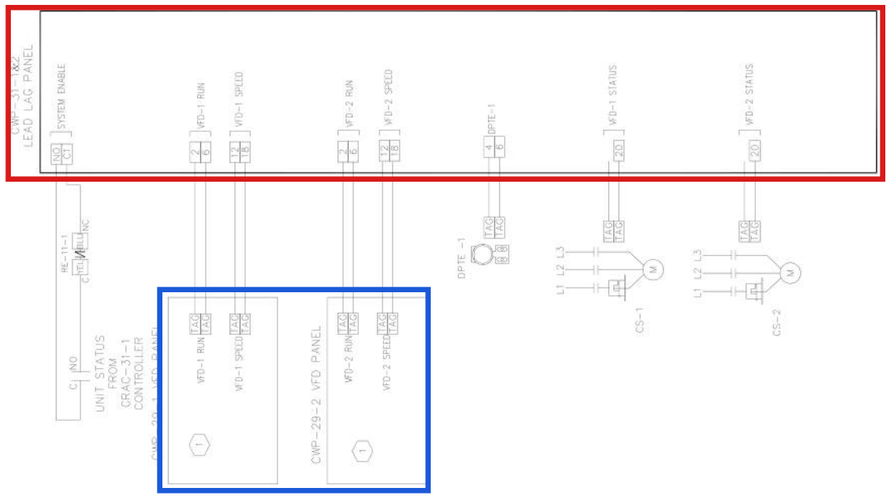
CONDENSER WATER PUMPS(ELEC-1)

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VDF WIRING

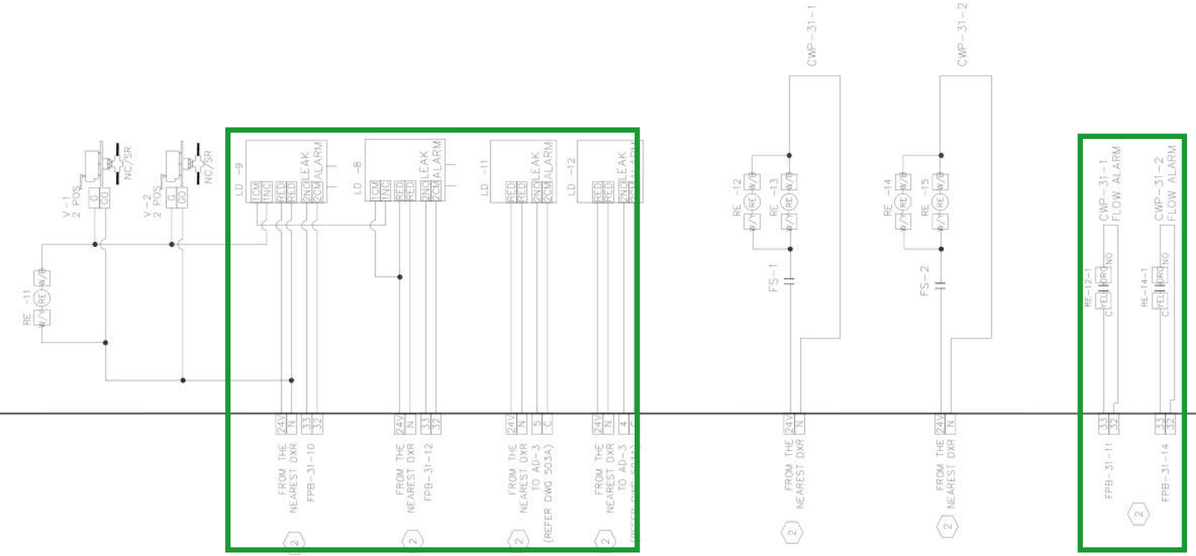


DRAWING NOTES:-

1. TERMINATION DETAILS TO BE FIELD DETERMINED.
2. REFER DWG 300 FOR THE NEAREST DXR DETAILS.
3. ALL THE POINTS ARE CONNECTED TO DXR PROGRAMMABLE I/O SOFTWARE.

FIELD WIRING

FIELD PANEL



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CONDENSER WATER PUMPS(ELEC-2)

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I: BOM - no flow switches, no panel, no horn/strobe, 1 DPTE/floor

Control Device		Qty	Product Number	Manufacturer	Document Number	Description
Field Mounted Devices						
CS	1-2	6	H608	VERIS	1006cut016	CUR SW SPLTCOR-ADJ SETPT W/LED
DPTE	1	3	2301025PD5V11B	SETRA SYST	N/A	DP TRAN,WET,25PSI,4-20MA,W/ 5VALVE MAN
LD	1-4	12	LT410	LIEBERT	N/A	POINT LEAK DETECTOR SENSOR
RE	1-5	15	RIBU1C	FUNCTIONAL DEVICES	1208cut013	RIB 120VAC 24VAC/DC SPDT
V						SEE VALVE SUBMITTAL

CONDENSER WATER PUMPS - SEQUENCE OF OPERATION:

a) CIRCULATING PUMP OPERATION SHALL BE INTERLOCKED WITH THE AC UNITS. UPON ACTIVATION OF ANY UNIT, THE LEAD CIRCULATING PUMP SHALL START AND RUN CONTINUOUSLY. THE PUMPS SHALL BE PROVIDED WITH AN AUTOMATIC RESTART AFTER A POWER OUTAGE.

b) PROVIDE FOR LEAD/LAG SEQUENCING IN ORDER TO HAVE ONE PUMP RUNNING AND THE OTHER AS STAND-BY. PUMPS [illegible] ADJUSTABLE) FOR EVEN WEAR.

J: 3c loss of flow -> standby + local audio/visual alarm

c) A FLOW SENSOR (ON DISCHARGE OF EACH PUMP) SHALL PROVE FLOW BEFORE STARTING ANY AIR CONDITIONING UNITS. A LOSS OF FLOW AS INDICATED BY THE FLOW SENSOR ON THE RUNNING PUMP SHALL SWITCH OPERATION TO THE STAND-BY PUMP AND ANNUNCIATE AN AUDIO/VISUAL ALARM LOCALLY.

d) PROVIDE A LEAK DETECTOR AT THE AUXILIARY DRAIN PAN OF EACH PUMP. LEAK DETECTION SHALL ANNUNCIATE AN AUDIO/VISUAL ALARM LOCALLY AND THROUGH THE LANDLORD BMS. IF MOISTURE IS DETECTED IN THE PAN OF THE OPERATING PUMP, THE PUMP SHALL STOP, AND THE STAND-BY PUMP SHALL START.

K: 3d leak -> local A/V + landlord BMS

e) SYSTEM DIFFERENTIAL PRESSURE WILL BE MAINTAINED BY VARIABLE SPEED DRIVES PROVIDED FOR EACH PUMP. THE PUMP SPEED SHALL MODULATE AS REQUIRED TO MAINTAIN A SYSTEM DIFFERENTIAL PRESSURE SETPOINT (ADJUSTABLE) AS SENSED BY A PIPE MOUNTED DIFFERENTIAL PRESSURE SENSOR. DIFFERENTIAL PRESSURE SENSOR TO BE MOUNTED 3/4 DOWNSTREAM IN THE LONGEST PIPING DISTRIBUTION RUN.

f) PROVIDE CONDENSER WATER PUMP SYSTEM WITH A 24 HOUR, 7 DAY TIMER TO INSURE PUMP OPERATION NOT LESS THAN 1 HOUR EVERY 4 HOURS.

L: 3f 24/7 timer, 1 h every 4 h

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7TS 29-31 SNOWFLAKE
NEW YORK, NY

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CONDENSER WATER PUMPS(BOM&SEQ)

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MECHANICAL SPECIFICATIONS

PART 1- GENERAL

1.01 GENERAL
A. THE LATEST EDITION OF AIA DOCUMENTS A201 GENERAL CONDITIONS OF THE CONTRACT FOR CONSTRUCTION, OR AS REQUIRED BY THE ARCHITECTURAL DOCUMENTS AND/OR THE STRUCTURAL ENGINEERS DOCUMENTS ARE PART OF THE CONTRACT.
B. BIDDERS SHALL VISIT AND CAREFULLY EXAMINE THE AREA AFFECTED BY THIS WORK TO FAMILIARIZE THEMSELVES WITH THE EXISTING CONDITIONS AND THE DIFFICULTIES THAT WILL AFFECT THE EXECUTION OF THIS WORK BEFORE SUBMITTING PROPOSALS. SUBMISSION OF A PROPOSAL WILL BE CONSTRUED AS EVIDENCE THAT SUCH AN EXAMINATION HAS BEEN MADE AND LATER CLAIMS WILL NOT BE RECOGNIZED FOR EXTRA LABOR, EQUIPMENT, OR MATERIALS REQUIRED BECAUSE OF DIFFICULTIES ENCOUNTERED WHICH COULD HAVE BEEN FORESEEN HAD SUCH AN EXAMINATION BEEN MADE. ANY DISCREPANCIES SHALL BE BROUGHT TO THE ENGINEERS ATTENTION PRIOR TO BID. IF DISCREPANCIES ARE NOT RESOLVED TO CONTRACTORS SATISFACTION THEY SHALL BE QUALIFIED IN THEIR BID SUBMISSION.
C. THIS CONTRACTOR SHALL REVIEW ALL CONSTRUCTION DOCUMENTS ASSOCIATED WITH THIS PROJECT INCLUDING GENERAL CONSTRUCTION, DEMOLITION, ARCHITECTURAL, MECHANICAL, ELECTRICAL, PLUMBING AND SPRINKLER PLANS AND SPECIFICATIONS. ALL WORK REQUIRED IN THE BID WHICH IS INDICATED OR IMPLIED TO BE PERFORMED BY THIS TRADE IN OTHER SECTIONS OF THE WORK SHALL BE INCLUDED IN THEIR BID. IF A CONFLICT OCCURS IN THE BID SPECIFICATIONS AND/OR ON THE DRAWINGS, THE MORE STRINGENT SITUATION SHALL APPLY.
D. COORDINATE ALL WORK OF THE SECTION WITH EXISTING CONDITIONS AND THE WORK OF OTHER TRADES. THE CONTRACTOR SHALL THOROUGHLY ACQUAINT HIMSELF WITH THE WORK INVOLVED AND SHALL VERIFY AT THE BUILDING ALL MEASUREMENTS NECESSARY FOR THE PROPER INSTALLATION OF THE WORK, OBTAINING THE SAME WHEN NECESSARY FROM THE OTHER CONTRACTORS AND SECTIONS. CONTRACTOR SHALL ALSO BE PREPARED TO PROMPTLY FURNISH TO OTHER CONTRACTORS ANY INFORMATION RELATING TO THE WORK OF THIS SECTION NECESSARY FOR THE PROPER INSTALLATION OF OTHER CONTRACTS AND SHALL COOPERATE TO SECURE THE BEST PROGRESS OF, AND HARMONY BETWEEN, THE WORK OF THE DIFFERENT CONTRACTS AND SECTIONS IN THE INTERESTS OF THE INSTALLATION AS A WHOLE. CONFER WITH OTHER CONTRACTORS AND ENGINEER FOR ADJACENT WORK TO THIS SECTION AND ARRANGE TO HAVE VISIBLE PORTIONS OF WORK FIT AND HARMONIZE IN A MANNER SATISFACTORY TO THE OWNER'S REPRESENTATIVE.
E. THE SPECIFICATIONS ARE ACCOMPANIED BY DRAWINGS INDICATING THE GENERAL LOCATION OF EQUIPMENT AND CONNECTIONS THERETO. UNLESS SPECIFICALLY DIMENSIONED, LOCATIONS OF EQUIPMENT AND ROUTINGS ARE APPROXIMATE. SCALES ON DRAWINGS ARE INDICATED FOR BIDDING PURPOSES ONLY. DRAWINGS SHALL NOT BE SCALED FOR CONSTRUCTION AND MANUFACTURING DETAILS. CERTAIN SYSTEMS ARE DIAGRAMMATIC AND GIVE THE GENERAL ARRANGEMENT ONLY. NO ADDED COMPENSATION WILL BE PERMITTED FOR VARIATIONS DUE TO FIELD CONDITIONS. EXACT LOCATIONS AND ARRANGEMENTS SHALL BE DETERMINED IN THE FIELD ON THE BASIS OF DETAILS INDICATED ON APPROVED SHOP DRAWINGS, AND SUPPLEMENTARY INFORMATION ISSUED BY THE ENGINEER, AND SHALL PROVIDE FOR OPERATING EFFICIENCY, NEATNESS OF APPEARANCE, AND EASE OF MAINTENANCE.
F. GUARANTEE: THE CONTRACTOR SHALL GUARANTEE AND SERVICE THE ENTIRE INSTALLATION FOR A PERIOD OF ONE YEAR FROM THE DATE OF THE FINAL ACCEPTANCE OF THE INSTALLATION. THE CONTRACTOR SHALL, DURING THE PERIOD OF THE GUARANTEE, REPLACE OR REPAIR AT HIS OWN EXPENSE ANY PIECE OF EQUIPMENT AND/OR MATERIAL WHICH IS FOUND TO BE DEFECTIVE. THE REPLACEMENT OR REPAIR SHALL BE PERFORMED THE SAME DAY OF NOTIFICATION IN AN EMERGENCY FASHION WHEN NOTIFIED BY THE OWNER OR AUTHORIZED REPRESENTATIVE. THE CONTRACTOR SHALL ALSO REPAIR ALL DAMAGE TO SURROUNDING WORK CAUSED BY THE FAILURE, REPAIR OR REPLACEMENT OF DEFECTIVE EQUIPMENT. ALL REFRIGERATION COMPRESSORS SHALL HAVE A FACTORY GUARANTEE INCLUDING PARTS AND LABOR FOR FIVE YEARS TOTAL. THE FINAL ACCEPTANCE WILL BE MADE AFTER THE CONTRACTOR HAS ADJUSTED HIS EQUIPMENT, BALANCED THE VARIOUS SYSTEMS, DEMONSTRATED THAT IT FULFILLS THE REQUIREMENTS OF THE DRAWINGS AND SPECIFICATION, AND HAS FURNISHED ALL THE REQUIRED CERTIFICATES OF INSPECTION AND APPROVALS.
G. EQUIPMENT AND MATERIALS: MOST ITEMS OF MECHANICAL AND ELECTRICAL EQUIPMENT AND MATERIAL ARE NOTED ON THE DRAWINGS OR IN THESE SPECIFICATIONS WITH A MANUFACTURER'S NAME AND CATALOG NUMBER. THIS DESIGNATION IS USED TO SET THE STANDARD FOR CONSTRUCTION, PERFORMANCE, OPERATION AND APPEARANCE. PRODUCTS OF OTHER MANUFACTURERS WILL BE CONSIDERED AND RULED UPON BY THE ENGINEER. THE SUBMISSION OF A SUBSTITUTION IMPLIES THAT THE ITEM HAS ALL NECESSARY UNDERWRITERS' LABORATORIES, BOARD OF STANDARDS AND APPEALS, NEW YORK STATE MEA, NATIONAL ELECTRICAL CODE, NEW YORK CITY ELECTRICAL CODE AND NEW YORK CITY ELECTRICAL ADVISORY BOARD, ETC. APPROVALS. SHOULD THE ITEM BE FOUND NOT TO HAVE SUCH APPROVAL, IT SHALL BE REPLACED BY THE CONTRACTOR AT NO ADDITIONAL COST TO THE OWNER.
H. SUBSTITUTIONS: DEVIATIONS FROM CONTRACT DOCUMENTS AND SUBSTITUTION OF MATERIALS OR EQUIPMENT FOR THOSE SPECIFIED SHALL BE REQUESTED INDIVIDUALLY IN WRITING. FURNISH INFORMATION AS REQUIRED TO DEMONSTRATE THAT THE ARTICLE, MATERIAL, PRODUCT OR PROCESS TO BE USED IS EQUALLY COMPARABLE TO THAT SPECIFIED IN QUALITY, FINISH, DESIGN, EFFICIENCY, DURABILITY AND GENERAL APPEARANCE, AND HAS BEEN ELSEWHERE DEMONSTRATED TO BE SERVICEABLE FOR THE PURPOSES FOR WHICH IT IS INTENDED. IF TESTS OR DEMONSTRATIONS ARE REQUIRED BY THE OWNER'S REPRESENTATIVES, THE COST OF SUCH TESTS OR DEMONSTRATIONS SHALL BE BORNE BY THE CONTRACTOR. DESCRIBE REASON FOR CHANGE, CONNECTIONS TO ADJACENT MATERIALS, ELECTRICAL SERVICES, SERVICE ACCESS REQUIREMENTS, DIFFERENCES IN OPERATING CHARACTERISTICS OR CYCLES AND ALL OTHER SIGNIFICANT CONSIDERATIONS OF DEVIATION. CONTRACTOR TO ASSUME FULL RESPONSIBILITY FOR SAFETY, COORDINATION WITH OTHER TRADES, OPERATION AND PERFORMANCE OF ALTERED SYSTEM.
I. THIS CONTRACTOR IS TO OBTAIN A COPY OF THE BUILDING RULES AND REGULATIONS PRIOR TO BID SUBMISSION. ALL WORK MUST BE INSTALLED IN ACCORDANCE WITH THE BUILDING RULES AND REGULATIONS. DETERMINE REQUIREMENTS AND THE EXTENT OF PREMIUM TIME WORK REQUIRED BY BUILDING, FOR THE PURPOSE OF THE BID ASSUME ANY NOISY WORK (E.G., CHOPPING, CORE DRILLING, WELDING, BRAISING, SOLDERING, ETC.) AND BASE BUILDING SYSTEM INTERRUPTIONS ARE TO BE PERFORMED OUTSIDE NORMAL BUSINESS HOURS.
J. REMOVAL, TEMPORARY CONNECTIONS AND RELOCATION OF CERTAIN EXISTING WORK WILL BE NECESSARY FOR THE INSTALLATION OF THE NEW SYSTEMS. ALL EXISTING CONDITIONS ARE NOT COMPLETELY DETAILED ON THE DRAWINGS. THE CONTRACTOR SHALL SURVEY THE SITE AND MAKE ALL NECESSARY CHANGES REQUIRED BASED ON EXISTING CONDITIONS FOR PROPER INSTALLATION OF WORK.
K. ALL NECESSARY CUTTING AND PATCHING IN FLOOR SLABS, ROOF SLABS, WALLS, AND CEILINGS FOR THE HVAC WORK SHALL BE PERFORMED BY THIS CONTRACTOR. RESTORE TO MATCH EXISTING CONDITIONS.
L. WHERE PIPE AND/OR DUCTWORK PENETRATE WALLS, THE SPACE BETWEEN THE INSULATION AND THE WALL SHALL BE CAULKED WITH NON-COMBUSTIBLE MATERIAL IN AN APPROVED MANNER. ALL PIPING AND/OR DUCTWORK TO BE INSTALLED ABOVE HUNG CEILING UNLESS OTHERWISE NOTED ON DRAWINGS. THE CONTRACTOR SHALL COORDINATE WITH ARCHITECTURAL DRAWINGS FOR ALL CEILING ELEVATIONS.
M. ACCESS DOORS IN FINISHED CONSTRUCTION: THE CONTRACTOR SHALL PREPARE A LIST OF ALL ACCESS DOORS (MINIMUM 18"X18") REQUIRED FOR OPERATION AND MAINTENANCE OF ALL CONCEALED EQUIPMENT AND OTHER DEVICES. WHERE SHALL BE SUPPLIED TO THE GENERAL CONTRACTOR FOR INSTALLATION. THE COST TO FURNISH AND INSTALL ACCESS DOORS SHALL BE INCLUDED IN THIS CONTRACTORS BID. THIS CONTRACTOR IN ADVANCE OF CEILING INSTALLATIONS SHALL SUITABLY FIELD TAG AND IDENTIFY ALL CONCEALED EQUIPMENT, VALVES, DAMPERS, ETC., WHICH REQUIRE ACCESS DOOR PROVISIONS.
N. NEW DUCTWORK SHALL ARRIVE ON THE CONSTRUCTION SITE SEALED AND REMAIN PROTECTED FROM DEBRIS THROUGHOUT CONSTRUCTION PRIOR TO FINAL INSTALLATION. AIR DISTRIBUTION ACCESSORIES AND INTERNAL COMPONENTS OF ALL HVAC EQUIPMENT SHALL BE SEALED AND PROTECTED FROM DEBRIS WHILE ON THE CONSTRUCTION SITE PRIOR TO FINAL CONNECTION AND START-UP.
O. ALL VOLATILE ORGANIC COMPOUND (VOC) LIMITS OF ADHESIVES, SEALANTS AND SEALANT PRIMERS MUST COMPLY WITH SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD) RULE #1168, AMENDMENT DATE OF JANUARY 7, 2005.

1.02 SCOPE OF WORK
A. THE CONTRACTOR SHALL FURNISH AND INSTALL AN HVAC SYSTEM COMPLETE WITH ALL EQUIPMENT, DUCTWORK, PIPING, INSULATION, CONTROLS, ACCESSORIES AND ASSOCIATED WORK IN ACCORDANCE WITH THE NEW YORK CITY BUILDING CODE. ALL NATIONAL, STATE AND LOCAL AUTHORITIES HAVING JURISDICTION, BUILDING MANAGEMENT, DESIGN DRAWINGS AND THIS SPECIFICATION.
B. THE WORK SHALL INCLUDE ALL LABOR, MATERIALS, EQUIPMENT, HOISTING AND RIGGING, BREAKDOWN AND SETUP OF EQUIPMENT FOR INSTALLATION, SCAFFOLDING, AND SERVICES TO COMPLETE THE SYSTEM AND PROVIDE THE OWNER WITH A FULLY OPERATIONAL SYSTEM. ANY EQUIPMENT, PARTS, MATERIALS, ACCESSORIES, OR LABOR THAT IS NECESSARY FOR PROPER PERFORMANCE OF THE MECHANICAL WORK ALTHOUGH NOT SPECIFICALLY MENTIONED HEREIN OR SHOWN ON THE DRAWINGS, SHALL BE FURNISHED AND INSTALLED WITHOUT ADDITIONAL COST WHEN INSTALLATION OF A PART OF ANY SYSTEM (PLUMBING, HEATING, AIR CONDITIONING, ELECTRICAL OR OTHERWISE) REQUIRES A SHUTDOWN OF ANY OPERATING SYSTEM, CONNECT THE PARTIAL SYSTEM ONLY AFTER NOTIFICATION TO AND WITH APPROVAL OF THE OWNER. COORDINATE ACTIVITIES CLOSELY WITH THOSE SUBCONTRACTORS SO THE OPERATION IS RESTRICTED TO AS SHORT AN INTERVAL AS POSSIBLE AND "OUT OF SERVICE" TIME OF THESE FACILITIES IS KEPT TO A MINIMUM. ANY SHUTDOWN OF THE ELECTRICAL SYSTEM WILL BE DONE OUTSIDE OF HOURS AS APPROVED BY OWNER.
C. IT IS IMPERATIVE THAT EXISTING SYSTEMS BE MAINTAINED IN CONTINUOUS OPERATION DURING THE COURSE OF CONSTRUCTION; IF SHUTDOWNS ARE REQUIRED TO PERMIT THE DISCONNECTION AND REMOVAL OR RECONNECTION OF EXISTING WORK, OR FINAL CONNECTION TO BE MADE TO AN EXISTING SYSTEM, THEY SHALL OCCUR ONLY DURING OFF-HOURS AND ONLY AFTER PROPER PERMISSION HAS BEEN OBTAINED FROM BUILDING MANAGEMENT.
D. THE BUILDING MANAGEMENT REQUIRES NOT LESS THAN SEVEN DAYS NOTICE FOR SHUTDOWN OF ANY BUILDING SYSTEM.
E. MAKE AN ACCURATE TAKE-OFF ALL EXISTING EQUIPMENT, DUCTWORK, PIPING, CONDUIT, PANELBOARDS, WIRING DEVICES, AND OTHER ACCESSORIES BEING REMOVED DURING DEMOLITION AND INCLUDE THE COST FOR DISCONNECTING AND REMOVAL OF STATED EQUIPMENT, ETC. INTO THE BASE BID. REMOVALS SHALL BE AS SPECIFIED AND/OR AS INDICATED ON THE DRAWINGS. IN CERTAIN CASES, EQUIPMENT OR MATERIALS DESIGNATED FOR REMOVAL SHALL REMAIN THE PROPERTY OF THE OWNER AND SHALL BE TURNED OVER AT LOCATIONS IN THE BUILDING AS DIRECTED BY THE OWNER.
F. PLAN INSTALLATION OF NEW WORK AND CONNECTIONS TO EXISTING WORK TO INSURE MINIMUM INTERFERENCE WITH REGULAR OPERATION OF EXISTING FACILITIES. ALL SYSTEM SHUTDOWNS AFFECTING OTHER AREAS SHALL BE COORDINATED WITH BUILDING MANAGEMENT.
G. THIS TENANT/OWNER SHALL PROCURE THE SERVICES OF A THIRD PARTY INSPECTION COMPANY TO PERFORM ALL SPECIAL INSPECTIONS IN ACCORDANCE WITH THE NEW YORK CITY BUILDING CODE. SECURE ALL REQUIRED PERMITS AND APPROVALS AND TRANSMIT SAME TO THE OWNER. CONTRACTOR SHALL BE RESPONSIBLE FOR ALL FEES.

1.03 SHOP DRAWINGS, EQUIPMENT SUBMISSION, MAINTENANCE MANUALS
A. SUBMIT ONE (1) REPRODUCIBLE AND ONE (1) PRINT OF THE SHEET METAL AND PIPING SHOP DRAWINGS, 3/8" SCALE, CERTIFIED BY ALL TRADES THAT COORDINATION HAS BEEN ESTABLISHED.
B. SUBMIT THREE (3) COPIES OF ALL SHEET METAL AND PIPING SHOP STANDARDS LEAKAGE TEST CERTIFICATION, AIR AND WATER BALANCING REPORTS, AND CERTIFIED EQUIPMENT CUTS WITH CONSTRUCTION WIRING DIAGRAMS, AND AUTOMATIC TEMPERATURE CONTROL SHOP DRAWINGS INCLUDING CONTROL AND POWER WIRING DIAGRAMS, SEQUENCE OF OPERATIONS AND ALL CUTS OF EQUIPMENT AND DEVICES.
C. SUBMIT FOUR (4) BOOK BOUND OPERATING AND SERVICE MANUALS WHICH SHALL INCLUDE COPIES OF ALL AS-BUILT SHOP DRAWINGS FOLDED AND PLACED INTO BINDER POCKETS, AS-BUILT DRAWINGS IN ELECTRONIC FORMAT, COPIES OF REVIEWED EQUIPMENT CUTS FOR INSTALLED EQUIPMENT, COPIES OF EQUIPMENT START UP CHECKLISTS, AIR AND WATER BALANCING REPORTS, LEAK TESTS, HYDROSTATIC TESTS, WATER TREATMENT AND CHEMICAL CLEANING CERTIFICATION. CONTRACTOR SHALL INSTRUCT OWNERS PERSONNEL ON THE OPERATION OF ALL HVAC SYSTEMS.
D. AS WORK PROGRESSES AND FOR DURATION OF CONTRACTOR, MAINTAIN COMPLETE AND SEPARATE SET OF PRINTS OF CONTRACT DRAWINGS AT THE JOB SITE. RECORD WORK COMPLETED AND ALL CHANGES FROM ORIGINAL CONTRACT DRAWINGS CLEARLY AND ACCURATELY INCLUDING WORK INSTALLED AS A MODIFICATION OR ADDITION TO THE ORIGINAL DESIGN. RECORD VALVE TAGS AS THEY ARE INSTALLED. FINAL SUBMISSION OF REPRODUCIBLE AS-BUILT DRAWINGS ARE TO BE SIGNED AND CERTIFIED BY INSTALLING CONTRACTOR THAT THIS IS THE AS-BUILT CONDITION OF THE WORK. AS-BUILT SHOP DRAWINGS SHALL BE SUBMITTED IN DRAWING AND ELECTRONIC FORMAT (AUTOCAD 2007 MINIMUM).

PART 2- PRODUCT/APPLICATION

2.01 DUCTWORK
A. PROVIDE ALL SUPPLY, RETURN, EXHAUST, AND OUTSIDE AIR SHEET METAL DUCTWORK, FITTINGS, DAMPERS, TURNING VANES, ACCESS DOORS, PLENUMS, FLEXIBLE CONNECTIONS, AND SUPPORTS AND PERFORM LEAK TEST PER LATEST SMACNA STANDARDS AND NFPA90A AS MODIFIED BY N.Y.C. BUILDING CODE. ALL DUCTWORK JOINTS SHALL BE SEALED AIR TIGHT WITH APPROVED DUCT SEALANT, SIMILAR TO 3M-900.
B. ALL LOW PRESSURE DUCTS EXPOSED IN OCCUPIED AREAS, OTHER THAN MECHANICAL AND FAN ROOMS FABRICATED WITH HEMMED "S" SLIPS. REINFORCE JOINTS OF DUCTS OVER 30" WIDE WITH FLAT BARS OR FLAT BARS AND 3/8" RODS FOR DUCTS OVER 54" WIDE. TOP JOINT WITH BAR SKIP UNDER 31" WIDTH AND REINFORCED BAR SKIP FOR 31" AND LARGER IN WIDTH.
C. ROUND DUCTS SPIRAL LOCK, G.I. COMPANY, SHEET METAL PRODUCTS, UNITED SHEET METAL, PACIFIC AIR PRODUCTS, OR AS APPROVED. ROUND DUCTS OVER 60" WITH BUTT WELDED, LONGITUDINAL SEAMS, AND FLANGE JOINTS.
D. FITTINGS IN ROUND DUCTS SHALL BE NO LIGHTER THAN 20 GAUGE, AND WELDED. G. 1. COMPANY, SHEET METAL PRODUCTS, UNITS SHEET METAL, PAACH TEE TAKE-OFFS MADE WITH "CON-T" TYPE CONICAL TEE FITTINGS. WHERE MAIN DUCT REDUCES IN SIZE AFTER TAKE-OFF, USE "CON-T" OR TURNS, AND 3-PIECE FOR 45 DEGREE TURNS.
E. CONTRACTOR SHALL ADHERE TO THE FULL INSIDE CROSS SECTIONAL DUCTWORK AREAS SHOWN ON THE DRAWINGS AND PROVIDE ALL TRANSITIONS AND OFFSETS AS REQUIRED TO MEET FIELD CONDITIONS, ACCOMMODATE EQUIPMENT MAINTENANCE REQUIREMENTS AND COORDINATE WITH ALL TRADES. ALL FIELD CONDITIONS WHICH REQUIRE MODIFIED TRANSITIONS WILL NOT BE APPROVED WITHOUT PRIOR ENGINEER APPROVAL THROUGH SHOP DRAWING OR RFI.
F. FOR DUCTS WITH ACOUSTICAL LINING THE SIZES SHOWN ON THE PLAN SHALL BE THE CLEAR INSIDE DIMENSIONS.
G. ALL OPEN-ENDED RETURN, TRANSFER OR EXHAUST DUCTS SHALL BE PROVIDED WITH WIRE MESH SCREENS.
H. NEW AND EXISTING DUCTWORK TO BE REUSED SHALL HAVE PRESSURE CLASSIFICATION, SEALING REQUIREMENTS AND LEAKAGE TESTING AS LISTED BELOW UNLESS OTHERWISE SPECIFIED OR SHOWN ON THE DRAWINGS:
1.4" CLASS: ALL SUPPLY DUCTWORK FROM DISCHARGE OF AIR UNITS TO INLETS OR TERMINAL BOXES. SEAL CLASS A, LEAKAGE CLASS 6 (RECTANGULAR) OR CLASS 3 (ROUND). PROVIDE TDF FLANGE CONNECTIONS FOR ALL SYSTEM 4 PRESSURE CLASS AND ABOVE.
2. 2" CLASS: ALL OTHER LOW PRESSURE DUCTWORK. SEAL CLASS B, LEAKAGE CLASS 24 (RECTANGULAR) OR CLASS 12 (ROUND).
3. LEAKAGE TESTING:
A) FOR 4" CLASS DUCTWORK, ALL DUCTWORK SHALL BE TESTED TO VERIFY THAT ACTUAL AIR LEAKAGE IS BELOW ALLOWABLE AMOUNTS AS DEFINED BY ASHRAE 90.1 OR THE ENERGY CODE AS INDICATED ON THE PROJECT'S COMCHECK COMPLIANCE CERTIFICATE, WHICHEVER IS MORE STRINGENT. TESTING SHALL INCLUDE ALL NEW DUCTWORK UPSTREAM OF FAN POWERED TERMINAL UNITS INCLUDING, BUT NOT LIMITED TO MAINS, BRANCHES, AND THE PRIMARY AIR INLET BRANCHES TO EACH FP-VAV. WORK SHALL BE WITNESSED BY THE BUILDING'S CHIEF ENGINEER OR AUTHORIZED REPRESENTATIVE AND A COPY OF THE CERTIFICATION SHALL BE SUBMITTED TO THE BUILDING MANAGEMENT.
B) ALL NEW LOW PRESSURE DUCTWORK (2" CLASS) SHALL BE TESTED ON AN AS-NEEDED BASIS AT THE ENGINEERS DISCRETION OR IF BALANCING AIR QUANTITIES CAN NOT BE MET. IF SPECIMEN FAILS TO MEET ALLOTTED LEAKAGE LEVEL, THE CONTRACTOR SHALL MODIFY TO BRING IT INTO COMPLIANCE AND SHALL RETEST IT UNTIL ACCEPTABLE LEAKAGE IS DEMONSTRATED. TESTS AND NECESSARY REPAIRS SHALL BE COMPLETED PRIOR TO CONCEALMENT OF DUCTS.
F. MATERIALS:
1.SHEETMETAL: HOT-DIPPED GALVANIZED SHEETMETAL WITH G60 COMMERCIAL COATING ACCORDING TO ASTM A653 & A924 FOR ALL DUCTWORK UNLESS OTHERWISE SPECIFIED.
2.ALUMINUM: ALLOY 3003-H14, OF THICKNESS REQUIRED BY THE SMACNA DUCT CONSTRUCTION STANDARDS. PROVIDE FOR ALL DUCTWORK EXPOSED TO WEATHER AND MOISTURE INCLUDING OUTSIDE AIR DUCTS WITHIN 10 FEET OF LOUVERS AND TOILET ROOMS EQUIPPED WITH BATHS OR SHOWERS.
3.FLEXIBLE CONNECTIONS AT FANS SHALL BE NEOPRENE COATED, FLAME RETARDANT GLASS FABRIC (COMPLYING WITH NFPA 90), 30 OUNCES PER SQ. YD. WITH SEWED AND CEMENTED SEAMS.
G. PROVIDE MANUAL BALANCING DAMPERS AS REQUIRED TO PROPERLY BALANCE THE AIR DISTRIBUTION SYSTEM AS SHOWN ON DRAWINGS AND AS LISTED BELOW:
1.ALL SUPPLY AIR MAIN BRANCHES FROM TRUNK, EACH SPLIT, AND ALL SUB-BRANCHES FROM MAINS SHALL HAVE BALANCING DAMPERS.
2.EXHAUST AND RETURN MAIN BRANCHES FROM TRUNK, EACH SPLIT AND ALL SUB-BRANCHES FROM MAINS SHALL HAVE BALANCING DAMPERS.
3.IF DAMPER IS NOT ACCESSIBLE, OR IS LOCATED ABOVE A PLASTER OR DRYWALL CEILING, PROVIDE A REMOTE DAMPER ACTUATOR AND LOCATE AS MAGNETIC AND GIVE THE GENERAL ARRANGEMENT ONLY. NO ADDED COMPENSATION WILL BE PERMITTED FOR VARIATIONS DUE TO FIELD CONDITIONS. EXACT LOCATIONS AND ARRANGEMENTS SHALL BE DETERMINED IN THE FIELD ON THE BASIS OF DETAILS INDICATED ON APPROVED SHOP DRAWINGS, AND SUPPLEMENTARY INFORMATION ISSUED BY THE ENGINEER, AND SHALL PROVIDE FOR OPERATING EFFICIENCY, NEATNESS OF APPEARANCE, AND EASE OF MAINTENANCE.
H. FIRE DAMPERS AND FIRE/SMOKE DAMPERS:
1.PROVIDE ALL FIRE DAMPERS, SMOKE DAMPERS, COMBINATION FIRE/SMOKE DAMPERS, SMOKE DETECTORS, AND ASSOCIATED CONTROLS AND ALARMS AS REQUIRED BY CODE.
2.DAMPERS SHALL GUARANTEE AND SERVICE THE ENTIRE INSTALLATION FOR A PERIOD OF ONE YEAR FROM THE DATE OF THE FINAL ACCEPTANCE OF THE INSTALLATION. THE CONTRACTOR SHALL, DURING THE PERIOD OF THE GUARANTEE, REPLACE OR REPAIR AT HIS OWN EXPENSE ANY PIECE OF EQUIPMENT AND/OR MATERIAL WHICH IS FOUND TO BE DEFECTIVE. THE REPLACEMENT OR REPAIR SHALL BE PERFORMED THE SAME DAY OF NOTIFICATION IN AN EMERGENCY FASHION WHEN NOTIFIED BY THE OWNER OR AUTHORIZED REPRESENTATIVE. THE CONTRACTOR SHALL ALSO REPAIR ALL DAMAGE TO SURROUNDING WORK CAUSED BY THE FAILURE, REPAIR OR REPLACEMENT OF DEFECTIVE EQUIPMENT. ALL REFRIGERATION COMPRESSORS SHALL HAVE A FACTORY GUARANTEE INCLUDING PARTS AND LABOR FOR FIVE YEARS TOTAL. THE FINAL ACCEPTANCE WILL BE MADE AFTER THE CONTRACTOR HAS ADJUSTED HIS EQUIPMENT, BALANCED THE VARIOUS SYSTEMS, DEMONSTRATED THAT IT FULFILLS THE REQUIREMENTS OF THE DRAWINGS AND SPECIFICATION, AND HAS FURNISHED ALL THE REQUIRED CERTIFICATES OF INSPECTION AND APPROVALS.
3.FIRE DAMPER SHALL BE FUSIBLE LINK TYPE (165 DEGREE F.), TYPE B SHUTTER OUT OF THE AIR STREAM AS MANUFACTURED BY POTTORFF MODEL VFD-1D (1-1/2 HR RATED) OR MODEL VFD-3D (3HR RATED) AS REQUIRED OR APPROVED EQUAL.
4.COMBINATION SMOKE/FIRE DAMPERS SHALL BE CLASS II, REMOTE RESETTABLE, MULTI-BLADE TYPE WITH FIRESTAT HEAT SENSOR DEVICE AND 120-VOLT ACTUATOR MOUNTED OUT OF THE AIR STREAM, WITH DAMPER OPERATOR AND BLADE POSITION INDICATOR SWITCHES. PROVIDE MOTOR MOUNT BRACKET STRENGTHENER FOR DAMPERS OVER 10" IN HEIGHT. PROVIDE WELDED VERTICAL STIFFENER AT EACH CORNER TO PREVENT DAMPER MISALIGNMENT.
5.COMBINATION FIRE/SMOKE DAMPERS SHALL BE MANUFACTURED BY POTTORFF MODEL FSD-142 (1-1/2 HR RATED) OR MODEL FSD-342 (3HR RATED) AS REQUIRED OR APPROVED EQUAL.
6.THE HVAC CONTRACTOR SHALL PROVIDE ALL DEVICES, RELAYS, END SWITCHES, E/P SWITCHES, CONTROL COMPONENTS, AIR PIPING, POWER WIRING, CONTROL WIRING AND INTERLOCK WIRING AS REQUIRED TO ACCOMPLISH THE SEQUENCE OF OPERATION FOR THESE DAMPERS.
7.DUCTWORK SHALL BE TRANSITIONED LARGER AT ALL FIRE AND FIRE/SMOKE DAMPERS SUCH THAT THE NET FREE AREA OF THE DUCTWORK IS NOT COMPROMISED.
8.CONTRACTOR TO PROVIDE A 12"X12" ACCESS DOOR IN THE DUCTWORK FOR ACCESS TO INTERNAL COMPONENTS OF THE DAMPER.
I. PROTECTION PANS AND DRIP PANS:
1.PROVIDE PROTECTION PANS UNDER NEW CEILING MOUNTED AIR CONDITIONING UNITS, PIPES PASSING THROUGH SWITCHGEAR ROOMS OR OVER ELECTRIC EQUIPMENT. THE PANS SHALL BE CONSTRUCTED OF GALVANIZED STEEL, SUITABLE REINFORCEMENT TO PREVENT SAGGING.
2.EDGES OF PANS SHALL BE TURNED UP 2" ON ALL SIDES WITH CORNERS SEALED TO MAKE PAN WATERTIGHT.
3.PAN SHALL BE SUPPORTED BY PIPE HANGERS AND SHALL DRAIN CLEAR OF ELECTRICAL EQUIPMENT.
4.PROVIDE A 3/4" DRAIN OPENING FOR EACH PAN TERMINATING ABOVE NEAREST VENTILATION OR FLOOR DRAIN. SLOPE AND DRAIN ALL DUCTS EXPOSED TO MOISTURE; CONSTRUCT OF ALUMINUM AND DO NOT INTERNALLY LINE.
J. SLOPE AND DRAIN ALL DUCTS EXPOSED TO MOISTURE, CONSTRUCT OF ALUMINUM AND DO NOT INTERNALLY LINE
K. AUTOMATIC VOLUME DAMPERS: PROVIDE WEATHERPROOF COMPONENTS FOR DAMPERS IN A MOISTURE ENVIRONMENT.

2.02 GRILLES, REGISTERS AND DIFFUSERS
A. PROVIDE ALL AIR OUTLETS AND RETURNS OF THE TYPE AND SIZES, AS SELECTED AND INDICATED ON DRAWING. ALL DUCTED RETURN AND EXHAUST GRILLES SHALL HAVE OPPOSED BLADE DAMPERS UNLESS OTHERWISE NOTED. PROVIDE FRAMES AND MOUNTING TYPES AS REQUIRED TO MATCH SURROUNDING CEILING CONSTRUCTION. FINISHES TO BE SELECTED BY THE ARCHITECT.
B. ALL CEILING TYPE AIR DIFFUSERS SHALL BE PROVIDED WITH EQUALIZING DEFLECTOR.
C. A SCHEDULE OF DIFFUSERS, GRILLES AND REGISTERS WITH MANUFACTURERS MODELS, SIZES, ACCESSORIES, FINISHES, ETC., SHALL BE SUBMITTED FOR APPROVAL AND NO EQUIPMENT SHALL BE RELEASE FOR FABRICATION AND DELIVERY
D. DIFFUSERS SHOWN ON DIFFUSER SCHEDULE SHALL BE CHANGED TO MATCH EXISTING DIFFUSER TYPE WHERE EXISTING DIFFUSERS ARE REMAINING.
E. ALL UNUSED PATTERNS SHALL BE PROVIDED WITH PATTERN CONTROL VANES. ALL ADJUSTABLE PATTERN DEFLECTORS SHALL BE FIELD ADJUSTED TO OPTIMIZE AIR DISTRIBUTION PREVENTING DRAFT CONDITIONS. CONTRACTOR SHALL PLAN FOR A SECOND COMFORT FIELD ADJUSTMENT PER OWNER/ENGINEER DIRECTION.
F. ALL UNUSED PATTERNS OF SUCH LINEAR DIFFUSERS SHALL BE USED AS RETURN GRILLE AND SHOULD BE PROVIDED WITH LIGHT SHIELDS.
G. ALL LINEAR DIFFUSERS AND DIFFUSERS IN INACCESSIBLE CEILINGS SHALL BE PROVIDED WITH A REMOTE OPERATED OPPOSED BLADE DAMPER AND A 3-FOOT (MINIMUM) FLEXIBLE ADJUSTMENT CABLE WITH 1/8-INCH KEY OPERATOR.

2.03 PIPING
A. PROVIDE PIPING WHICH IS SCHEMATICALLY INDICATED AND SIZED ON DRAWINGS. PIPING TO BE INSTALLED TO MEET SPECIFIED HEADROOM OR FIELD CONDITIONS AND SHALL CONFORM TO LATEST ASME CODES FOR PRESSURE PIPING. PIPE MATERIALS AND FITTING MATERIALS SHALL BE AS NOTED IN THE PIPING SCHEDULES SHOWN ON DRAWINGS.
B. PIPING, FITTINGS, AND ALL PIPE APPURTENANCES SHALL BE SUITABLE FOR THE PRESSURE AND TEMPERATURE OF SERVICE.
C. PROVIDE DIELECTRIC FITTINGS TO CONNECT DIFFERENT PIPING MATERIALS.
D. PROVIDE AIR VENTS AT EACH HIGH POINT AND VALVED DRAIN CONNECTION AT EACH LOW POINT.
E. PIPING SHALL BE INSTALLED WITH PROPER ANCHORS AND EXPANSION/CONTRACTION DEVICES SUCH AS LOOPS OR APPROVED EXPANSION JOINTS TO PREVENT UNDUE STRAINS ON PIPING OR APPARATUS CONNECTED TO THE PIPING, AS REQUIRED.
F. SUPPORT PIPING WITH HANGERS EQUIPPED WITH INSULATION SADDLES FROM APPROVED CONCRETE INSERTS, EXPANSION SHIELDS, BEAM CLAMPS, AND/OR SUPPLEMENTARY STEEL ANGLES, PLATES, AND CHANNELS. CONTRACTOR SHALL SUBMIT METHOD OF PIPING SUPPORT SIGNED AND SEALED BY A LICENSED PROFESSIONAL ENGINEER FOR REVIEW.
G. UNIONS WITH REMOVABLE SECTIONS OF PIPING SHALL BE INSTALLED AT ALL EQUIPMENT TO PERMIT EASE OF DISCONNECTION FOR EQUIPMENT SERVICE/REMOVALS WITHOUT REMOVAL OF MAJOR PORTIONS OF CONNECTED PIPING.
H. PROVIDE TEES IN PIPING SYSTEM FOR TESTING AND BALANCING,

BALANCED WHEEL. THE HOUSING AND WHEEL SHALL BE DESIGNED FOR QUIET LOW VELOCITY OPERATION. THE FAN HOUSING SHALL BE REMOVABLE FROM THE UNIT WITHOUT DISCONNECTING THE SUPPLY AIR DUCTWORK FOR SERVICING OF THE FAN MOTOR. THE FAN MOTOR SHALL BE AN ECM-2 MICROPROCESSOR CONTROLLED DC TYPE MOTOR WITH INTERNAL PROGRAMMING FACTORY SET FOR THE SPECIFIC UNIT AND FEATURING SOFT START/STOP AND DELAY OFF FEATURE FOR MAXIMUM EFFICIENCY AND QUIET OPERATION. AIR FLOW RATES SHALL BE VARIED ACCORDING TO THE STAGING OF THE UNIT. THERE WILL FURTHER BE PROVISIONS FOR ADJUSTING THE AIR DELIVERY OF THE MOTOR AND BLOWER BY +/- 15% FROM THE RATED AIRFLOW.

VII. ELECTRICAL: CONTROLS AND SAFETY DEVICES WILL BE FACTORY WIRED AND MOUNTED WITHIN THE UNIT. CONTROLS SHALL INCLUDE FAN RELAY, COMPRESSOR CONTACTOR, 24V (50 VA) TRANSFORMER, REVERSING VALVE AND SOLID STATE LOCKOUT CONTROLLER (UPM). THE UPM CONTROLLER SHALL INCLUDE THE FOLLOWING FEATURES:
1. DIAGNOSTIC LED'S
2. LOW PRESSURE BYPASS TIME DELAY
3. ANTI-SHORT CYCLE TIME DELAY
4. RANDOM START TIME DELAY AND ONE TIME INTELLIGENT RESET.

VIII. A LOCK-OUT INDICATING TERMINAL SHALL BE PROVIDED IN THE LOW VOLTAGE CIRCUIT. SAFETY DEVICES INCLUDE A LOW PRESSURE CUTOUT SET AT 40 PSIG FOR LOSS OF CHARGE PROTECTION AND A HIGH PRESSURE CUTOUT CONTROL SET AT 600 PSIG. THE ECM MOTOR INTERFACE BOARD SHALL PROVIDE A SCREW TYPE TERMINAL BLOCK FOR THERMOSTAT CONNECTION, LED'S TO INDICATE THERMOSTAT STATUS AND AIR DELIVERY. IT SHALL ALSO PROVIDE A MEANS OF CHANGING THE MOTOR PROGRAM TO ANY OF UP TO FOUR PRE-PROGRAMMED OPTIONS. DIRECT WIRING OF THE MOTOR CONTROL HARNESS TO THE THERMOSTAT IS UNACCEPTABLE. A TERMINAL BLOCK WITH SCREW TERMINALS SHALL BE PROVIDED FOR CONTROL WIRING. A CONDENSATE OVERFLOW SWITCH SHALL BE FACTORY INSTALLED TO STOP COMPRESSOR OPERATION IF DRAIN PAN OVERFLOW IS EMINENT. AN ENERGY MANAGEMENT RELAY TO ALLOW UNIT CONTROL BY AN EXTERNAL SOURCE SHALL BE FACTORY INSTALLED.

IX. PIPING: SUPPLY, RETURN WATER AND CONDENSATE DRAIN CONNECTIONS SHALL BE BRASS FEMALE PIPE THREAD FITTINGS AND MOUNTED FLUSH TO CABINET EXTERIOR.

X. PROVIDE THE FOLLOWING ACCESSORIES:
1. HOT GAS BYPASS - HOT GAS BYPASS VALVING SHALL BE FACTORY MOUNTED AND INSTALLED INTEGRAL TO THE UNIT. THE HOT GAS BYPASS VALVE SHALL RUN OFF REFRIGERANT PRESSURE AND ALLOW UNIT TO RUN ON LOW LOAD CONDITIONS.
2. EXTRA QUIET SOUND PACKAGE - THE UNIT SHALL BE PROVIDED WITH A FACTORY BUILT AND TESTED EXTRA QUIET SOUND PACKAGE CONSISTING OF ½" INCH HIGH DENSITY CLOSED CELL FOAM INSULATION WRAP ON THE COMPRESSOR AND AN INSULATED DIVIDER PANEL BETWEEN THE AIR- HANDLING SECTION AND THE COMPRESSOR SECTION TO MINIMIZE THE TRANSMISSION OF COMPRESSOR NOISE. ALL BULK HEADS ARE TO BE PROVIDED WITH 1 RUBATEX INSULATION. THE CABINET INTERIOR SHALL BE INSULATED WITH 2" THICK, MULTIDENSITY COATED GLASS FIBER. THE UNIT SHALL ALSO HAVE A HEAVY GAUGE STEEL COMPRESSOR BASE PAN WITH WELDED HEAVY GAGE MOUNTING RAILS FOR COMPRESSOR MOUNTING. THE COMPRESSORS ARE TO BE MOUNTED ON RESILIENT RUBBER GROMMETS FOR VIBRATION DAMPING.
3. LOW WATER TEMPERATURE PACKAGE FOR UNIT OPERATION DOWN TO 45F EWT IN THE COOLING MODE WITHOUT THE USE OF A WATER REGULATING VALVE CONSISTS OF:
1. AN EXTERNALLY EQUALIZED TXV FOR HEAD PRESSURE CONTROL.
2. WRAPPED CONDENSER.
3. LOW PRESSURE TIME DELAY RELAY

F. CEILING MOUNTED CONTROL UNIT.

I. PROVIDE A SELF-CONTAINED, NON-DUCTED AIR PURIFICATION UNIT DESIGNED FOR FLUSH INSTALLATION IN A STANDARD 48" X 24" SUSPENDED "T-GRID" CEILING. UNIT SHALL BE CAPABLE OF DELIVERING 500-600 CFM WITH ALL CLEAN FILTERS INSTALLED AND SHALL INCLUDE AN INTEGRAL FAN ASSEMBLY, MULTI-STAGE FILTRATION, GAS-PHASE ADSORPTION MEDIA, BIPOLAR IONIZATION DEVICE, CONTROLS, AND MONITORING COMPONENTS. UNIT SHALL BE SUITABLE FOR CONTROLLING PARTICULATE MATTER, ODORS, AND VOCS IN OCCUPIED SPACES SUCH AS OFFICES AND FOOD PANTRY AREAS. THE AIR PURIFICATION PROCESS SHALL INCLUDE: A 1-INCH MERV 13 PREFILTER; TWO (2) 2-INCH HONEYCOMB ACTIVATED CARBON PANELS WITH BS-145XL MEDIA FOR GAS-PHASE REMOVAL; A 4-INCH MERV 13 PRIMARY PARTICULATE FILTER; AND ONE (1) BIPOLAR IONIZATION MODULE. UNIT SHALL BE CONSTRUCTED OF 5052 ALUMINUM INTERNALLY LINED WITH 1-INCH ACOUSTICAL INSULATION AND SEALED TO PREVENT AIR BYPASS. PROVIDE WIRELESS REMOTE CONTROL, 0-10 VDC BMS FAN SPEED CONTROL, DIRTY FILTER INDICATION WITH DIFFERENTIAL PRESSURE SWITCHES, AND A UNIT-MOUNTED DISCONNECT. ELECTRICAL SERVICE SHALL BE 120 OR 240 VAC, SINGLE PHASE, 60 HZ.

2.08 MOTOR STARTERS, CONTROL DEVICES AND MOTORS

A. MECHANICAL CONTRACTOR TO FURNISH AND INSTALL STARTERS FOR POWER WIRING BY THE ELECTRICAL CONTRACTOR.

B. MOTOR STARTERS SHALL BE CUTLER HAMMER, WESTINGHOUSE OR ALLEN-BRADLEY MANUFACTURER, SUITABLE FOR WALL OR ANGLE IRON FRAME MOUNTING.

C. ALL STARTERS FOR MOTORS LESS THAN 1/2 HP SHALL BE 120 VOLT, SINGLE PHASE, 60 CYCLE, A.C. SERVICE. MANUAL STARTERS WITH OVERLOAD PROTECTION AND CUTOUT TYPE DISCONNECT SWITCH OR BREAKER MAY BE USED TO CONTROL SUCH MOTORS, EXCEPT WHERE INTERLOCKS OR AUTOMATIC CONTROLS ARE REQUIRED. IN SUCH CASES, MAGNETIC ACROSS-THE-LINE STARTERS SHALL BE FURNISHED.

D. ALL STARTERS FOR MOTORS 1/2 HP TO 75 HP SHALL BE COMBINATION FUSED DISCONNECT, MAGNETIC ACROSS-THE-LINE TYPE WITH FUSIBLE SWITCH. STARTERS 75 HP AND GREATER SHALL BE SOLID STATE ELECTRONIC SOFT START TYPE STARTERS.

E. CONTROLLERS FOR CONDENSATE PUMPS, DUPLEX AIR COMPRESSOR, SUMP AND EJECTOR PUMPS, ETC., SHALL BE FACTORY MOUNTED AND WIRED AS PART OF THE WORK OF THE HEATING, VENTILATING AND AIR CONDITIONING SECTION.

F. ALL MAGNETIC STARTERS SUBJECT TO MANUAL START AND IN DIRECT VIEW OF THE MOTORS THEY CONTROL SHALL HAVE MOMENTARY CONTACT START AND STOP BUTTONS AND PILOT LIGHT BUILT INTO COVER. ALL MAGNETIC STARTERS SUBJECT TO ELECTRICAL INTERLOCK OR AUTOMATIC CONTROL SHALL HAVE HAND-OFF-AUTOMATIC SWITCHES AND PILOT LIGHT BUILT INTO COVER.

G. WHERE STARTERS ARE NOT IN SIGHT OF MOTORS THEY CONTROL, A LOCAL DISCONNECT SWITCH WILL BE PROVIDED BY THE ELECTRICAL CONTRACTOR.

H. PROVIDE ALL STARTERS WITH TRANSFORMERS BUILT INTO EACH STARTER CASING FOR CONTROL CIRCUIT. TRANSFORMERS SHALL SERVE ALL CONTROL CIRCUITS. EACH STARTER SUBJECT TO ELECTRICAL INTERLOCK AND/OR AUTOMATIC CONTROL SHALL HAVE THE NECESSARY AUXILIARY CONTACTS. ONE SET OF TERMINALS SHALL BE PROVIDED FOR EACH CONTROL CIRCUIT. CONTROL CENTERS SHALL BE PROVIDED WITH CONTROL TERMINAL BLOCKS. PROVIDE THREE SETS OF NORMALLY CLOSED OR NORMALLY OPEN CONTACTS.

I. ALL MAGNETIC STARTERS SHALL HAVE THERMAL OVERLOAD IN EACH PHASE LEG AND LOW VOLTAGE PROTECTION.

J. ALL PARTS SUBJECT TO WEAR, ARCING, ETC., SHALL BE REPLACEABLE.

K. ALL WIRING, STARTERS, SWITCHES, ETC., SHALL BE IN FULL ACCORDANCE WITH ALL LOCAL INSURANCE UNDERWRITERS CODE REQUIREMENTS.

L. FURNISH DETAILED COMPOSITE WIRING DIAGRAMS FOR THOSE INSTALLING ELECTRICAL WORK, AND FURNISH SUCH OTHER INFORMATION NECESSARY TO ASSURE THE PROPER CONNECTION, OPERATION AND CONTROL OF MOTORIZED EQUIPMENT, INCLUDING INTERLOCKS, AUTOMATIC OR SAFETY CONTROLS AND AUXILIARY CIRCUITS.

M. FURNISH THE PERTINENT INFORMATION SUCH AS STARTING TORQUE REQUIREMENTS OF HIGH INERTIA EQUIPMENT, SO THAT THE PROPER TYPE STARTER MAY BE PROVIDED BY THE STARTER MANUFACTURER. ALL INFORMATION IS SUBJECTED TO THE REVIEW OF THE ENGINEER.

N. PROVIDE LAMINATED NAME PLATE ATTACHED TO EACH STARTER AND VFD IDENTIFYING THE SYSTEM IT SERVES.

O. STARTERS AND VFD'S SHALL BE PROVIDED WITH ENCLOSURES RATED NEMA 1 FOR INDOOR APPLICATIONS, NEMA 3R WITH ADDITIONAL GASKETING FOR WEATHERPROOF RAINTIGHT OUTDOOR ENCLOSURE OR INDOOR ENVIRONMENTS SUBJECT TO MOISTURE.

P. MOTORS SHALL BE HIGH EFFICIENCY, COMPLY WITH NEMA MG-1 STANDARD AND MEET THE 1992 EPA ENERGY EFFICIENCY ACT AND UTILITY COMPANY REBATE REQUIREMENTS.

Q. PROVIDE VARIABLE FREQUENCY DRIVES (VFD) AS MANUFACTURED BY GENERAL ELECTRIC, MAGNATEK, ABB, OR APPROVED EQUAL BY THE ENGINEER FOR CONTROL OF FANS AND/OR PUMPS AS SHOWN ON THE PLANS AND AS SPECIFIED HEREIN. VFD DISTORTION FACTOR SHALL NOT EXCEED 3% THD (VOLTAGE) AT POINT OF COMMON COUPLING, AS DEFINED BY IEEE 519.1992 AND IN NO CASE SHALL THE CURRENT THD EXCEED 10%. VFDS SHALL INCLUDE THE FOLLOWING:
1. PWM TECHNOLOGY INCORPORATING IGBT.
2. 40 CHARACTER FULL ENGLISH DIGITAL DISPLAY. CODES ARE NOT ACCEPTABLE.
3. DC LINE CHOKE.
4. MANUAL BYPASS AND MANUAL BYPASS CONTACTORS.
5. THREE SETS OF NORMALLY CLOSED OR NORMALLY OPEN CONTACTS.
6. CIRCUIT BREAKER DISCONNECT.
7. VFD DRIVE SERVICE SWITCH.
8. SPEED CONTROL DIAL.
9. THERMAL MOTOR OVERLOADS.
10. 3% AC LINE REACTOR PRE-WIRED AND INSTALLED WITHIN VFD ENCLOSURE.
11. FACTORY START-UP SERVICE INCLUDING COMPONENT TESTING, FIELD CHECK OF CONTROL CONNECTION, AND DOCUMENTS STATING THAT ALL WORK AND DRIVE FUNCTIONS ARE DEEMED ACCEPTABLE.
12. PROGRAMMING OF ALL DRIVE PARAMETERS PARTICULAR TO THIS INSTALLATION.
13. 2 YEAR SITE PARTS AND LABOR WARRANTY AFTER START-UP.

R. VARIABLE FREQUENCY DRIVE MOTORS SHALL COMPLY WITH NEMA MG-1 PART 31.40.4.2 STANDARD SUITABLE FOR VFD OPERATION. CONTRACTOR TO COORDINATE VFD AND MOTOR MANUFACTURERS.

S. ALL VFD DRIVES FOR ALL EQUIPMENT SHALL BE OF THE SAME MANUFACTURER. MECHANICAL CONTRACTOR SHALL COORDINATE VFD DRIVE MANUFACTURER WITH EACH EQUIPMENT VENDOR.

2.09 AUTOMATIC TEMPERATURE CONTROL

A. GENERAL
1. PROVIDE ALL CONTROL, POWER, AND INTERLOCK WIRING INCLUDING CONDUITS AND INSTALL PER THE NEW YORK CITY AND NATIONAL ELECTRIC CODE. SUBMIT TERMINAL TO TERMINAL WIRING DIAGRAM, SEQUENCE OF OPERATION AND CUTS OF ALL COMPONENTS FOR APPROVAL. PROVIDE ALL RELAYS, SWITCHES, DAMPERS AND ACTUATORS, PNEUMATIC EQUIPMENT, PILOT POSITIONERS THERMOSTATS, PANELS, LIMIT SAFETIES, TRANSFORMERS, TIME CLOCKS, CONTROL VALVES AND OTHER DEVICES TO ACCOMPLISH THE DESIRED SEQUENCE OF OPERATION.
2. FURNISH AND INSTALL AS HEREIN SPECIFIED, A COMPLETE AUTOMATIC TEMPERATURE CONTROL SYSTEM OF THE DDC TYPE WITH NATIVE BACNET DEVICES, MANUFACTURED BY SIEMENS INDUSTRY INC. 1000 DEERFIELD PARKWAY, BUFFALO GROVE, IL 60089, PHONE # (212) 258-4000. THE ATC CONTRACTOR SHALL BE AN INDEPENDENT CONTRACTOR NOT AFFILIATED WITH THE MECHANICAL CONTRACTOR.
3. ALL TEMPERATURE CONTROL SYSTEMS AND COMPONENTS ARE TO BE FULLY MODULATING TYPE, EXCEPT WHERE NOTED OTHERWISE.
4. IF NEW WORK IS TO CONNECT TO AN EXISTING SYSTEM, THE PROPOSED NEW SYSTEM TO BE INSTALLED SHALL BE FULLY COMPATIBLE WITH THE EXISTING SYSTEM. THE MANUFACTURER OF THE PROPOSED NEW SYSTEM SHALL PROVIDE ALL REQUIRED INTERFACES OR GATEWAYS TO ENSURE THAT THEIR SYSTEM IS FULLY COMPATIBLE.
5. WHEN CONNECTING TO AN EXISTING BMS WORKSTATION THE CONTRACTOR SHALL UPDATE THE EXISTING BMS WORKSTATION AND GRAPHICS WITH THE NEW SYSTEMS INSTALLED/MONITORED AS PART OF THIS PROJECT.
6. TRAINING
A) THE BMS CONTRACTOR SHALL PROVIDE A MINIMUM OF (8) HOURS TRAINING FOR FACILITY STAFF ON-SITE. TRAINING CAN BE PERFORMED IN SEPARATE (4) HOUR INTERVALS AT THE DISCRETION OF THE OWNER.
B) THE BMS CONTRACTOR SHALL PROVIDE HARDCOPY OF AS-BUILT DRAWINGS AND REVIEW ALL MAINTENANCE REQUIREMENTS AND PROCEDURES FOR ALL EQUIPMENT
7. ALL CONTROLS MUST BE THE PRODUCT OF ONE MANUFACTURER. ALL AUTOMATIC CONTROL VALVES AND DAMPER OPERATORS SHALL BE MANUFACTURED BY THE TEMPERATURE CONTROL MANUFACTURER.
8. THE MANUFACTURER OF THE AUTOMATIC CONTROL EQUIPMENT SHALL SUBMIT THE FOLLOWING FOR APPROVAL: A SCHEMATIC DIAGRAM OF EACH CONTROL SYSTEM WHICH SHALL INDICATE THE PROPER SEQUENCE OF OPERATION AND RANGE OF THE CONTROLS FOR ALL CYCLES, PROVIDE TERMINAL POINT TO TERMINAL POINT ELECTRICAL WIRING DIAGRAMS FOR APPROVAL, A COMPLETE DESCRIPTION OF THE AUTOMATIC OPERATION OF EACH SYSTEM WHERE THE DESCRIPTION INCLUDES THE DUTY OF EACH THERMOSTAT, VALVE, SWITCH, ETC., INCORPORATED IN THE CONTROL SYSTEM WITH A SCHEDULE AND ILLUSTRATION OF ALL CONTROL INSTRUMENTS AND EQUIPMENT INCLUDING CONTROL PANELS AND DEVICES FOR EACH SYSTEM.
9. INDIVIDUAL SMOKE DETECTORS SHALL BE INSTALLED (PROVIDED BY ELECTRICAL CONTRACTOR) IN THE RETURN DUCT OF ALL AIR HANDLING SYSTEMS SHARING A COMMON CEILING OR DUCT PLENUM AS REQUIRED BY CODE.
10. FOR AIR DISTRIBUTION SYSTEMS 2,000 CFM OR LARGER, INSTALL SMOKE DETECTORS (PROVIDED BY ELECTRICAL CONTRACTOR) IN MAIN SUPPLY DUCT (DOWNSTREAM OF AIR FILTERS AND AHEAD OF ANY BRANCH CONNECTIONS) AND MAIN RETURN DUCT (UPSTREAM OF ANY FILTERS AND BEFORE RETURN AIR IS DILUTED WITH OUTDOOR AIR).
11. PROVIDE SMOKE DAMPERS (N.C.) IN THE SUPPLY (AHEAD OF ANY BRANCH CONNECTIONS) AND RETURN DUCTS FOR SUPPLY AIR SYSTEMS 15,000 CFM OR LARGER AND SERVING MORE THAN THE FLOOR ON WHICH THE SYSTEM IS LOCATED. PROVIDE END SWITCHES IN SMOKE DAMPERS FOR CONNECTION TO FIRE ALARM SYSTEM AND HVAC CONTROLS.
12. ALL SMOKE DETECTORS SHALL BE TIED TO THE BUILDING FIRE ALARM SYSTEM. A SIGNAL FROM THE BUILDING FIRE ALARM SYSTEM SHALL AUTOMATICALLY SHUT DOWN SYSTEM FANS AND CLOSE ALL ASSOCIATED SMOKE DAMPERS AND COMBINATION FIRE/SMOKE DAMPERS. SIGNAL, INTERLOCK WIRING, POWER WIRING AND FINAL CONNECTIONS WILL BE PROVIDED BY ELECTRICAL CONTRACTOR.
13. FIRE/SMOKE DAMPERS SHALL REMAIN OPEN AT ALL TIMES DURING NORMAL OPERATION AND SHALL BE CONTROLLED BY THE AUTOMATIC TEMPERATURE CONTROL SYSTEM FOR THE HVAC SYSTEM CONTROL. FIRE/SMOKE DAMPERS SHALL BE UNDER THE FULL CONTROL OF THE FIRE ALARM SYSTEM. CONTROL CONTRACTOR SHALL COORDINATE WITH THE ELECTRIC CONTRACTOR FOR ALL CONTROL WIRING, DEVICES, AND INTERLOCK WIRING TO OPEN, CLOSE AND PROVE FIRE SMOKE DAMPER POSITIONS.
14. ALL AUTOMATIC TEMPERATURE CONTROL SYSTEM SHALL COMPLY WITH THE NEW YORK CITY ENERGY CONSERVATION CODE REQUIREMENTS. ALL HVAC SYSTEM CONTROLS SHALL COMPLY WITH THE FOLLOWING REQUIREMENTS:
A) HEATING AND COOLING TO EACH ZONE SHALL BE CONTROLLED BY A THERMOSTAT. A MINIMUM OF ONE HUMIDITY CONTROL DEVICE SHALL BE INSTALLED PER HUMIDIFICATION OR DEHUMIDIFICATION SYSTEM.
B) ALL ZONE THERMOSTAT SHALL OPERATE WITH A MINIMUM OF 5°F DEADBAND BETWEEN HEATING AND COOLING WITH SETPOINT OVERLAP CAPABILITY.
C) ALL ZONE THERMOSTATS SHALL BE OPERATED VIA THERMOSTATIC SETBACKS CONTROLS OPERATED VIA AN AUTOMATIC TIME CLOCK OR PROGRAMMABLE CONTROL SYSTEM.
D) ALL CONTROLS SHALL HAVE THE ABILITY TO SETBACK TEMPERATURE DOWN TO 55°F OR UP TO 85°F.
E) CONTROLS SHALL BE CAPABLE OF AUTOMATICALLY STARTING AND STOPPING THE SYSTEM FOR SEVEN DIFFERENT DAILY SCHEDULES PER WEEK, CAPABLE OF HAVING SETTING SAVED IN MEMORY FOR 10 HOURS DURING A LOSS OF POWER AND A MANUAL SYSTEM "ON" OVERRIDE FOR UP TO TWO (2) HOURS OR AN OCCUPANCY SENSOR.
F) CONTROL SYSTEM SHALL AUTOMATICALLY RESET SUPPLY AIR TEMPERATURE IN RESPONSE TO BUILDING LOAD OR OUTSIDE AIR TEMPERATURE.
G) THE CONTRACTOR SHALL PROVIDE TESTING TO ENSURE PER OPERATION, CALIBRATION AND ADJUSTMENT OF CONTROLS.
H) OFF-HOURS CONTROLS WITH SETBACK AND/OR SHUTDOWN CAPABILITIES. FOR SPECIFIC OCCUPANCIES AND CONDITIONS, EACH SPACE-CONDITIONING SYSTEM MUST BE PROVIDED WITH CONTROLS THAT CAN AUTOMATICALLY SHUT-OFF THE EQUIPMENT DURING UNOCCUPIED HOURS. THE CONTROL DEVICE SHALL BE AN AUTOMATIC WITH SWITCH DEVICE. THIS CAN BE ACCOMPLISHED WITH A 7-DAY PROGRAMMABLE THERMOSTAT WITH BACKUP CAPABILITIES THAT STORES THE DEVICE'S SCHEDULE FOR AT LEAST 7 DAYS AND THE TIME AND DATE FOR AT LEAST 72 HOURS IF POWER IS LOST.

B. ELECTRIC WIRING:
1. ALL ELECTRIC WORK (EXCEPT FOR MOTOR FEEDERS, WIRING BETWEEN MOTORS, MOTOR CONTROLLERS, FEEDER PANELS, FUSES, CIRCUIT BREAKERS AND BUS BARS) REQUIRED FOR THE AUTOMATIC TEMPERATURE CONTROL SYSTEM SHALL BE PROVIDED BY THIS CONTRACTOR. WORK SHALL INCLUDE BUT NOT BE LIMITED TO TIME SWITCHES, DAMPER MOTORS, DAMPER SWITCHES, ELECTRIC THERMOSTAT, ELECTRIC RELAYS, E/P SWITCHES, INTERLOCKING WIRING, WIRE, CONDUIT, ETC.
1. ALL CONTROL POWER, WIRING AND TRANSFORMERS FOR DAMPERS, ACUATORS, VAV BOXES, CONTROL PANLES, ETC. TO BE PROVIDED BY THE CONTROLS CONTRACTOR FROM A SOURCE DESIGNATED BY THE ELECTRICAL CONTRACTOR. CIRCUITS FOR CONTROL DEVICES HAVE BEEN DESIGNATED IN THE ELECTRICAL PANEL SCHEDULES.
2. THE CONTROL MANUFACTURER SHALL INCLUDE WIRING DIAGRAMS IN HIS SHOP DRAWINGS SUBMITTALS FULLY COORDINATED WITH THE ELECTRICAL CONTRACTORS WORK. IT SHALL BE THE AUTOMATIC TEMPERATURE CONTROL CONTRACTORS RESPONSIBILITY TO PROVIDE ALL WIRING AND CONDUIT AS REQUIRED TO ACHIEVE THE FUNCTION CALLED FOR IN THESE SPECIFICATIONS, CONFORMING WITH LOCAL CODES FOR MATERIAL AND INSTALLATION. THE ELECTRICAL SPECIFICATION FOR THE PROJECT ELECTRICAL WORK IS TO BE FOLLOWED

G. CONTROL PANELS SHALL BE NEMA 1 FOR INDOOR APPLICATIONS, NEMA 3R WITH ADDITIONAL GASKETING FOR WEATHERPROOF RAINTIGHT OUTDOOR ENCLOSURE OR INDOOR ENVIRONMENTS SUBJECT TO MOISTURE. THEY SHALL BE PROVIDED WITH WELDED ANGLE BRACKETS AND A BAKED PRIME COAT ENAMEL FINISH. THE PANEL DOORS SHALL BE HINGED LOCKING TYPE. CONTROL PANELS SHALL CONTAIN ALL CENTRAL CONTROL DEVICES, SUCH AS CONTROLLERS, RELAYS, SWITCHES, PILOT LIGHTS, TERMINAL BLOCKS, AND ALL OTHER ACCESSORIES AS REQUIRED FOR A WORKABLE ENVIRONMENTAL CONTROL SYSTEM. ALL COMPONENTS WITHIN THE CONTROL PANELS SHALL BE PRE-WIRED TO NUMBERED TERMINAL TRIPS, READY FOR FIELD CONNECTION FOR FIELD MOUNTED CONTROL COMPONENTS. PROVIDE ENGRAVED NAMEPLATES TO LABEL THE CONTROLLED EQUIPMENT. PROVIDE A PLASTIC LAMINATED CONTROL SCHEMATIC DRAWING HUNG NEXT TO EACH CONTROL PANEL.

H. THE SYSTEM INSTALLATIONS SHALL BE SUPERVISED BY THE AUTOMATIC CONTROL MANUFACTURER, WHO SHALL COORDINATE WITH AND INSTRUCT PIPING OR SHEET METAL TRADES AS TO TEES OR TAPPINGS TO BE INSTALLED IN PIPING OR EQUIPMENT AND OPENINGS THAT ARE REQUIRED IN SHEET METAL FOR THE SETTING AND INSTALLATIONS OF CONTROL DEVICES THEREIN BY THESE TRADES.

I. ALL ROOM THERMOSTATS/SENSORS AND SWITCH LOCATIONS SHALL BE SUBMITTED FOR REVIEW BY THE ARCHITECT AND ENGINEER PRIOR TO INSTALLATION WHETHER THE DEVICES ARE SHOWN ON PLANS OR NOT.

J. ALL ROOM THERMOSTATS/SENSORS SHALL MATCH THE BUILDING STANDARD.

K. AUTOMATIC VALVES:
1. ALL AUTOMATIC CONTROL AND ISOLATION VALVES SHALL BE OF THE ELECTRONIC TYPE, UNLESS OTHERWISE SPECIFIED, QUIET IN OPERATION, AND SHALL BE ARRANGED TO SPRING RETURN FAIL SAFE, IN A NORMALLY OPEN OR NORMALLY CLOSED POSITION. CONTROL VALVES SHALL BE FULLY PROPORTIONING AND ISOLATION VALVES SHALL BE 2-POSITION.
2. VALVES TO HAVE ADJUSTABLE OPERATING RANGES AND STARTING POINTS TO PROVIDE FLEXIBILITY OF ADJUSTMENT IN SEQUENCING AND THROTTLING. MODULATING VALVES SHALL BE PROVIDED WITH PILOT POSITIONERS
2. VALVES SHALL BE SIZED BY THE TEMPERATURE CONTROL MANUFACTURER AND GUARANTEED TO MEET THE HEATING OR COOLING REQUIREMENTS AS SPECIFIED. CONTROL VALVES SHALL BE SUITABLE FOR PRESSURE CONDITIONS AND CLOSE AGAINST 110% OF PUMP DIFFERENTIAL PRESSURE.
3. ALL VALVE BODIES SHALL HAVE THE SAME PRESSURE CHARACTERISTICS AS THE PIPE IN WHICH IT IS INSTALLED.
4. VALVES 2 INCHES AND SMALLER UNLESS OTHERWISE SPECIFIED SHALL HAVE BRONZE BODIES WITH SCREWED CONNECTIONS. VALVES SHALL BE FISHER TYPE ED, WARREN TYPE 20/70, K&M SERIES GCG, OR AS APPROVED.
5. VALVES BETWEEN 2-1/2" AND 4 INCH UNLESS OTHERWISE SPECIFIED, SHALL HAVE CAST IRON OR CARBON STEEL BODIES WITH FLANGED CONNECTIONS IN ACCORDANCE WITH THE PIPING SPECIFICATIONS. VALVES SHALL BE FISHER STYLE ED, WARREN TYPE 20/70 OR 1800 SERIES GCG, K&M SERIES GCG OR AS APPROVED.
6. AUTOMATIC CONTROL VALVES EXPOSED TO THE ELEMENTS SHALL HAVE ELECTRONIC ACTUATORS WITH ALL REQUIRED ACCESSORIES.

I. AUTOMATIC DAMPERS:
1. PROVIDE CONTROLS FOR ALL THE AUTOMATIC DAMPERS, AS SPECIFIED IN THE DUCTWORK SECTION, AND SHOWN ON THE DRAWINGS.
2. CONTROL MOTORS OR ACTUATORS SHALL BE OF THE ELECTRONIC TYPE, UNLESS OTHERWISE NOTED, OF APPROPRIATE SIZE AND QUANTITIES TO PROVIDE TWO-POSITION OR PROPORTIONING CONTROL ACTION AS SPECIFIED. PROPORTIONING TYPE SHALL BE EQUIPPED WITH PILOT TYPE POSITIONERS. PILOT POSITIONERS SHALL BE SELECTED FOR VARIED SPRING RANGES AND ADJUSTABLE WITHOUT DISMANTLING POSITIONER AND CONTROL MOTOR.
3. AUTOMATIC DAMPERS EXPOSED TO THE ELEMENTS SHALL HAVE ELECTRIC ACTUATORS WITH ALL REQUIRED ACCESSORIES.

I. SEQUENCES OF OPERATION - FURNISH AND ACQUINT ALL DEVICES AS REQUIRED TO PERFORM THE FOLLOWING SEQUENCES OF OPERATION:
1. SERIES FAN POWERED VARIABLE AIR VOLUME (VAV) BOXES:
a) ELECTRICALLY CONTROLLED FPVAV BOXES WITH FAIL IN PLACE DAMPERS SHALL BE CONTROLLED FROM A UNIT MOUNTED DDC CONTROLLER.
b) A DDC SOFTWARE PROGRAM SHALL START THE FPVAV BOXES INDIVIDUALLY WHEN THE RESPECTIVE BASE BUILDING AC SYSTEM IS IN OPERATION.
c) PROVIDE ROOM THERMOSTATS (SENSORS), ONE PER VAV BOX, LOCATED AS INDICATED ON THE DRAWINGS TO MAINTAIN A 75°F SPACE TEMPERATURE (ADJUSTABLE).
(1) WHEN THE SPACE TEMPERATURE RISES, THE CONTROLLER SHALL MODULATE THE PRIMARY AIR VALVE TO ITS MAXIMUM POSITION AS REQUIRED TO MAINTAIN THE SPACE TEMPERATURE SETPOINT.
(2) AS THE SPACE TEMPERATURE DECREASES, THE TERMINAL UNIT DAMPER SHALL MODULATE TO ITS MINIMUM POSITION OF 30%, UNLESS OTHERWISE NOTED.
(3) WHEN THE SPACE TEMPERATURE CONTINUES TO DECREASE, THE OUTPUT OF THE CONTROLLER SHALL MODULATE THE HEATING COIL CONTROL VALVE TO MAINTAIN SPACE TEMPERATURE. THIS CONTRACTOR SHALL PROVIDE THE CONTROL VALVE, ACTUATOR AND CONTROLS.
(4) THE VALVE SHALL FAIL TO THE FULL HEAT POSITION.
d) AS THE ROOM TEMPERATURE VARIES BETWEEN +/-1°F OF THE THERMOSTAT SETPOINT, THE CONTROLLER RESETS AIR FLOW BETWEEN MAXIMUM AND MINIMUM.
e) DURING WARM-UP, THE FPVAV SHALL REMAIN UNDER CONTROL.
f) WHENEVER THE FPVAV IS OFF, THE TERMINAL UNIT DAMPER SHALL MODULATE TO ITS MINIMUM FLOW POSITION.
g) WHEN THE FPVAV IS OFF, THE SET POINT OF THE CONTROLLER SHALL BE LOWERED TO 55 DEGREES F ADJUSTABLE. IF THE TEMPERATURE FALLS BELOW SET POINT, THE CONTROLLER SHALL START THE FAN AND MODULATE THE REHEAT COIL TO MAINTAIN SPACE TEMPERATURE SET POINT.
h) MECHANICAL CONTRACTOR TO COORDINATE THE PROVISION OF ALL POWER, CONTROL WIRING, TRANSFORMERS, DISCONNECT SWITCHES, ETC. WITH A.T.C. CONTRACTOR AS REQUIRED.
i) THERMOSTATS/SENSORS:
(1) THERMOSTATS/SENSORS SHALL BE PROVIDED FOR EACH VAV BOX.
(2) ALL ROOM THERMOSTATS/SENSORS SHALL MATCH THE BUILDING STANDARD.
j) DURING AUTOMATIC FIRE/SMOKE PURGE, THE VAV BOX DAMPER SHALL OPEN.
k) FOR ALL BOXES SERVING AN ENCLOSED ROOM GREATER THAN 500 SQ FT, SENSORS SHALL MONITOR THE CO2 LEVEL IN THE SPACE AND REPORT TO THE BMS SYSTEM. BMS SHALL BE ALARMED IF CO2 LEVEL RISES ABOVE ITS ADJUSTABLE SETPOINT.
l) VAV BOXES CONTROLLED BY MULTIPLE ROOM SENSORS SHALL BE CAPABLE OF BEING CONTROLLED BASED ON EITHER THE WORST-CASE TEMPERATURE OR AN AVERAGE OF THE MULTIPLE SENSORS (OPERATOR SELECTABLE VIA THE BMS) AS REQUIRED.
m) CONTROLLERS AND ACTUATORS SHALL BE SHIPPED TO THE TERMINAL BOX MANUFACTURER FOR FACTORY MOUNTING. STEEL CONTROL ENCLOSURES TO BE PROVIDED BY BOX MANUFACTURER.
n) EACH BOX SHALL BE CONNECTED TO THE LANDLORD BMS AND BE CAPABLE OF PERFORMING THE FOLLOWING MINIMUM REMOTE FUNCTIONS:
(1) COOLING SETPOINT ADJUSTMENT.
(2) INDEX BOX TO MINIMUM OR MAXIMUM FLOW QUANTITY.
(3) MONITOR ZONE TEMPERATURE, SUPPLY AIR TEMPERATURE AND PRIMARY AIRFLOW THROUGH VAV BOX.
(4) MONITOR FAN STATUS
(5) FAN COMMAND ON/OFF

2. EXHAUST/TRANSFER FANS:
a) FANS SERVING PANTRIES, TOILET ROOMS, ETC. SHALL OPERATE CONTINUOUSLY DURING OCCUPIED PERIODS. TIME SCHEDULE SHALL MATCH THE BUILDING HOURS OF OPERATION AND BE ADJUSTABLE.
b) FANS SERVING IDF ROOMS, AV CLOSETS ETC. SHALL BE STARTED AND STOPPED VIA A WALL MOUNTED THERMOSTAT.

3. CONDENSER WATER PUMPS
a) CIRCULATING PUMP OPERATION SHALL BE INTERLOCKED WITH THE AC UNITS. UPON ACTIVATION OF ANY UNIT, THE LEAD CIRCULATING PUMP SHALL START AND RUN CONTINUOUSLY. THE PUMPS SHALL BE PROVIDED WITH AN AUTOMATIC RESTART AFTER A POWER OUTAGE.
b) PROVIDE FOR LEAD/LAG SEQUENCING IN ORDER TO HAVE ONE PUMP RUNNING AND THE OTHER AS STAND-BY. PUMPS SHALL BE CYCLED ON A 150 HOUR OPERATING SCHEDULE (ADJUSTABLE) FOR EVEN WEAR.
c) A FLO

DUCTWORK INSULATION

DUCTWORK LOCATION	FLAME SPREAD RATING	SMOKE DEVELOPED INDEX RATING	INSULATION MEAN TEMPERATURE RATING	MINIMUM INSTALLED INSULATION VALUE
CONDITIONED SPACE	25	50	75°F	R-6
UNCONDITIONED SPACE	25	50	75°F	R-8

DUCTWORK INSULATION NOTES:

1. REFER TO MECHANICAL SPECIFICATION DRAWINGS FOR ADDITIONAL INFORMATION.
2. ALL MINIMUM REQUIRED DUCTWORK INSULATION VALUES COMPLY WITH THE 2020 NYCECC.

BRANCH DUCT SIZING SCHEDULE

DUCT SIZE (INSIDE CLEAR DIMMENSIONS)	CFM RANGE
12x6, 15x5	UP TO 175
14x6, 18x5	180-245
12x8, 16x6, 20x5	250-285
18x6, 24x5	290-335
14x8, 20x6, 26x5	340-375
16x8, 24x6	380-435
14x10, 18x8	440-515
16x10, 20x8	520-585
22x8	590-645
18x10, 24x8	655-715
20x10, 26x8	720-825
22x10, 28x8	830-895
20x12, 24x10, 32x8	900-1,005

BRANCH DUCT SCHEDULE NOTE:

1. THIS SCHEDULE APPLIES TO BRANCH DUCTS THAT ARE NOT SIZED ON PLANS.
2. ALL SUPPLY DUCTWORK SHOWN TO HAVE ROUND DIMENSIONS SHALL BE SPIRAL ROUND DUCTWORK.

MINIMUM PIPE INSULATION

FLUID TEMPERATURE RANGE	CONDUCTIVITY RANGE (K-VALUE)	INSULATION MEAN TEMPERATURE RATING	NOMINAL PIPE DIAMETER (IN INCHES)				
			< 1	1 TO < 1.5	1.5 TO < 4	4 TO < 8	8+
			INSULATION THICKNESS REQUIRED (IN INCHES)				
141°F - 200°F	0.25 - 0.29	125°F	1.5	1.5	2.0	2.0	2.0
105°F - 140°F	0.21 - 0.28	100°F	1.0	1.0	1.5	1.5	1.5
40°F - 60°F	0.21 - 0.27	75°F	0.5	0.5	1.0	1.0	1.0
< 40°F	0.20 - 0.26	50°F	0.5	1.0	1.0	1.0	1.5

PIPE INSULATION NOTES:

1. REFER TO MECHANICAL SPECIFICATION DRAWINGS FOR ADDITIONAL INFORMATION.
2. ALL MINIMUM REQUIRED PIPING INSULATION THICKNESS COMPLY WITH THE 2020 NYCECC.

FAN POWERED BOX SCHEDULE

DESIGNATION	BOX TYPE	FAN DATA						BOX SIZE			PRIMARY AIRFLOW RANGE (CFM)			MINIMUM STATIC PRESS. PS (IN. W.G.)	RADIATED SOUND LEVEL AT MAX CFM & PS=3" (NC)	DISCHARGE SOUND LEVEL AT MAX CFM & PS=3" (NC)	HOT WATER HEATING COIL							MANUFACTURER/ MODEL	TYPE OF CONTROLS	REMARKS
		CFM	ESP (IN W.G.)	V/Ø/Hz	HP	MCA	MOP	UNIT	INLET (INCH)	OUTLET (INCH)	MIN	MAX	HEATING				CAPACITY (MBH)	NO. OF ROWS	GPM	EWT (°F)	LWT (°F)	EAT (°F)	LAT (°F)	WATER △P (FT. HD)			
FPB-B	SERIES	325/600	0.27	277/1/60	1/3	3.2	15	B	6" / 8" Ø	20.5x 12.5"	100/150	325/500	100/150	0.04 / 0.06	13 / 20	11 / 23	-	-	-	-	-	-	-	-	TITUS/ DTFS-F	DDC	-
FPB-HW-B	SERIES	500	0.28	277/1/60	1/3	3.2	15	B	8" Ø	20.5x 12.5"	150	500	150	0.06	20	23	10.4	1	0.7	160	130	68	87	0.1	TITUS/ DTFS-F	DDC	-
FPB-C	SERIES	900	0.32	277/1/60	1/3	3.2	15	C	10" Ø	20.5x 12.5"	270	900	270	0.11	20	23	-	-	-	-	-	-	-	-	TITUS/ DTFS-F	DDC	-
FPB-HW-C	SERIES	900	0.32	277/1/60	1/3	3.2	15	C	10" Ø	20.5x 12.5"	270	900	270	0.11	20	23	12.2	1	0.7	160	125	68	80	0.1	TITUS/ DTFS-F	DDC	-
FPB-D	SERIES	1500	0.31	277/1/60	1/2	5.1	15	D	12" Ø	25x 17.5"	450	1500	450	0.08	24	27	-	-	-	-	-	-	-	-	TITUS/ DTFS-F	DDC	-
FPB-HW-D	SERIES	1500	0.31	277/1/60	1/2	5.1	15	D	12" Ø	25x 17.5"	450	1500	450	0.08	24	27	23.3	1	1.2	160	119	68	82	0.34	TITUS/ DTFS-F	DDC	-
FPB-E	SERIES	1700	0.33	277/1/60	3/4	6.9	15	E	14" Ø	25x 17.5"	510	1700	510	0.06	29	28	-	-	-	-	-	-	-	-	TITUS/ DTFS-F	DDC	-
FPB-HW-E	SERIES	1700	0.33	277/1/60	3/4	6.9	15	E	14" Ø	25x 17.5"	510	1700	510	0.06	29	28	24.9	1	1.2	160	119	68	81	0.36	TITUS/ DTFS-F	DDC	-

FAN COIL UNIT SCHEDULE:

1. ALL PERFORMANCE BASED ON TESTS CONDUCTED IN ACCORDANCE WITH ASHRAE 130-2009 AND AHRI 880-2008.
2. ALL NC LEVELS DETERMINED USING AHRI 885-2008 APPENDIX E.
3. THE MINIMUM SUPPLY CIRCUIT AMPACITY (MCA) AND MAXIMUM OVER-CURRENT PROTECTION (MOP) RATINGS WERE CALCULATED IN ACCORDANCE WITH UL STANDARDS BASED ON MOTOR FULL LOAD CURRENT.
4. CONTROLS SHALL BE PROVIDED BY OTHERS, SHIPPED TO FACTORY AND FACTORY MOUNTED.
5. FAN POWER BOX TO BE PROVIDED WITH ECM MOTOR AND MULTI-POINT CENTER AVERAGING INLET VELOCITY SENSOR CAPABLE OF OPERATING AT VELOCITIES AS LOW AS 250 FPM.
6. OPTIONAL REMOTE PWM ALLOWS FOR A 0-10V SIGNAL TO MODULATE FAN RPM TO MIN/MAX, IF BMS NEEDS TO STOP THE FAN OPTIONAL CONTROL SEQUENCE IS AVAILABLE WITH 0-2V SINGAL ACTING AS A STOP SIGNAL AND 2_1-10V SIGNAL AS A FAN MIN/MAX SETTING. REVIEW SPECIFICATIONS FOR EXACT LOCATIONS AND CONTROLS NECESSARY.
7. PROVIDE FACTORY MOUNTED BRACKETS AND TOGGLE DISCONNECT SWITCH.
8. PROVIDE FILTER RACK WITH TWO (2) FACTORY SUPPLIED FILTERS FOR RETURN INLET (ONE (1) FOR CONSTRUCTION USE AND ONE FOR OPERATION).
9. COORDINATION TO SUBMIT INDIVIDUAL BOX SCHEDULE FOR REVIEW DURING THE SHOP DRAWING PROCESS. ACCESS SIDE SHALL BE INDICATED AND COORDINATED WITH THE DESIGN AND SHEETMETAL SHOP DRAWINGS.
10. DRAWINGS ARE TEMPERATURE IS BASED ON 49°F, RETURN AIR BASED ON 70°F.
11. PROVIDE 277 VOLT TO 24 VOLT TRANSFORMER.
12. U.O.N. EACH BOX SHALL HAVE A MINIMUM AIRFLOW SETPOINT OF 25% ON PRIMARY AIR INLET.
13. UNITS SHALL BE CONSTRUCTED OF A MINIMUM 20-GAUGE GALVANIZED STEEL.
14. SEE SPECS FOR MORE INFORMATION.

NOTE: ALL FAN POWERED BOXES LABEL FPB-HW ON SHEETS SHALL BE PROVIDED WITH HOT WATER HEATING COIL.

AIR OUTLET SCHEDULE

DESIGNATION	DESCRIPTION	NECK SIZE	FRAME SIZE	CFM RANGE	MAX. NC	MANUFACTURER/ MODEL	COMMENTS
A	PLAQUE TYPE SUPPLY DIFFUSER	8"Ø	24"x24"	0-200	<20	TITUS/ OMNI	LAY-IN COORD. W/ CEILING TYPE
	PLAQUE TYPE SUPPLY DIFFUSER	10"Ø	24"x24"	201-350	<20	TITUS/ OMNI	LAY-IN COORD. W/ CEILING TYPE
	PLAQUE TYPE SUPPLY DIFFUSER	12"Ø	24"x24"	351-450	<20	TITUS/ OMNI	LAY-IN COORD. W/ CEILING TYPE
	PLAQUE TYPE SUPPLY DIFFUSER	14"Ø	24"x24"	451-600	<20	TITUS/ OMNI	LAY-IN COORD. W/ CEILING TYPE
	PLAQUE TYPE SUPPLY DIFFUSER	15"Ø	24"x24"	601-750	<20	TITUS/ OMNI	LAY-IN COORD. W/ CEILING TYPE
B	PLAQUE TYPE SUPPLY DIFFUSER	8"Ø	12"x12"	0-200	<20	TITUS/ OMNI	SURFACE MOUNT, G.C. TO CUT CEILING TILE AS REQUIRED
C	LINEAR SUPPLY DIFFUSER	(1) 1¼" SLOT		0-50	22	TITUS FL-15-HT	VERTICAL/HORIZONTAL MOUNT CONTINUOUS LINEAR
D	LINEAR SUPPLY DIFFUSER	(2) ¾" SLOT		0-37	22	TITUS ML-38	VERTICAL/HORIZONTAL MOUNT CONTINUOUS LINEAR
F	SUPPLY REGISTER	12"x6"	14"x8"	0-200	19	TITUS/ 300RL	SURFACE MOUNTED
	SUPPLY REGISTER	14"x6"	16"x8"	201-300	20	TITUS/ 300RL	SURFACE MOUNTED
	SUPPLY REGISTER	20"x6"	22"x8"	301-450	23	TITUS/ 300RL	SURFACE MOUNTED
	SUPPLY REGISTER	22"x22"	24"x24"	750-1000	<25	TITUS/ 272RL	LAY-IN, DOUBLE DEFLECTION COORD. W/ CEILING TYPE
G	SUPPLY REGISTER	16"x4"	-	0-200	22	TITUS/ S300FL	SPIRAL DUCT MOUNTED
	SUPPLY REGISTER	22"x4"	-	201-300	23	TITUS/ S300FL	SPIRAL DUCT MOUNTED
	SUPPLY REGISTER	32"x4"	-	301-400	23	TITUS/ S300FL	SPIRAL DUCT MOUNTED
H	CEILING RETURN/ EXHAUST GRILLE	22"x22"	24"x24"	0-1570	<20	TITUS/ 355RL	½" SPACING, 35° DEFLECTION PROVIDE LIGHT SHIELD
I	TRANSFER DUCT RETURN GRILLE	PER PLANS	PER PLANS	0-1200	<20	TITUS/ 355RL	½" SPACING, 35° DEFLECTION
J	CEILING EXHAUST GRILLE	10"x10"	12"x12"	0-325	<20	TITUS/ 350RL	¾" BLADE SPACING, 35° DEFLECTION
K	CEILING EXHAUST GRILLE	12"x12"	24"x24"	0-500	<20	TITUS/ 350FL	3/4" BLADE SPACING, 35° DEFLECTION

AIR OUTLET SCHEDULE NOTES (TYPICAL FOR EACH OUTLET):

1. INSTALL FOUR (4) WAY DIFFUSER UNLESS OTHERWISE NOTED. PROVIDE BLANK OFF BAFFLES FOR DIFFUSERS SHOWN TO HAVE 2-WAY AND 3-WAY PATTERNS. INCREASE NECK SIZES AS REQUIRED TO COMPENSATE FOR BLANKED-OFF DIFFUSERS.
2. DIFFUSERS SHALL BE SUITABLE FOR THE TYPE OF CEILING CONSTRUCTION BEING INSTALLED IN.
3. DIFFUSERS THAT SERVE AREAS WITHOUT HUNG CEILINGS SHALL BE SUITABLE FOR DUCTWORK MOUNTING.
4. ALL ADJUSTABLE AIR OUTLET PATTERN DEFLECTORS SHALL BE FIELD ADJUSTED TO OPTIMIZE AIR DISTRIBUTION PREVENTING DRAFT CONDITIONS. CONTRACTOR SHALL PLAN FOR A SECOND COMFORT FIELD ADJUSTMENT PER OWNER/ENGINEER DIRECTION.
5. PROVIDE CABLE OPERATED DAMPERS FOR LINEAR DIFFUSERS AND DIFFUSERS IN INACCESSIBLE CEILINGS.
6. NON-ACTIVE LENGTHS OF LINEAR DIFFUSERS TO BE USED AS RETURN. PROVIDE LIGHT SHIELDS.
7. FINISHES SHALL BE AS SPECIFIED BY THE ARCHITECT.
8. SEE EQUIPMENT SPECIFICATIONS FOR ADDITIONAL INFORMATION.

EXHAUST FAN SCHEDULE

DESIGNATION	SERVICE	CFM	STATIC PRESSURE (IN W.G.)	ELECTRICAL V/Ø/Hz	MOTOR (HP)	MCA	MOCP	DRIVE (BELT OR DIRECT)	VFD	CONTROL (SWITCH OR THERMOSTAT)	WEIGHT	MANUFACTURER/ MODEL	COMMENTS
TF-31-1	F&B PANTRY 31.61	100	0.25	115/60/1	1/15	1.6	15	DIRECT	NO	SWITCH	27	GREENHECK/ SQ-70-VG	-
EF-30-1	DISH RM 30.13	500	0.4	115/60/1	1/10	4.75	15	DIRECT	NO	SWITCH	49	GREENHECK/ CSP-A1050-VG	-

EXHAUST FAN SCHEDULE NOTES (TYPICAL FOR EACH FAN):

1. PROVIDE FACTORY SUPPLIED VARIABLE SPEED SWITCH MOUNTED TO FAN CASING.
2. PROVIDE INTEGRAL DISCONNECT SWITCH.
3. SEE EQUIPMENT SPECIFICATIONS FOR ADDITIONAL INFORMATION.

WATER COOLED COMPUTER ROOM AIR CONDITIONING UNIT SCHEDULE

DESIGNATION	AREA SERVED	CAPACITY			SPACE CONDITION			AIR FLOW				INFRARED HUMIDIFIER		CONDENSER SECTION					ELECTRICAL V/Ø/Hz	UNIT FLA	UNIT MCA	UNIT MOP	OPERATING WEIGHT (LBS) (EVAP / COND)	MANUFACTURER/ MODEL (EVAP / COND)	COMMENTS
		NOMAL TONS	TOTAL (MBH)	SENSIBLE (MBH)	DB (°F)	WB (°F)	% RH	CFM	ESP	VFD	LAT (DB/WB °F)	LBS/HR	FLA	DESIGNATION	GPM	MAX. PD (FT)	EWT (°F)	LWT (°F)							
CRAC-29-1	IDF-29.35	3.0	37.5	37.5	75	62.5	50	1500	0.5	NO	57.0/42.1	5	8.2	WCCU-29-1	9.0	10.7	85	95	460/3/60	16.5	20.6	25	205 / 150	ABOVE AIR/ WCH-011A / CCW-011B	WALL MOUNTED UNIT
CRAC-30-1	IDF-30.29	3.0	37.5	37.5	75	62.5	50	1500	0.5	NO	57.0/42.1	5	8.2	WCCU-30-1	9.0	10.7	85	95	460/3/60	16.5	20.6	25	205 / 150	ABOVE AIR/ WCH-011A / CCW-011B	WALL MOUNTED UNIT
CRAC-31-1	IDF-31.44	3.0	37.5	37.5	75	62.5	50	1500	0.5	NO	57.0/42.1	5	8.2	WCCU-31-1	9.0	10.7	85	95	460/3/60	16.5	20.6	25	205 / 150	ABOVE AIR/ WCH-011A / CCW-011B	WALL MOUNTED UNIT

WATER COOLED COMPUTER ROOM AIR CONDITIONING UNIT SCHEDULE NOTES (TYPICAL FOR EACH UNIT)

1. PROVIDE SPRING TYPE VIBRATION ISOLATION HANGERS FOR EACH UNIT.
2. PROVIDE MC-SERIES ADVANCED MICROPROCESSOR W/ ALARMS & BACNET COMPATIBILITY.
3. PROVIDE FACTORY SUPPLIED COMBINATION STARTER AND DISCONNECT SWITCH.
4. PROVIDE AUXILIARY STAINLESS STEEL DRAIN PAN WITH LEAK DETECTION.
5. PROVIDE LOW ENTERING WATER TEMP KIT.
6. UNITS TO BE RATED FOR SYSTEM PRESSURE.
7. PROVIDE HOT GAS BYPASS.
8. FACTORY INSTALLED ACCESSORIES: CONDENSATE PUMP, RECEIVER, COMPRESSOR SOUND JACKET, FIRESTAT, SMOKE DETECTOR, HUMIDIFIER.
9. SEE EQUIPMENT SPECIFICATIONS FOR ADDITIONAL INFORMATION.

PIPING FITTING SCHEDULE

PIPE MATERIAL	PIPE SIZE	JOINT TYPE	FITTING MATERIAL	FITTING CLASS
COPPER TUBING HARD DRAWN	4" & SMALLER	BRAZING	WROUGHT COPPER	450 PSIG AT 100°F TO 200°F, 150 PSIG AT 250°F
RED BRASS	ALL SIZES	THREADED	CAST BRONZE	125 AND 250 PSIG

PIPING FITTING NOTES:

1. REFER TO SPECIFICATIONS FOR MORE INFORMATION.

AIR PURIFICATION SYSTEM SCHEDULE

DESIGNATION	SERVING	SUPPLY AIR FLOW CFM	FAN			STAGES OF FILTRATION				ELECTRICAL		UNIT WEIGHT (LBS)	UNIT DIMENSIONS (LxWxH)	MANUFACTURER/ MODEL	COMMENTS
			AMP	RPM	HP	1	2	3	4	V /Ø/ Hz	FLA				
BAS-30-1,2	FOOD PREP/ SERVING PT	500	1.5	1725	1/8	1" PREFILTER	(2) 2" CARBON MEDIA	4" PRIMARY PARTICULATE FILTER	(2) BI-POLAR IONIZATION MODULES	115/1/60	5.0	90	48x24x12	BIOCLIMATIC/ MC 880	- -

BIOCLIMATIC SYSTEM SCHEDULE NOTES:

1. UNIT SHALL BE A SELF-CONTAINED FLUSH MOUNTED SYSTEM FOR INSTALLATION INTO "T" GRID CEILING, 48"X24" MODULE1"
2. ACOUSTICAL FOAM INSULATION
3. SYSTEM SHALL DELIVER SPECIFIED CFM WITH ALL FILTERS INSTALLED IN UNIT
4. ALL METAL TO METAL SURFACES SEALED WITH SILICONE SEALANT
5. SOUND LEVEL NOT TO EXCEED 53.5 DB @ 4.5 FT.
6. FAN SHALL BE DWDI FORWARD CURVE WHEEL, TWIN UNITS
7. TWO (2) AIR DIFFERENTIAL PRESSURE SWITCHES FOR PREFILTER AND PRIMARY FILTER CHANGE
8. DIRTY FILTER LIGHT
9. EQUIPMENT SHALL BE ETL LISTED
10. UNIT MOUNTED CONTROLS FOR STANDALONE AND ON/OFF OPERATION
11. UNIT MANUFACTURER TO PROVIDE ONE FACTORY STARTUP AND ONE ANNUAL FOLLOW UP INSPECTION WITHIN 1 YEAR OF STARTUP.
12. UNIT INSTALLATION AND ALL WIRING IS BY OTHERS
13. MEDIA SHIPPED SEPARATELY FOR FIELD INSTALLATION BY OTHERS
14. VIBRATION ISOLATORS AND MOUNTING HARDWARE BY OTHERS
15. MINIMUM 36" CLEARANCE REQUIRED FOR FILTER CHANGEOUT
16. UNITS NOT TO EXCEED 48"L)24"W)X14"H DIMENSIONS
17. ONE (1) SPARE MEDIA FILTER SET PER UNIT SHALL BE PROVIDED.

PUMP SCHEDULE

DESIGNATION	SERVICE	TYPE	GPM	TDH (FT)	RPM	NPSH	PUMP EFFICIENCY	BHP	MOTOR HP	VFD	ELECTRICAL V/Ø/Hz	WEIGHT (LBS.)	MANUFACTURER/ MODEL	COMMENTS
CWP-29-1,2	CONDENSER WATER	INLINE, MULTISTAGE	10	35	2,845	3.85	84	-	0.5	YES	208/1/60	73	GRUNDFOS/ CRE3-00L2VFD	TWO PUMPS PRIMARY-STANDBY
CWP-30-1,2	CONDENSER WATER	INLINE, MULTISTAGE	10	35	2,845	3.85	84	-	0.5	YES	208/1/60	73	GRUNDFOS/ CRE3-00L2VFD	TWO PUMPS PRIMARY-STANDBY
CWP-31-1,2	CONDENSER WATER	INLINE, MULTISTAGE	10	35	2,845	3.85	84	-	0.5	YES	208/1/60	73	GRUNDFOS/ CRE3-00L2VFD	TWO PUMPS PRIMARY-STANDBY

PUMP SCHEDULE NOTES (TYPICAL FOR EACH PUMP):

1. PROVIDE SPRING TYPE VIBRATION AT PUMP.
2. PROVIDE COMBINATION DISCONNECT SWITCH AND STARTER AS REQUIRED.
3. PROVIDE DRIP PAN UNDER IN-LINE PUMPS WITH LEAK DETECTOR.
4. PUMPS TO BE RATED FOR 363 PSI.
5. MECHANICAL CONTRACTOR TO VERIFY SUPPLY AND RETURN DIFFERENTIAL PRESSURE OF EXISTING RISERS IN FIELD AT POINT OF CONNECTION AND SUBMIT PRESSURE READINGS WITH PUMP SHOP SUBMITTALS FOR REVIEW AND RECORD.
6. PROVIDE PERMANENT MAGNET ECM MOTOR (IE5 EFFICIENCY) WITH INTEGRAL VFD.
7. MECHANICAL CONTRACTOR TO PROVIDE NON OVERLOADING PUMP CURVE FOR REVIEW DURING THE SHOP DRAWING PROCESS. FOR VFD APPLICATIONS PROVIDE PARALLEL PUMP SPEED CURVES IN 10% INCREMENTS.
8. INSTALLATION SHALL BE IN STRICT ACCORDANCE WITH THE MANUFACTURERS INSTRUCTIONS.
9. SEE EQUIPMENT SPECIFICATIONS FOR ADDITIONAL INFORMATION.

DEHUMIDIFICATION UNIT SCHEDULE - ADD ALT

DESIGNATION	AREA SERVED	CAPACITY	SPACE CONDITION			FILTER	EFFICIENCY (PW/SQKWH)	BTU (TOTAL)	ELECTRICAL V/Ø/Hz	UNIT MCA	UNIT MOP	OPERATING WEIGHT (LBS)	MANUFACTURER/ MODEL	COMMENTS
		TOTAL (MBH)	WATER REMOVAL (@ 80°F / 60RH%)	MIN TEMP (°F)	MAX TEMP (°F)	% RH								
DU-29-1	IDF-29.35	37.5	200	56	95	50	MERV-13	5.7	12,800	120/1/60	15	25	130	QUEST/ 205 DUAL	ADD ALT
DU-30-1	IDF-30.29	37.5	200	56	95	50	MERV-13	5.7	12,800	120/1/60	15	25	130	QUEST/ 205 DUAL	ADD ALT
DU-31-1	IDF-31.44	37.5	200	56	95	50	MERV-13	5.7	12,800	120/1/60	15	25	130	QUEST/ 205 DUAL	ADD ALT

DEHUMIDIFICATION UNIT SCHEDULE NOTES (TYPICAL FOR EACH UNIT)

1. PROVIDE SPRING TYPE VIBRATION ISOLATION HANGERS FOR EACH UNIT.
2. PROVIDE AUXILIARY STAINLESS STEEL DRAIN PAN WITH LEAK DETECTION.
3. TO BE INSTALLED ACCESSORIES: CONDENSATE PUMP

PIPING MATERIAL SCHEDULE

SERVICE	SIZE	MATERIAL	WEIGHT	STANDARD	JOINT TYPE
COLD CONDENSATE DRAINS, MISCELLANEOUS DRAINS AND OVERFLOWS	2" & BELOW	HARD COPPER	TYPE 'L'	ASTM A88	BRAZE
HOT WATER TO 220°F	4" & BELOW	HARD COPPER	TYPE 'K'	ASTM B88	BRAZE
CONDENSER WATER	4" & BELOW	HARD COPPER	TYPE 'K'	ASTM B88	BRAZE
AUTOMATIC AIR VENT	ALL	HARD COPPER	TYPE 'K'	ASTM B88	BRAZE
COLD WATER MAKEUP AND FILL	4" & BELOW	HARD COPPER	TYPE 'L'	ASTM B88	BRAZE
PIPING TO MISC. GAUGES	ALL	RED BRASS	STANDARD	ASTM B43	THREADED

PIPING MATERIAL NOTES:

1. REFER TO SPECIFICATIONS FOR MORE INFORMATION.

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NO.	DATE	ISSUE
3	07/01/2026	ISSUE FOR ADD ALTERNATE
	05/22/2026	ISSUE FOR FILING
1	04/22/2026	ADDENDUM #1
	04/03/2026	ISSUE FOR PRICING

REVISIONS

PROJECT:
SNOWFLAKE
7 TIMES SQUARE, NEW YORK, NY
FLOORS 29, 30, 31

DRAWING TITLE:
MECHANICAL SCHEDULES

SEAL & SIGNATURE

DATE: 05.22.2026
SCALE: N.T.S.
PROJECT NO. ----
DRAWN: CHECKED:
DRAWING NO. M-401.00

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CMTA
A LEGENCE Company
Formerly AMA Group

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CMTA Project No.:	B069-02-039	**Control No.:**	**H-02.00**
CMTA Project Name:	Boston Properties @ 7 Times Square - Snowflake Floors 29-31	**Page:** 1 **of** 1	

☐ **NO EXCEPTIONS TAKEN** No further review of submittal required.	☐ **REVISE AND RESUBMIT** Revise as noted and resubmit for further review.
☒ **MAKE CORRECTIONS AS NOTED** Incorporate corrections in work; resubmission is not required. Where corrections cannot be complied with, revise the exceptions and resubmit.	☐ **EXAMINED** Submittal received, formal review by others and/or formal review by this office not required. Submittal has been examined for information only.

Corrections or comments made on the shop drawings during this review do not relieve contractor from compliance with requirements of the drawings and specifications. This check is only for review of general conformance with the design concept of the project and general compliance with the information given in the contract documents. The contractor is responsible for: confirming and correlating all quantities and dimensions; selecting fabrication processes and techniques of construction; coordinating his work with that of all other trades; and performing his work in a safe and satisfactory manner.

Date:	6/15/2026	**Engineer**:	Mike Sisti
SUBMITTAL DESCRIPTION:	Pump Cuts		

COMMENTS:

CMTA Comments:

1. Provide vibration isolation for pumps and piping within 50 feet or 50 pipe diameters, whichever is greater. See specifications for details.

2. Provide valve and trim as per the details on the design documents.

3. Pumps shall meet all pressure and flow characteristics as per the design documents.

4. Contractor to provide the differential pressure at the tie in to the risers in full flow condition, as per the design documents, and provide to the manufacturer to ensure the proper head is obtained to ensure full flow throughout the system.

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GRUNDFOS

Company name:
Created by:
Phone:

Date: 03/04/2026

Qty.	Description
1	**CRE 3-2 A-FGJ-A-E-HQQE**



Note! Product picture may differ from actual product

Product No.: On request

Vertical, multistage centrifugal pump with inlet and outlet ports on same the level (inline). The pump head and base are in cast iron – all other wetted parts are in stainless steel. A cartridge shaft seal ensures high reliability, safe handling, and easy access and service. Power transmission is via a rigid split coupling. Pipe connection is via combined DIN-ANSI-JIS flanges.

The pump is fitted with a 1-phase, fan-cooled, permanent-magnet, synchronous motor. The motor efficiency is classified as IE5 in accordance with IEC 60034-30-2.

The motor includes a frequency converter and PI controller in the motor terminal box. This enables continuously variable control of the motor speed, which again enables adaptation of the performance to a given requirement. The operating panel on the motor terminal box features a four-inch TFT display, push-buttons and the Grundfos Eye indicator.

The display gives an intuitive and user-friendly interface to all functions.
The push-buttons are used to navigate through the menu structure to access pump and performance data on site and enable setting of required setpoint as well as setting of pump to "Min." or "Max." operation or to "Stop".

The Grundfos Eye indicator on the operating panel provides visual indication of pump status:

- "Power on": Motor is running (rotating green indicator lights) or not running (permanently green indicator lights)
- "Warning": Motor is still running (rotating yellow indicator lights) or has stopped (permanently yellow indicator lights)
- "Alarm": Motor has stopped (flashing red indicator lights).

Communication with the pump is also possible by means of Grundfos GO Remote (accessory). The remote control enables further settings as well as reading out of a number of parameters such as "Actual value", "Speed", "Power input" and total "Power consumption".

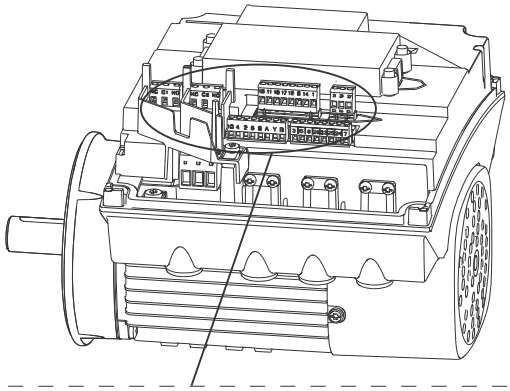
The terminal box has a number of inputs and outputs enabling the motor to be used in advanced applications where many inputs and outputs are required:

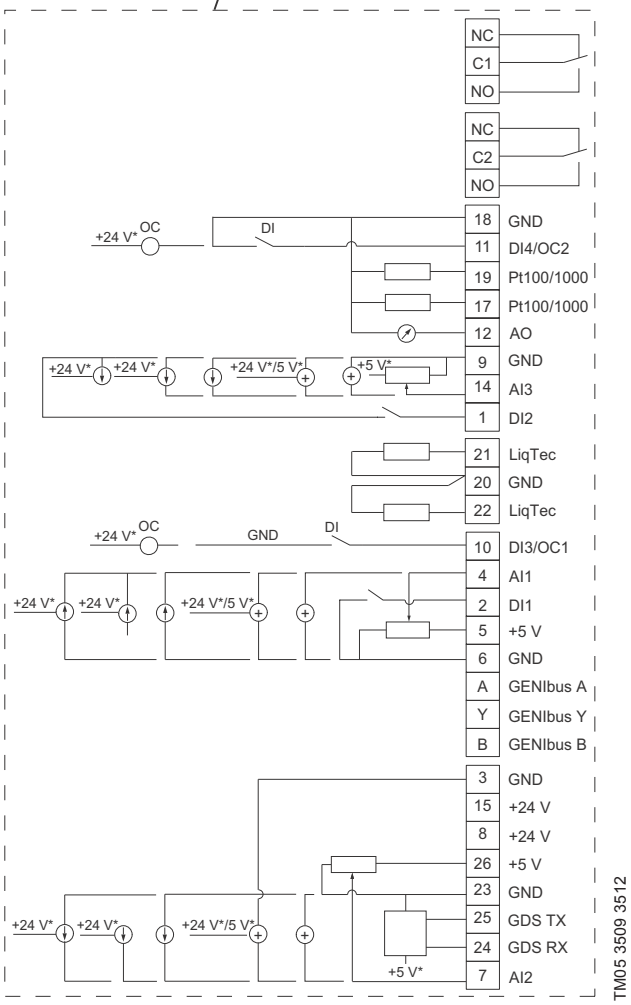
- two dedicated digital inputs
- three analog inputs, 0(4)-20 mA, 0-5 V, 0-10 V, 0.5-3.5 V
- 5 V voltage supply to potentiometer and sensor
- one analog output, 0-10 V, 0(4)-20 mA
- two configurable digital inputs or open-collector outputs
- two Pt100/Pt1000 inputs
- LiqTec, dry-running protection sensor input
- Grundfos Digital Sensor input and output
- 24 V voltage supply for sensors
- two signal-relay outputs (potential-free contacts)
- GENIbus connection
- interface for Grundfos CIM fieldbus module.

Further product details

An external sensor can be connected if controlled pump operation based on for example flow, differential pressure or temperature is required.

The operating panel on the motor terminal box features a four-inch TFT display, push-buttons and the Grundfos Eye indicator.





* If you use an external supply source, there must be a connection to GND.

Fig. 8 Connection terminals, CRE, CRIE, CRNE, CRKE, SPKE, MTRE and CME pumps

Terminal	Type	Function
NC	Normally closed contact	Signal relay 1 (LIVE or PELV)
C1	Common	
NO	Normally open contact	
NC	Normally closed contact	Signal relay 2 (PELV only)
C2	Common	
NO	Normally open contact	
18	GND	Ground
11	DI4/OC2	Digital input/output, configurable. Open collector: Max. 24 V resistive or inductive.
19	Pt100/1000 input 2	Pt100/1000 sensor input
17	Pt100/1000 input 1	Pt100/1000 sensor input
12	AO	Analog output: 0-20 mA / 4-20 mA 0-10 V
9	GND	Ground
14	AI3	Analog input: 0-20 mA / 4-20 mA 0-10 V
1	DI2	Digital input, configurable
21	LiqTec sensor input 1	LiqTec sensor input (white conductor)
20	GND	Ground (brown and black conductors)
22	LiqTec sensor input 2	LiqTec sensor input (blue conductor)
10	DI3/OC1	Digital input/output, configurable. Open collector: Max. 24 V resistive or inductive.
4	AI1	Analog input: 0-20 mA / 4-20 mA 0.5 - 3.5 V / 0-5 V / 0-10 V
2	DI1	Digital input, configurable
5	+5 V	Supply to potentiometer and sensor
6	GND	Ground
A	GENIbus, A	GENIbus, A (+)
Y	GENIbus, Y	GENIbus, GND
B	GENIbus, B	GENIbus, B (-)
3	GND	Ground
15	+24 V	Supply
8	+24 V	Supply
26	+5 V	Supply to potentiometer and sensor
23	GND	Ground
25	GDS TX	Grundfos Digital Sensor output
24	GDS RX	Grundfos Digital Sensor input
7	AI2	Analog input: 0-20 mA / 4-20 mA 0.5 - 3.5 V / 0-5 V / 0-10 V

TYPICAL WIRING DIAGRAM AND CONFIGURATION SETUP FOR CONTROL BY BMS
-START/STOP - DI1 - SET TO "EXTERNAL STOP"
-SPEED - AI1 - SET TO "SETPOINT INFLUENCE" (WHEN USING THE DONGLE) / "EXT. SETPOINT INFLUENCE" (WHEN USING THE PUMP INTERFACE)
-FEEDBACK - AO - SET TO "ACTUAL SPEED"
-RUN STATUS - OC1 - SET TO "PUMP RUNNING"
-ALARM - SIGNAL RELAY 1 - SET TO "ALARM"

The total leakage current of all the electrical equipment in the installation must be taken into account. You find the leakage current of the motor in the sections on leakage current and leakage current (AC).

This product can cause a direct current in the protective earth conductor.

Overvoltage and undervoltage protection

Overvoltage and undervoltage may occur in case of unstable power supply or faulty installation. The motor stops if the voltage falls outside the permissible voltage range. The motor restarts automatically when the voltage is again within the permissible voltage range. Therefore, no additional protection relay is required.

The motor is protected against transients from the power supply according to EN 61800-3. In areas with high lightning intensity, we recommend external lightning protection.

Overload protection

If the upper load limit is exceeded, the motor automatically compensates for it by reducing the speed and stops if the overload condition persists.

The motor remains stopped for a set period. After this period, the motor automatically attempts to restart. The overload protection prevents damage to the motor. Consequently, no additional motor protection is required.

Overtemperature protection

The electronic unit has a built-in temperature sensor as an additional protection. When the temperature rises above a certain level, the motor automatically compensates for it by reducing the speed and stops if the temperature keeps rising. The motor remains stopped for a set period. After this period, the motor automatically attempts to restart.

Protection against phase unbalance

Three-phase motors must be connected to a power supply with a quality corresponding to IEC 60146-1-1, class C, to ensure correct motor operation at phase unbalance. This also ensures long life of the components.

Related information

9.2.1.2 Leakage current

9.2.2.2 Leakage current (AC)

5.5 Connection terminals

The descriptions and terminal overviews in this section apply to both single- and three-phase motors.

For maximum torques, see the section on other technical data.

Related information

9.4 Other technical data

5.5.1 Connection terminals, CRE, CRIE, CRNE, SPKE and MTRE pumps

The CRE, CRIE, CRNE, SPKE and MTRE pumps have a number of inputs and outputs enabling the pumps to be used in advanced applications where many inputs and outputs are required.

The pumps have the following connections:

- three analog inputs
- one analog output
- two dedicated digital inputs
- two configurable digital inputs or open-collector outputs
- Grundfos Digital Sensor input and output
- two Pt100/1000 inputs
- two LiqTec sensor inputs
- two signal relay outputs
- GENIbus connection.

See the figure below.

Digital input 1 is factory-set to be start-stop input where open circuit results in stop. A jumper is factory-fitted between terminals 2 and 6. Remove the jumper if digital input 1 is to be used as external start-stop or any other external function.

DANGER
Electric shock
Death or serious personal injury

- Make sure that the wires to be connected to the connection groups below are separated from each other by reinforced insulation in their entire lengths.

- Inputs and outputs

All inputs and outputs are internally separated from the mains-conducting parts by reinforced insulation, and galvanically separated from other circuits. All control terminals are supplied with protective extra-low voltage (PELV), thus ensuring protection against electric shock.

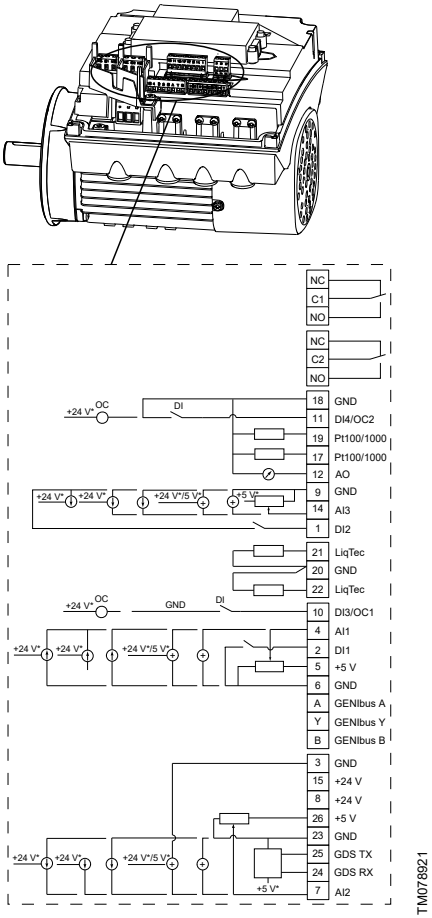
- Signal relay outputs
  - Signal relay 1:

    LIVE:

    You can connect supply voltages up to 250 VAC. PELV: The output is galvanically separated from other circuits. Therefore, you can connect the supply voltage or protective extra-low voltage to the output as desired.
  - Signal relay 2:

    PELV:

    The output is galvanically separated from other circuits. Therefore, you can connect the supply voltage or protective extra-low voltage to the output as desired.
- Mains supply (terminals N, PE, L or L1, L2, L3, PE).



Connection terminals, CRE, CRIE, CRNE, SPKE and MTRE pumps

*If you use an external supply source, there must be a connection to GND.

Terminal	Type	Function
2	DI1	Digital input, configurable
5	+5 V	Supply to potentiometer and sensor
6	GND	Ground
A	GENIbus, A	GENIbus, A (+)
Y	GENIbus, Y	GENIbus, GND
B	GENIbus, B	GENIbus, B (-)
3	GND	Ground
15	+24 V	Supply
8	+24 V	Supply
26	+5 V	Supply to potentiometer and sensor
23	GND	Ground
25	GDS TX	Grundfos Digital Sensor output
24	GDS RX	Grundfos Digital Sensor input
7	AI2	Analog input: 0-20 mA / 4-20 mA 0.5 - 3.5 V / 0-5 V / 0-10 V

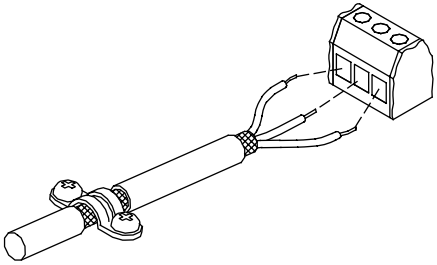
5.6 Signal cables

Use screened cables with a cross-sectional area of minimum 0.5 mm^2 and maximum 1.5 mm^2 for the external on/off switch, digital inputs, setpoint and sensor signals.

The wires in the motor terminal box must be as short as possible.

5.6.1 Connecting signal cables

1. Connect the screens of the cables to the frame at both ends with good connection. The screens must be as close as possible to the terminals.



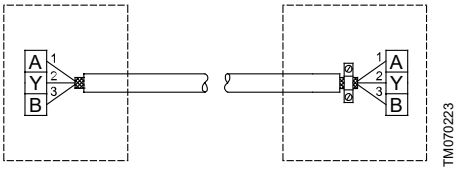
2. Connect the signal cables to the terminals.
3. Tighten the terminal screws.

5.7 Bus connection cable

5.7.1 Connecting a 3-core bus cable, GENIbus

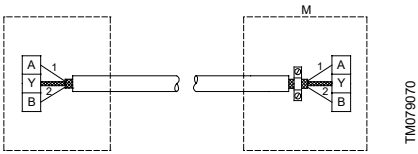
For the bus connection, use a screened 3-core cable with a cross-sectional area of minimum 0.5 mm^2 and maximum 1.5 mm^2.

- If the motor is connected to a unit with a cable clamp identical to the one on the product, connect the screen to the cable clamp.
- If the unit has no cable clamp, leave the screen unconnected at this end.



5.7.2 Replacing a motor

If a 2-core cable is used in the installation, connect it as shown in the figure below.



Connection with screened 2-core cable

Pos.	Description
M	Motor

If a screened 3-core cable is used in the installation, follow the instructions in the section on connecting a 3-core bus cable.

Related information

5.7.1 Connecting a 3-core bus cable, GENIbus

5.8 Signal relays

The motor has two outputs for potential-free signals via two internal relays. You can set the signal outputs to **Operation**, **Pump running**, **Ready**, **Alarm** and **Warning**.

The functions of the two signal relays are shown in the table below.

If you want to set the Pt100/1000 input for a feedback sensor, we recommend that you do this via the **Assisted pump setup** menu. See the section on assisted pump setup.

If you want to set a Pt100/1000 input for other purposes, you can do this manually.

You can set the analog inputs via the **Setup, analog input** menu. See the section on setup, analog input.

If you make the manual setting via the Grundfos GO, you need to enter the menu for the Pt100/1000 input under the **Settings** menu.

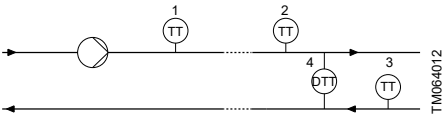
Function

The Pt100/1000 inputs can be set to the following functions:

- **Not active**
- **Feedback sensor**: The sensor is used for the selected control mode.
- **Ext. setpoint infl.**: See the section on the external setpoint function.
- **Other function**.

Measured parameter

Select one of the parameters listed to be measured in the system by the PT100/1000 sensor connected to the actual PT100/1000 input.



Overview of PT100/1000 sensor locations

Parameter	Pos.
Liquid temp.	1
Temperature 1	2
Temperature 2	3
Ambient temp.	Not shown

Measuring range

-50 to 204 °C.

Related information

6.8 Identification of the functional module
7.16 External setpoint function
7.46 Assisted pump setup
7.47 Setup, analog input
7.52 Factory settings

7.10 Digital inputs

Available inputs depend on the functional module fitted in the pump:

Function (terminal)	FM 200 [8] (standard)	FM 300 [8] (advanced)
Digital input 1, setup (2 and 6)	•	•
Digital input 2, setup (1 and 9)	-	•

8) See the section on the identification of the functional module.

To set a digital input, make the settings below.

Function

Select one of the following functions:

- **Not active**: When set to **Not active**, the input has no function.
- **External stop**: When the input is deactivated (open circuit), the pump stops.
- **Min.** (minimum speed): When the input is activated, the pump runs at the set minimum speed.
- **Max.** (maximum speed): When the input is activated, the pump runs at the set maximum speed.
- **User-defined speed**: When the input is activated, the motor runs at a speed set by the user.
- **External fault**: When the input is activated, a timer is started. If the input is activated for more than 5 seconds, the pump stops and a fault is indicated. This function depends on input from external equipment.
- **Alarm resetting**: When the input is activated, a possible fault indication is reset.
- **Dry running**: When this function is selected, lack of inlet pressure or water shortage (dry running) can be detected, and if that occurs, the pump stops. The pump cannot restart as long as the input is activated. This requires the use of an accessory, such as the following:
  - a pressure switch installed on the inlet side of the pump
  - a float switch installed on the inlet side of the pump.
- **Accumulated flow**: When this function is selected, the accumulated flow can be registered. This requires the use of a flowmeter that can give a feedback signal as a pulse per defined volume of water. See the section on pulse flowmeter setup.
- **Predefined setpoint digit 1** (applies only to digital input 2): When digital inputs are set to predefined setpoint, the pump operates according to a setpoint based on the combination of the activated digital inputs. See the section on predefined setpoints.

- **Active output**: When the input is activated, the related digital output is activated. See the section on digital inputs/outputs. This is done without any changes to pump operation.
- **Local motor stop**: When the input is activated, the given pump in a multipump system stops without affecting the performance of the other pumps in the system.

The priority of the selected functions in relation to each other is listed in the section on priority of settings.

A stop command always has the highest priority.

Activation delay

Select the activation delay (T1).

It is the time between the digital signal and the activation of the selected function.

Range: 0-6000 seconds.

Duration timer mode

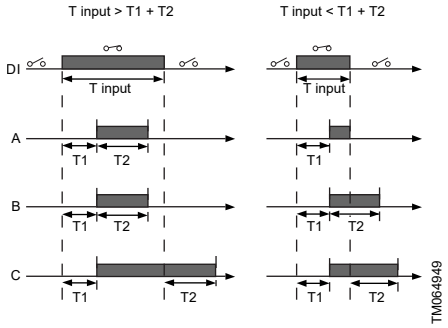
Select the mode. See the figure below.

- **Not active**
- active with interrupt (mode A)
- active without interrupt (mode B)
- active with after-run (mode C).

Select the duration time (T2).

It is the time that, together with the mode, determines how long the selected function is active.

Range: 0 to 15,000 seconds.



Duration timer function of digital inputs

Pos.	Description
DI	Digital input
A	Mode A
B	Mode B
C	Mode C

Related information

6.8 Identification of the functional module
7.1 Priority of settings
7.11 Digital inputs/outputs
7.17.1 Predefined setpoints
7.23 Pulse flowmeter setup
7.52 Factory settings

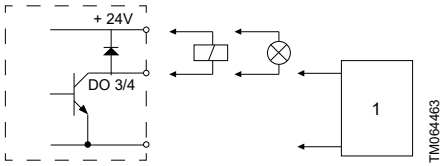
7.11 Digital inputs/outputs

Available inputs/outputs depend on the functional module fitted in the pump:

Function (terminal)	FM 200 [9] (standard)	FM 300 [9] (advanced)
Digital input/output 3, setup (6 and 10)	•	•
Digital input/output 4, setup (11 and 18)	-	•

9) See the section on the identification of the functional module.

You can select if the interface is to be used as input or output. The output is an open collector and you can connect it to, for example, an external relay or controller such as a PLC.



Example of configurable digital inputs/outputs

Pos.	Description
1	External controller

To set a digital input/output, make the settings below.

Mode

You can set the digital input/output 3 and 4 to act as digital input or digital output:

- **Digital input**
- **Digital output**.

Function

You can set the digital input/output 3 and 4 to the functions stated in the table below:

When you enable the LiqTec function, it stops the pump if dry running occurs. Restart the pump manually if it stops due to dry running.

Dry running detection delay

You can set a detection delay to make sure the pump is given a chance to start up before the LiqTec function stops it due to dry running.

Range: 0-254 seconds.

Related information

7.52 Factory settings

7.20 Stop function (Low-flow stop function)

You can set **Low-flow stop function** to the following values:

- **Not active**
- **Energy-optimal mode**
- **High-comfort mode**
- **User-defined mode** (**Customised operating mode**).

When the low-flow stop function is active, the flow is monitored. If the flow becomes lower than the set minimum flow (Q_{min}), the pump changes from continuous operation at constant pressure to start-stop operation and stops if the flow reaches zero.

The advantages of enabling **Low-flow stop function** are the following:

- There is no unnecessary heating from the pumped liquid.
- The wear of the shaft seals is reduced.
- There is reduced noise from operation.

The disadvantages of enabling **Low-flow stop function** may be the following:

- The delivered pressure is not completely constant as it fluctuates between the start and stop pressures.
- The frequent starts and stops of the pump may in some applications cause acoustic noise.

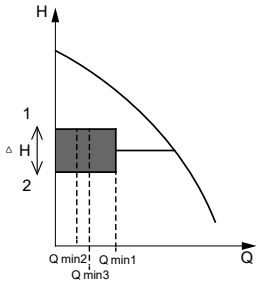
The impact of the above disadvantages very much depends on the setting selected for the stop function.

The **High-comfort mode** setting minimises pressure fluctuations and acoustic noise.

Select **Energy-optimal mode** if the main priority is to reduce the energy consumption as much as possible.

Possible settings of the stop function:

- **Energy-optimal mode**: The pump automatically adjusts the parameters for the stop function so that the energy consumption during the start-stop operation period is minimised. In this case, the stop function uses the factory-set values of the minimum flow (Q_{min1}) and other internal parameters. See the figure below.
- **High-comfort mode**: The pump automatically adjusts the parameters for the stop function so that the disturbances during the start-stop operation period are minimised. In this case, the stop function uses the factory-set values of the minimum flow (Q_{min2}) and other internal parameters. See the figure below.
- **User-defined mode** (**Customised operating mode**): The pump uses the parameters set for ΔH and minimum flow (Q_{min3}) respectively for the stop function. See the figure below.



Difference between start and stop pressures (ΔH) and minimum flow rate

Pos.	Description
1	Stop pressure
2	Start pressure

In start-stop operation, the pressure varies between the start and stop pressures. See the figure above.

In **User-defined mode** (**Customised operating mode**), ΔH is factory-set to 10 % of the actual setpoint. ΔH can be set within the range from 5 to 30 % of the actual setpoint.

The pump changes to start-stop operation if the flow becomes lower than the minimum flow rate.

The minimum flow rate is set in percentages of the nominal flow rate of the pump. See the pump nameplate.

In **User-defined mode** (**Customised operating mode**), the minimum flow rate is factory-set to 10 % of the nominal flow rate.

Low-flow stop function

A low flow can be detected in two ways:

1. One possibility is a built-in low-flow detection function that is active if none of the digital inputs are set for flow switch.
   - Low-flow detection function: The pump checks the flow regularly by reducing the speed for a short time. If there is no or only a small change in pressure, this means that there is low flow. The speed is increased until the stop pressure (actual setpoint + 0.5 × ΔH) is reached and

the pump stops. When the pressure falls to the start pressure (actual setpoint - 0.5 × ΔH), the pump restarts.

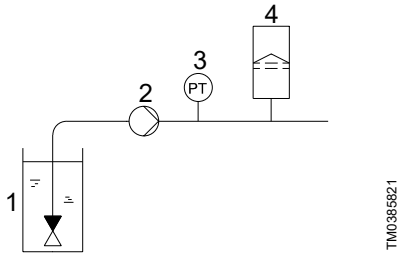
- If the flow rate is higher than the set minimum flow rate, the pump returns to continuous operation at constant pressure.
- If the flow rate is still lower than the set minimum flow rate (Q_{min}), the pump continues in start-stop operation until the flow rate is higher than the set minimum flow rate (Q_{min}). When the flow rate is higher than the set minimum flow rate (Q_{min}), the pump returns to continuous operation.

2. The other possibility is a flow switch connected to one of the digital inputs.
   - Flow switch: When the digital input is activated for more than 5 seconds due to low flow, the speed is increased until the stop pressure (actual setpoint + 0.5 × ΔH) is reached, and the pump stops. When the pressure falls to start pressure, the pump restarts. If there is still no flow, the pump quickly reaches the stop pressure and stops. If there is flow, the pump continues operating according to the setpoint.

Operating conditions for the low-flow stop function

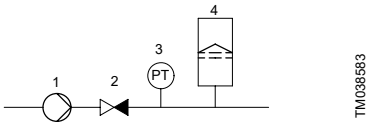
You can only use the stop function if the system incorporates a pressure sensor, a non-return valve and a diaphragm tank.

Always install the non-return valve before the pressure sensor and diaphragm tank.



Position of the non-return valve and pressure sensor in system with suction lift operation

Pos.	Description
1	Non-return valve
2	Pump
3	Pressure sensor
4	Diaphragm tank



Position of the non-return valve and pressure sensor in a system with a positive inlet pressure

Pos.	Description
1	Pump
2	Non-return valve
3	Pressure sensor
4	Diaphragm tank

Set minimum flow

Set the minimum flow rate (Q_{min}) in this display.
This setting determines at which flow rate the system is to change from continuous operation at constant pressure to start-stop operation. The setting range is 5 to 30 % of the rated flow rate.

Related information

7.52 Factory settings

7.21 Stop at min. speed

This stop function can be utilised in, for example, constant level applications where a boost of pressure is not needed. This stop function differs from the low-flow stop but the purpose is the same. The pump stops if there is no or low consumption.

This function monitors the speed of the pump. When the PI-controller has forced the speed of the pump to minimum according to the feedback value, the pump stops after a set period of time. It remains stopped until the feedback value starts to drop and the PI-controller starts the pump again.

- **Enable Stop at min. speed**
  Enables the function **Stop at min. speed**.
- **Delay**
  The delay time during which the pump must be running at minimum speed before it stops.
- **Restart speed**
  It is the speed in percentage when the pump must start again, hysteresis. It must be set higher than the minimum speed of the pump.

7.22 Pipe filling function

This function is typically used in pressure-boosting applications and ensures smooth startup of systems with, for instance, empty pipes.

4. Read the description for this control mode.
5. Select which analog input to use as sensor input.
6. Select the sensor function according to where the sensor is installed in the system. See the section on analog inputs.
7. Select the electrical input signal according to the sensor specifications.
8. Select the measuring unit according to the sensor specifications.
9. Set the minimum and maximum sensor range values according to the sensor specifications.
10. Set the desired setpoint.
11. Set the gain and integral time of the controller. See the section on the controller.
12. Type the desired pump name.
13. Check the summary of settings and confirm them.

On the advanced operating panel

1. Open the **Assist** menu.
2. Select **Assisted pump setup**.
3. Select the control mode **Const. pressure**.
4. Select which analog input to be used as sensor input.
5. Select the measured parameter which is to be controlled. See the section on analog inputs.
6. Select the measuring unit according to the sensor specifications.
7. Set the minimum and maximum sensor range values according to the sensor specifications.
8. Select electrical input signal according to the sensor specifications.
9. Set the desired setpoint.
10. Set the gain and integral time of the controller. See the section on the controller.
11. Check the summary of settings and confirm them by pressing [OK].

Related information

7.2 Setpoint
7.6 Control mode
7.8 Analog inputs
7.14 Controller settings

7.47 Setup, analog input

This menu is only available in the advanced operating panel.

This menu guides you through the following:

Setup, analog input

- Analog inputs 1 to 3.
- Pt100/1000 input 1 and 2.
- Adjusting the setpoint.
- Summary.

Related information

7.2 Setpoint
7.8 Analog inputs
7.9 Pt100/1000 inputs

7.48 Setting of date and time

This function is only available in the HMI 300 and HMI 301 operating panels.

The inputs and outputs available depend on the functional module fitted in the motor.

Functional module	Setting of date and time
FM110	-
FM310	●
FM311	●

The function guides you through the following settings:

- **Select date format**
- **Set date**
- **Select time format**
- **Set time**.

Related information

6.8 Identification of the functional module
7.32 Set date and time

7.49 Setup of multipump system

The function **Setup of multi-pump system** enables the control of two motors connected in parallel without external controllers. The pumps or motors in a system communicate with each other via the wireless GENIair connection or the wired GENI connection.

You can set a multipump system via the master motor which is the first selected motor.

If several pumps or motors in the system have sensors, they can all function as the master and take over the master function if the other fails. This provides additional redundancy in the multimotor system.

You can choose between the following multimotor functions:

Alternating operation

Alternating operation serves as a duty and standby operating mode, and is possible with two pumps or two motors of the same size and type connected in parallel. The main purpose of the function is to ensure an even amount of running hours, and to ensure that the other pump or motor starts if the duty pump or motor stops due to an alarm.

You can choose between two alternating operating modes:

- **Alternating operation, time**:

  The change from one pump or motor to the other is based on time.
- **Alternating operation, energy**:

  The change from one pump or motor to the other is based on energy consumption.

If the duty pump or motor fails, the other pump or motor starts.

Backup operation

Backup operation is possible with two motors of the same size and type connected in parallel. One motor is operating continuously. The backup motor is operated for a short time each day to prevent seizing up. If the duty motor stops due to a fault, the backup motor starts.

Cascade operation

This function is available with up to 4 motors installed in parallel. The motors must be of the same size and if used with pumps, the pumps must be of the same model.

- The performance is adjusted to the demand through cutting pumps in or out and through parallel control of the pumps in operation.
- The controller maintains a constant process value through continuous adjustment of the speed of the pumps.
- Pump changeover is automatic and depends on load, operating hours and fault detection.
- All pumps in operation run at the same speed.
- The number of pumps in operation also depends on the energy consumption of the pumps. If only one pump is required, two pumps run at a lower speed if this results in a lower energy consumption.
- If several pumps or motors in the system have a sensor, they can all function as the master and take over the master function if the other fails.

7.49.1 Alternating operation, time

The **Alternating operation, time** menu sets the interval of alternation between two pumps.

This setting is only available in alternating mode.

7.49.2 Time for pump changeover

The **Time for pump changeover** menu sets the time of day for pump changeover to take place.

This setting is only available in alternating operation.

7.49.3 Sensor to be used

This function defines the sensor for controlling the pump system.

Select **Master pump sensor** if the sensor is placed in a way where it can measure the output from all the pumps in the system, for example, in the manifold.

Select **Running pump sensor** if the sensor is placed on or across individual pumps, for example, if the sensor is installed behind non-return valves, or if it is not able to measure the output from all pumps.

This setting is only available in alternating operation and cascade operation.

7.49.4 Setting up a multipump system

You can set up a multipump system in the following ways:

- Grundfos GO and wireless motor connection
- Grundfos GO and wired motor connection
- HMI 300 or HMI 301 operating panel and wireless motor connection
- HMI 300 or HMI 301 operating panel and wired motor connection.

Related information

7.29 Pump number

7.40 Pump name

7.46 Assisted pump setup

7.49.4.1 Grundfos GO and wireless motor connection

1. Power on both motors.
2. Establish contact to one of the motors with the Grundfos GO.
3. Set the needed analog and digital inputs via the Grundfos GO, according to the connected equipment and the required functionality.
4. Assign a name to the motor using the Grundfos GO.
5. Disconnect the Grundfos GO from the motor.
6. Establish contact to the other motor.
7. Set the needed analog and digital inputs via the Grundfos GO, according to the connected equipment and the required functionality.
8. Assign a name to the motor using the Grundfos GO.
9. Select the **Assist** menu and **Setup of multi-pump system**.
10. Select the desired multimotor function.
11. Press the **Right** button to continue.
12. Set the time when the alternation between the two motors is to take place.

    This step applies only if you have selected **Alternating operation, time** and if the motors are fitted with the FM310 or FM311.
13. Press the **Right** button to continue.

14. Select **Radio** as the communication method between the two motors.
15. Press the **Right** button to continue.
16. Select pump 2 (motor 2).
17. Select the pump from the list.



Use **OK** or the **Radio communication** button to identify the pump.

18. Press the **Right** button to continue.
19. Confirm the setting by pressing **Send**.
20. When you finish the setup and the dialog box disappears, wait for the green indicator light in the middle of **Grundfos Eye** to light up.

7.49.4.2 Grundfos GO and wired motor connection

1. Connect the two motors with each other with a 3-core screened cable between the GENIbus terminals A, Y, B.
2. Power on both motors.
3. Establish contact to one of the motors with the Grundfos GO.
4. Set the required analog and digital inputs via the Grundfos GO, according to the connected equipment and the required functionality.
5. Assign a name to the motor using the Grundfos GO.
6. Assign motor number 1 to the motor.
7. Disconnect the Grundfos GO from the motor.
8. Establish contact to the other motor.
9. Set the analog and digital inputs according to the connected equipment and the required functionality by the Grundfos GO.
10. Assign a name to the motor using the Grundfos GO.
11. Assign motor number 2 to the motor.
12. Select the **Assist** menu and **Setup of multi-pump system** (**multimotor setup**).
13. Select the desired multimotor function.
14. Press the **Right** button to continue.
15. Set the time when the alternation between the two motors is to take place.



This step applies only if you have selected **Alternating operation, time** and if the motors are fitted with the FM310 or FM311.

16. Press the **Right** button to continue.
17. Select **Bus** as the communication method between the two motors.
18. Press the **Right** button to continue.
19. Select pump 2 (motor 2).
20. Select the additional motor from the list.



Use **OK** or the **Radio communication** button to identify the pump.

21. Press the **Right** button to continue.
22. Confirm the setting by pressing **Send**.
23. When you have finish the setup and the dialog box disappears, wait for the green indicator light in the middle of the **Grundfos Eye** to light up.

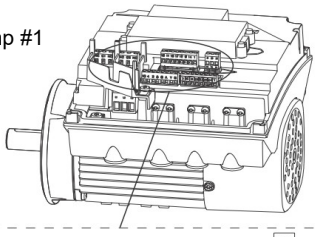
7.49.4.3 HMI 300 or HMI 301 and wireless motor connection

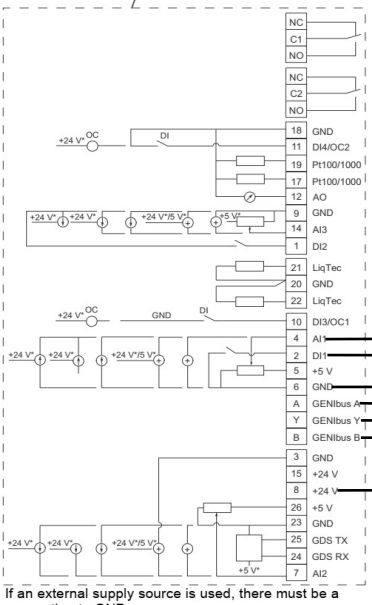
1. Power on both motors.
2. On both motors, set the analog and digital inputs according to the connected equipment and the required functionality.
3. Select the **Assist** menu on one of the motors and **Setup of multi-pump system**.
4. Press the **Right** button to continue.
5. Select **Wireless** as the communication method between the two motors.
6. Press the **Right** button to continue.
7. Select the desired multimotor function.
8. Press the **Right** button three times to continue.
9. Press **OK** to search for other motors.

   The green indicator light in the middle of the **Grundfos Eye** flashes on the other motors.
10. Press **OK** or the **Radio communication** button on the motor that is to be added to the multimotor system.
11. Press the **Right** button to continue.

Grundfos/PACO Permanent Magnet MLE motor (model H,I or J) (up to 15 HP)
Multi Pump Operation:
-Each MLE must be connected to a pressure sensor or differential pressure sensor
Each connected pump & MLE motor must be identical model and size
-All low voltage wiring requires 16ga twisted shielded pair
-All field wiring to be done by the contractor

Pump #1



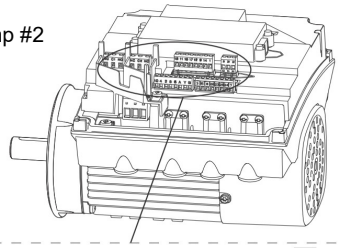


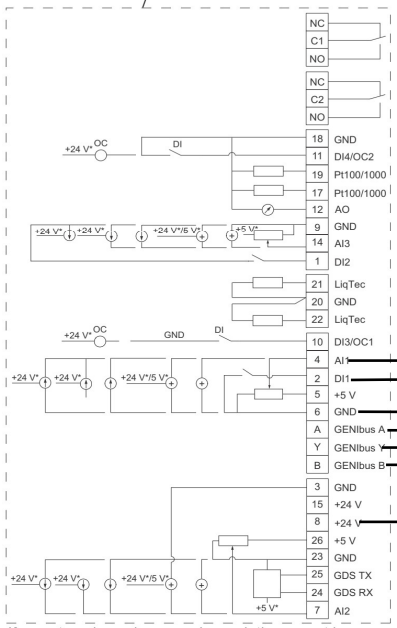
* If an external supply source is used, there must be a connection to GND.

Pressure sensor (Analog input):
0-20 mA / 4-20 mA
0.5 - 3.5 V / 0-5 V / 0-10 V

Bus Connection

Pump #2





* If an external supply source is used, there must be a connection to GND.

Start/Stop Relay (by others) - Normally open - 2-pole - Closed when AC units are in operaton. AC units to be wired in parallel.

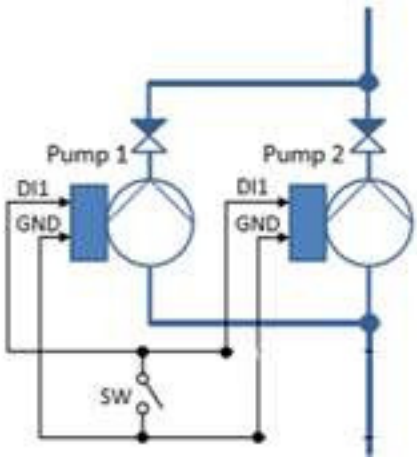
Pressure sensor (Analog input):
0-20 mA / 4-20 mA
0.5 - 3.5 V / 0-5 V / 0-10 V

Electrical Start/Stop Connections for Multi Pump

Remove the jumper (terminals 2 & 6), because we will use the digital input 1 as an external start-stop.

Terminal	Function
2	Digital Input 1
6	Ground

The multi pump system is to be controlled by an external, contractor provided start/stop switch as shown below. The switch can also be the a single or multiple wired in parallel normally open start/stop relay outputs from the system AC units.



The Digital Input 1 (terminal #2) terminals are to be wired together.
The Ground (terminal #6) terminals are to be wired together. (Must be #6 on both pumps)
The Switch (SW) is to be placed across these two wires.
The external switch (SW) will start the complete system, no matter which pump has the master (duty) role.
The digital input 1 on both of the pumps in the system is to be set to "Ext stop".
The factory setting of digital input 1 is "Ext stop".
The switch (SW) is to be closed for the pumps to run.

Connect the AYB terminals of pump number 1 to the AYB terminals of pump number 2.

Use a screened cable (16 ga twisted shielded pair) between the GENIbus terminals. Connect the screen of the cable to the frame.



Multi Pump Function

The multi pump function enables the control of two pumps connected in parallel without the use of external controllers.

The pumps in a multi pump system communicate with each other via the wireless GENIair connection or via the wired GENI connection.

Three different multi pump functions are available:

- Alternating operation, time – NOT RECOMMENDED
- Alternating operation, energy – RECOMMENDED SET-UP OPTION
- Back-up operation – NOT RECOMMENDED

Alternating operation functions as a duty/standby operating mode with only one pump in operation at a time.

Alternating operation is possible with two pumps of the same size, type, and connected in parallel.

Any pump can function as the master (duty) pump and take over the master (duty) pump function if the master (duty) pump should fail. This is only possible if each pump is configured with a dedicated pressure sensor.

Installation - Confirm

Each pump requires a non-return (check) valve in series with the pump.

Ensure the pumps are correctly installed in the system with non-return (check) valves.

Enable the Multi Pump Function

The main purpose of the multi pump function is to ensure an even amount of running hours and to ensure that the standby pump takes over if the running pump stops due to an alarm.

The operating pump can be alternated based on load, alternation takes place when the duty pump has consumed 20kWh.

Prepare the second pump.

Electrical Connections
Run the start-up guide / Assisted pump setup
Set Units; Display brightness / Blank display
Prime the pump

Give the second pump the pump number “2” using the graphical control panel

Home menu
Settings menu
Communication OK
Pump number OK
OK (change to Pump 2) OK
Home menu

Give the first pump the pump number “1” using the graphical control panel

Home menu
Settings menu
Communication OK
Pump number OK
OK (change to Pump 1) OK
Home menu

Go to the “setup of the multi-pump system” using pump number 1.

Home menu
Assist
Setup of the multi-pump system OK
>
Wired GENIbus OK > (Select the communication channel)
>
Alternating operation, energy OK > (Select the multi-pump function)

OK	(Search for other pumps)
OK >	(Pump 2 winks)
>	(See * below)
OK	(Confirm the setting)

* The pump changeover will be made automatically when the difference in energy consumption of the two pump is equal to the daily energy consumption of one pump running at the rated performance

Below is the graphical control panel displays for the two pumps.

Pump 1



Pump 2



The symbol at the end of the white arrow indicates master (duty) pump.
The symbol at the end of the yellow arrow indicates slave (standby) pump.
The symbol at the end of the red arrows indicates the pumps are turned off by the start stop button.

Operate the pumps in the multi-pump mode by pressing the start stop button on pump one and then press the start stop button on pump 2.

Pump 1 operates, and pump 2 is in standby mode.

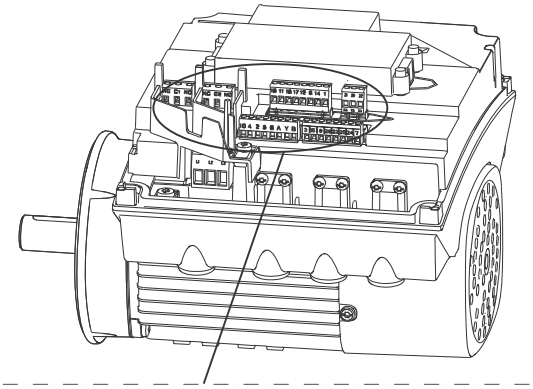
The graphical displays of pump 1 and pump 2 look as follows:

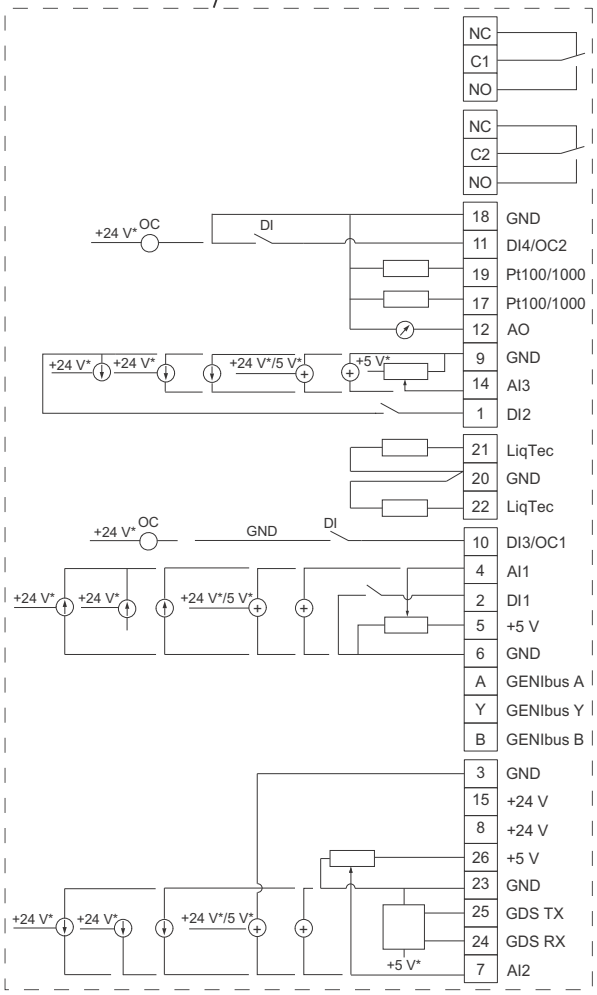
Pump 1



Pump 2







* If you use an external supply source, there must be a connection to GND.

Fig. 8 Connection terminals, CRE, CRIE, CRNE, CRKE, SPKE, MTRE and CME pumps

Terminal	Type	Function
NC	Normally closed contact	Signal relay 1 (LIVE or PELV)
C1	Common	
NO	Normally open contact	
NC	Normally closed contact	Signal relay 2 (PELV only)
C2	Common	
NO	Normally open contact	
18	GND	Ground
11	DI4/OC2	Digital input/output, configurable. Open collector: Max. 24 V resistive or inductive.
19	Pt100/1000 input 2	Pt100/1000 sensor input
17	Pt100/1000 input 1	Pt100/1000 sensor input
12	AO	Analog output: 0-20 mA / 4-20 mA 0-10 V
9	GND	Ground
14	AI3	Analog input: 0-20 mA / 4-20 mA 0-10 V
1	DI2	Digital input, configurable
21	LiqTec sensor input 1	LiqTec sensor input (white conductor)
20	GND	Ground (brown and black conductors)
22	LiqTec sensor input 2	LiqTec sensor input (blue conductor)
10	DI3/OC1	Digital input/output, configurable. Open collector: Max. 24 V resistive or inductive.
4	AI1	Analog input: 0-20 mA / 4-20 mA 0.5 - 3.5 V / 0-5 V / 0-10 V
2	DI1	Digital input, configurable
5	+5 V	Supply to potentiometer and sensor
6	GND	Ground
A	GENIbus, A	GENIbus, A (+)
Y	GENIbus, Y	GENIbus, GND
B	GENIbus, B	GENIbus, B (-)
3	GND	Ground
15	+24 V	Supply
8	+24 V	Supply
26	+5 V	Supply to potentiometer and sensor
23	GND	Ground
25	GDS TX	Grundfos Digital Sensor output
24	GDS RX	Grundfos Digital Sensor input
7	AI2	Analog input: 0-20 mA / 4-20 mA 0.5 - 3.5 V / 0-5 V / 0-10 V

7.44 "Multi-pump setup" ("Setup of multi-pump system")

Pump variant	"Multi-pump setup" ("Setup of multi-pump system")
CME	•
CRE, CRIE, CRNE, CRKE, SPKE, MTRE	•

The multipump function enables the control of two pumps connected in parallel without the use of external controllers. The pumps in a multipump system communicate with each other via the wireless GENIair connection or the wired GENI connection.

You can set a multipump system via the master pump, that is the first selected pump.

If two pumps in the system are configured with an outlet-pressure sensor, they can all function as master pumps and take over the master pump function if the other should fail. This provides additional redundancy in the multipump system.

The multipump functions are described in the following sections.

7.44.1 "Alternating operation"

Alternating operation functions as a duty/standby operating mode and is possible with two pumps of same size and type connected in parallel. The main purpose of the function is to ensure an even amount of running hours and to ensure that the standby pump starts if the duty pump stops due to an alarm.

Each pump requires a non-return valve in series with the pump.

You can choose between two alternating operation modes:

"ALTERNATING OPERATION, ENERGY" IS RECOMMENDED

- "Alternating operation, time"
  Change from one pump to the other is based on time.
- "Alternating operation, energy"
  Change from one pump to the other is based on energy consumption.

If the duty pump fails, the other pump starts.

7.44.2 "Backup operation"

Backup operation is possible with two pumps of same size and type connected in parallel. Each pump requires a non-return valve in series with the pump.

One pump is operating continuously. The backup pump is operated for a short time each day to prevent seizing up. If the duty pump stops due to a fault, the backup pump starts.

7.44.3 Setting up a multipump system

You can set a multipump system in the following ways:

- *Grundfos GO and wireless pump connection*
- *Grundfos GO and wired pump connection*
- *Advanced control panel and wireless pump connection*
- *Advanced control panel and wired pump connection*

See step-by-step descriptions below.

Grundfos GO and wireless pump connection

1. Power on both pumps.
2. Establish contact to one of the pumps with Grundfos GO.
3. Set the needed analog and digital inputs via Grundfos GO according to the connected equipment and the required functionality. See section *7.41 "Assisted pump setup"*.
4. Assign a pump name to the pump using Grundfos GO. See section *7.35 "Pump name"*.
5. Disconnect Grundfos GO from the pump.
6. Establish contact to the other pump.
7. Set the needed analog and digital inputs via Grundfos GO according to the connected equipment and the required functionality. See section *7.41 "Assisted pump setup"*.
8. Assign a pump name to the pump using Grundfos GO. See section *7.35 "Pump name"*.
9. Select the "Assist" menu and choose "Setup of multi-pump system".
10. Select the desired multipump function. See section *7.44.1 "Alternating operation"* and *7.44.2 "Backup operation"* above.
11. Press [>] to continue.
12. Set the time for pump changeover, that is the time at which the alternation between the two pumps is to take place.



This step applies only if you have selected "Alternating operation, time" and if the motors are fitted with FM 300.

13. Press [>] to continue.
14. Select "Radio" as the communication method to be used between the two pumps.
15. Press [>] to continue.
16. Press "Select pump 2".
17. Select the pump from the list.



Use the [OK] or button to identify the pump.

18. Press [>] to continue.
19. Confirm the setting by pressing [Send].
20. Press [Finish] in the "Setup complete" dialog box.
21. Wait for the green indicator light in the middle of Grundfos Eye to light up.

Snowflake- job review

Ruslan Zhdamarov <ruslan@northernwolvesac.com> Mon, Jun 29, 2026 at 8:00 AM
To: "Greda, Brian" <brian.greda@siemens.com>, "Jackson, Brendan" <brendan.jackson@siemens.com>
Cc: Kastriot Cikaqi <kastriot@northernwolvesac.com>

Hi Brendan and Brian,

Regarding the CHW loop setup for floors 29, 30, and 31,
Since each floor operates as an independent system serving only one data room AC unit, we can use the native multi-pump functionality built into the Grundfos CRE drives. See attached Manual. Please look at the "Setup of multipump system" wizard in the Grundfos manual (Section 7.49). By linking the two pump heads directly via their GENIbus communication terminals, they will automatically handle all duty rotation, staging, and standby redundancy on their own. With this setup and since pumps not going to base building BMS, i think lead/lag panel not necessary here.
Please review this approach and let me know your thoughts.

[Quoted text hidden]
--

NORTHERN WOLVES
AIR CONDITIONING

Best Regards,
Ruslan Zhdamarov - CEO/President
347-463-9248
NORTHERN WOLVES INC
55 9TH STREET, UNIT 55A-2, BROOKLYN, NY, 11215

2 attachments

98358864_09_2025_CRE__CRIE__CRNE__SPKE__MTRE__MTHE__CME_PUBTYPE_GB_online.pdf
2362K

H-02.00-B06902039-230000-001-1_Pumps Cuts.pdf
2800K

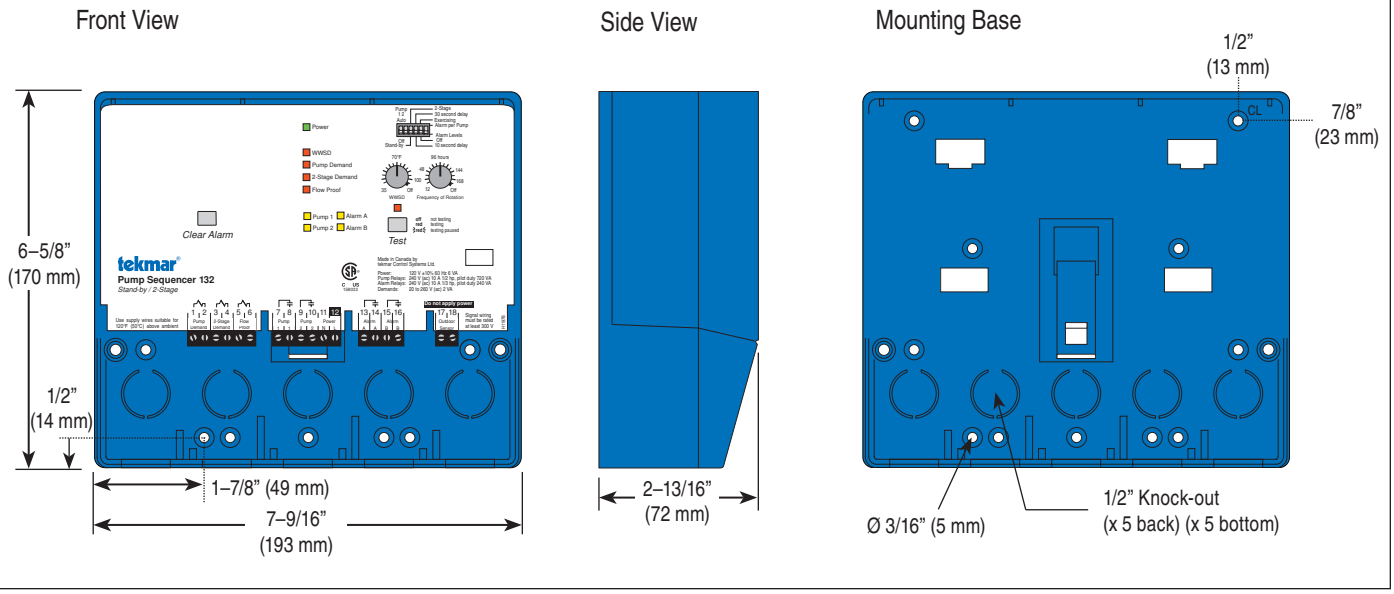
tekmar® Submittal
Pump Sequencer 132

Multi-Staging

C 132
01/11
Replaces: New

Job _____ Designer _____ Contact _____

The Pump Sequencer 132 is designed to operate 2 pumps by providing stand by or staging operation. It can be used in a wide range of applications, including residential installations using a backup system pump to commercial applications that require energizing a second stage for increased flow or head. The main function of the 132 is to provide lead lag capability to a standby pump setup, or to provide staging capability for a dual pump installation. The 132 can energize up to 2 pumps, as well as provide alert contacts that can be used in the event of a failure.



Specifications

Pump Sequencer 132 *Stand-by / 2-Stage*	
Literature	D132, A132, D001
Control	Microprocessor control. This is not a safety (limit) control
Packaged weight	2.9 lb. (1320 g)
Dimensions	6-5/8" H x 7-9/16" W x 2-13/16" D (170 x 193 x 72 mm)
Enclosure	Blue PVC plastic, NEMA type 1
Approvals	CSA C US, meets class B: ICES & FCC Part 15
Ambient conditions	Indoor use only, 32 to 120°F (0 to 50°C), RH ≤90% Non-condensing
Power supply	120 V ±10%, 50/60 Hz, 6 VA
Pump relays	240 V (ac) 10 A 1/2 hp
Alert relays	240 V (ac) 10 A 1/3 hp
Demands	20 to 260 V (ac) 2 VA
Sensors	NTC thermistor, 10 kΩ @ 77°F (25°C ±0.2°C) ß=3892
–Optional	Outdoor Sensor 070
Warranty	Limited 3 Year (See D132 for full warranty)

Energy Saving Features

- Warm Weather Shut Down

Additional Features

- Equal Run Time Rotation
- Lead lag operation
- 2 Stage capability
- Exercising
- Alert per pump or alert levels
- Adjustable flow proof delay
- Test sequence
- CSA C US certified
  (approved to applicable UL standards)

SPECIAL REQUIREMENTS

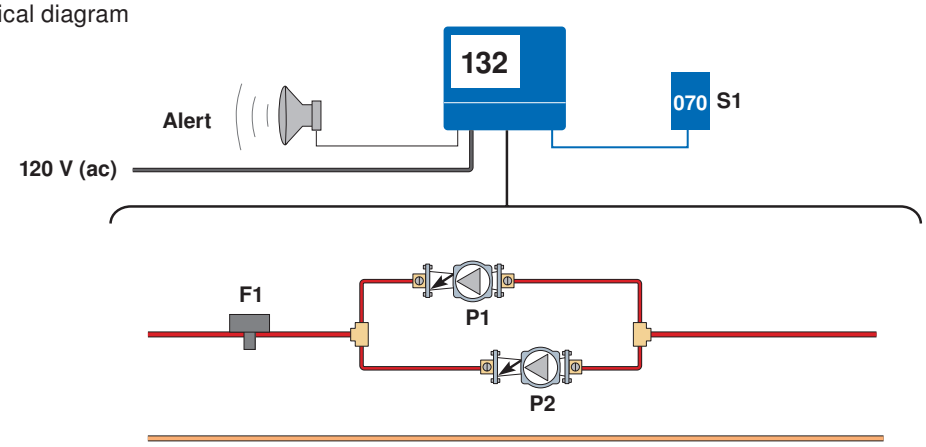
N / A

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Sample Application Drawing

Below is a sample application drawing for this product. This application may include other tekmar products that are required for installation. More sample applications can be found at www.tekmarcontrols.com.

Sample Mechanical diagram

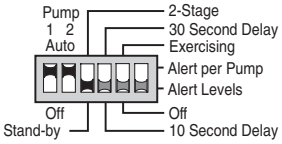


Sample Electrical diagram

Legend:

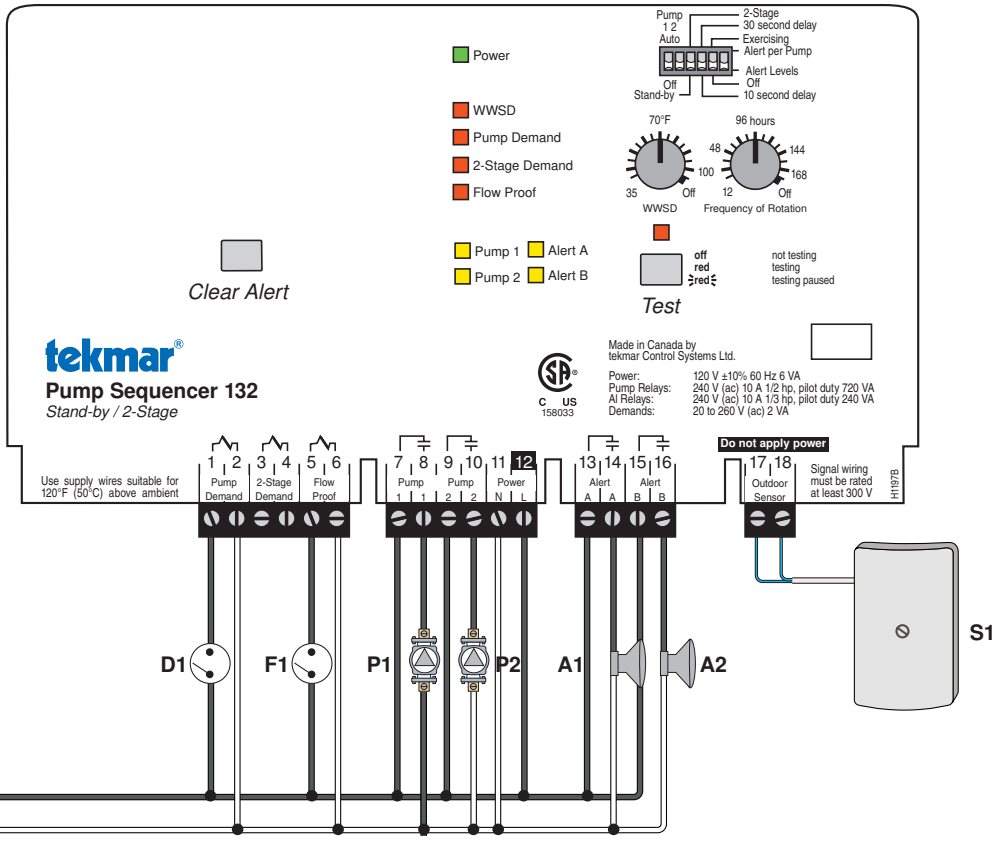
D1 = Pump Demand
F1 = Flow Switch
P1 = Pump 1
P2 = Pump 2
A1 = Alert 1
A2 = Alert 2
S1 = Outdoor Sensor

DIP Setting



= Required

= Optional



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tekmar® - Data Brochure

Pump Sequencer 132

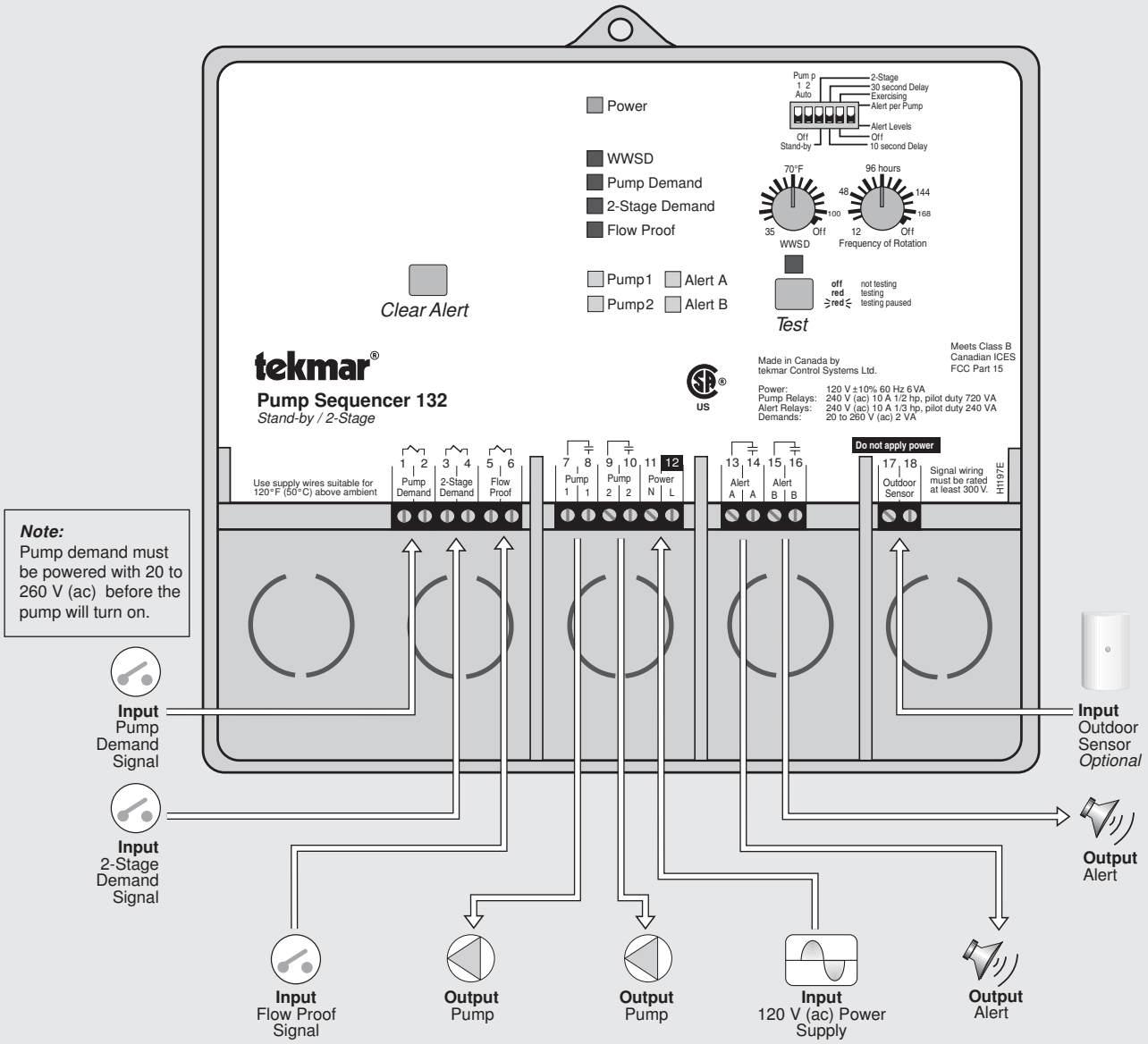
D 132

04/10

The Pump Sequencer 132 is designed to provide pump control for either stand-by or 2-stage operation. In stand-by mode, the 132 automatically switches over from the lead pump to the stand-by pump during a pump failure. In 2-stage mode, the 132 turns on the second stage pump to meet additional flow requirements.

Additional functions include:

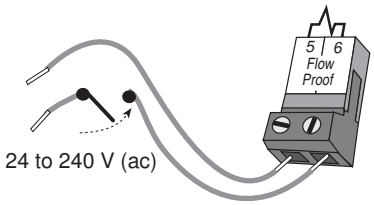
- Warm Weather Shut Down (WWSD)
- Exercising
- Equal Run Time Rotation
- Alert per Pump or Alert Levels
- Adjustable flow proof delay
- Test sequence to ensure proper component operation
- 120 V (ac) power supply
- CSA C US certified (approved to applicable UL standards)



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FLOW PROOF

A flow proof signal is required at all times during pump operation. A flow proof is generated by applying a voltage between 24 and 240 V (ac) across the *Flow Proof* terminals (5 and 6). Once voltage is applied, the *Flow Proof* light is turned on. Once a pump contact is turned on, a flow proof signal must be present before the flow proof delay has expired. A flow proof can come from a flow switch, pressure differential switch, current sensing switch, or power sensing switch.



FLOW PROOF DELAY *(30 second Delay / 10 second Delay)*

The 132 allows a time delay for detecting a flow proof signal once a pump contact is turned on. The amount of time is adjustable from 10 to 30 seconds through a DIP switch. If a flow proof signal is not present within the selected time, the control turns off the first pump contact and turns on the second pump contact. An alert contact is activated to indicate the failure.

FLOW PROOF TEST

The 132 has a flow proof test in order to determine if the flow proof device has failed. Once the pump contacts are turned off, a flow proof signal should not be present. If a flow proof signal is still present after 4 minutes, the control activates an alert contact and displays an error message. Refer to the *Error Messages* section at the back of this brochure.

ROTATION

The 132 has a function which automatically changes the operating sequence of the pumps based on *Equal Run Time Rotation*. *Equal Run Time Rotation* is based on pump running hours and allows for equal usage of both pumps. The 132 uses the *Frequency of Rotation* dial to set the rotation of the pumps.

The control rotates the operating sequence of the pumps when the pumps are off. In a constant circulation system where the lead pump runs continuously, the control waits for up to 12 hours to rotate the operating sequence of the pumps.

Note: *The* *Equal Run Time Rotation* function is reset by pressing the ***Clear Alert*** button.

EXERCISING

The 132 has a built-in pump exercising function. This function is only operational if the *Exercising / Off* DIP switch is set to Exercising. If a pump has not been operated at least once every three days, the control turns on the output for 10 seconds. This minimizes the possibility of a pump seizing during a long period of inactivity.

Note: The exercising function does not work if power to the control or pumps is off.

WARM WEATHER SHUT DOWN (WWSD)

When the 132 is used as a stand alone control, the pumps can be operated based on outdoor air temperature. The WWSD feature is only operational when an outdoor sensor is installed and the *WWSD* dial is not set to Off. When the outdoor air temperature rises above the WWSD setting, the 132 turns on the *WWSD* light. When the control is in Warm Weather Shut Down, the demand LED's are displayed if there is a demand. However, the control does not operate the pumps to satisfy these demands. If the 132 receives a pump demand from a reset control, the outdoor sensor should not be installed and the WWSD dial must be set to *Off*.

Section B: Pump Operation

Section B1
Stand-by

Section B2
2-Stage

Section B1: Stand-by

PUMP OPERATION

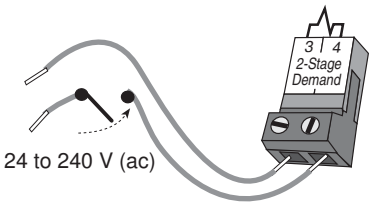
The lead pump contact closes whenever there is a pump demand and the 132 is not in WWSD. If a flow proof is not present after the flow proof delay has expired, the control turns off the lead pump contact and turns on the stand-by pump contact. An alert contact is activated to indicate the failure. The stand-by pump contact also turns off if a flow proof is not obtained once the flow proof delay has expired.

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Section B2: 2-Stage

2-STAGE DEMAND

A 2-stage demand is required in order for the 132 to provide additional flow. A 2-stage demand is generated by applying a voltage between 24 and 240 V (ac) across the *2-Stage Demand* terminals (3 and 4). Once voltage is applied, the *2-Stage Demand* light is turned on. A 2-stage demand can come from an additional pressure differential switch, a temperature differential device, or a setpoint control based on an outdoor air temperature.



Note: The 2-stage demand device should provide an appropriate differential to prevent the second stage pump from short cycling.

PUMP OPERATION

The first stage pump contact closes whenever there is a pump demand and the 132 is not in WWSD. The second stage pump contact closes whenever there is a pump demand, flow proof, 2-stage demand, and the 132 is not in WWSD.

If flow is not established by the first stage pump, the 132 turns off the first stage pump contact and turns on the second stage pump contact. An alert contact is activated to indicate the failure. The second stage pump contact also turns off if a flow proof is not obtained once the flow proof delay has expired.

Section C: Alert Operation

Section C1
Alert Levels

Section C2
Alert per Pump

There are two alert contacts (*Alert A and Alert B*) on the Pump Sequencer 132. These contacts are used to indicate either pump or control failures. The alert contacts have two modes of operation. The mode of operation for the alert contacts is selected using the *Alert per Pump / Alert Levels* DIP switch. When an alert contact is activated, refer to the *Error Messages* section of this brochure to determine the cause of the alert signal. To clear an alert, press the ***Clear Alert*** button.

Section C1: Alert Levels

When operating in the *Alert Levels* mode, the alert contacts are used to indicate the level of failure in the system.

ALERT A (Non-Critical)

The *Alert A* contact closes when a non-critical failure occurs. Non-critical failures include: a single pump failure, a control error, an outdoor sensor failure, or short cycling of the 2-stage demand. With these failures, it is still possible to establish flow in the system.

ALERT B (Critical)

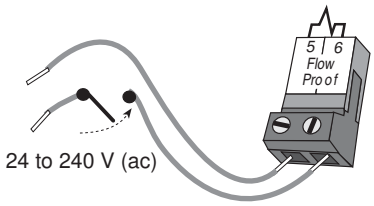
The *Alert B* contact closes when a critical failure occurs. Critical failures include: the failure of both pumps, or a failure of the flow proof device. With these failures, the control is no longer able to operate the pumps, and it is not possible to provide flow.

Section C2: Alert per Pump

When operating in *Alert per Pump* mode, the alert contacts are related to the pump contacts on the control. The *Alert A* contact closes anytime *Pump 1* fails, and the *Alert B* contact closes anytime *Pump 2* fails. The *Alert A* contact also closes to indicate an outdoor sensor failure, a control error, a flow proof device failure, or if the 2-stage demand is short cycling.

Flow Proof

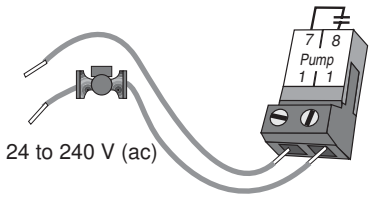
To generate a flow proof, a voltage between 24 V (ac) and 240 V (ac) must be applied across the *Flow Proof* terminals (5 and 6).



Output Connections

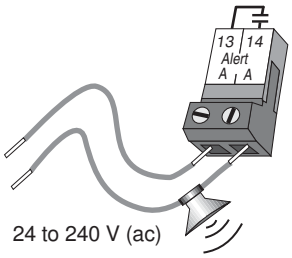
Pump Contacts

The *Pump 1* and *Pump 2* terminals (7, 8 and 9, 10) are isolated outputs in the 132. There is no power available on these terminals from the control. These terminals are to be used as a switch to either make or break power to the pump circuit. Since these are isolated contacts, they may switch a voltage between 24 V (ac) and 240 V (ac).



Alert Contacts

The *Alert A* and *Alert B* terminals (13, 14 and 15, 16) are isolated outputs in the 132. There is no power available on these terminals from the control. These terminals are to be used as a switch to either make or break power to the alert circuit. Since these are isolated contacts, they may switch a voltage between 24 V (ac) and 240 V (ac).



Outdoor Sensor Connection

Do not apply power to these terminals as this will damage the control.

Outdoor Sensor (Optional)

Connect the two wires from the Outdoor Sensor 070 to the *Outdoor Sensor* terminals (17 and 18). The outdoor sensor is used by the 132 to measure the outdoor air temperature.

Note: If an outdoor sensor is not installed, the *WWSD* dial must be set to Off.



STEP FIVE — TESTING THE WIRING

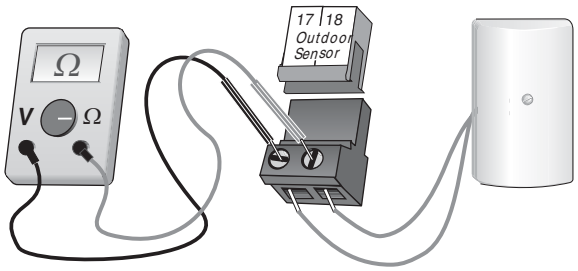
Each terminal block ***must be unplugged*** from its header on the control before power is applied for testing. To remove a terminal block, pull it straight down from the control.

The following tests are to be performed using standard testing practices and procedures, and should only be carried out by properly trained and experienced persons.

A good quality electrical test meter, capable of reading from at least 0 - 300 V (ac) and at least 0 - 2,000,000 Ohms, is essential to properly test the wiring and sensor.

Test The Sensor

In order to test the sensor, the actual temperature at the sensor location must be measured. A good quality digital thermometer with a surface temperature probe is recommended for ease of use and accuracy. Where a digital thermometer is not available, a spare sensor can be placed alongside the one to be tested and the readings compared. Test the sensor according to the instructions in the Data Brochure D 070.



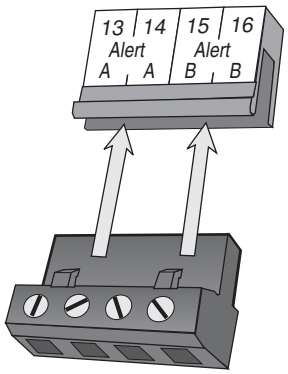
Connecting The Control

Make sure all power to the devices and terminal blocks is off, and remove any remaining jumpers from the terminals.

Reconnect the terminal blocks to the control by carefully aligning them with their respective headers on the control and then pushing the terminal blocks into the headers. The terminal blocks should snap firmly into place.

Install the supplied safety dividers between the unpowered sensor input and the powered 120 V (ac) or 24 V (ac) wiring chambers.

Apply power to the control. The operation of the control on power up is described in the *Sequence of Operation* section of this brochure.

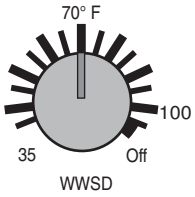


Settings

WARM WEATHER SHUT DOWN (WWSD)

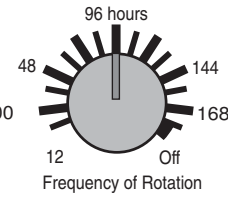
The *WWSD* dial is only operational when an outdoor sensor is installed. The WWSD dial can be set from 35 to 100°F or Off (2 to 38°C or Off). Refer to section A.

Note: If an outdoor sensor is not installed, the *WWSD* dial must be set to *Off*.



FREQUENCY OF ROTATION

The *Frequency of Rotation* dial can be set from 12 to 168 hours or Off. Refer to section A.



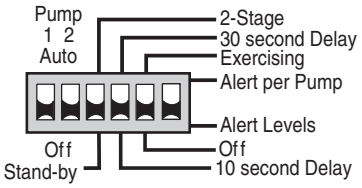
DIP SWITCH SETTINGS

Pump 1 Auto / Off

The *Pump 1 Auto / Off* DIP switch is used to select pump 1 operation. If the DIP switch is set to *Pump 1 Auto*, refer to section B.

Pump 2 Auto / Off

The *Pump 2 Auto / Off* DIP switch is used to select pump 2 operation. If the DIP switch is set to *Pump 2 Auto*, refer to section B.



2-Stage / Stand-by

The 2-Stage / Stand-by DIP switch is used to select which mode of pump control is required. If 2-Stage is selected, refer to section B2, and if Stand-by is selected, refer to section B1.

30 second Delay / 10 second Delay

The *30 second Delay / 10 second Delay* DIP switch is used to select what time delay is required to prove flow once a pump contact is turned on. Refer to section A.

Exercising / Off

The *Exercising / Off* DIP switch is used to select exercising. If the DIP switch is set to *Exercising*, refer to section A.

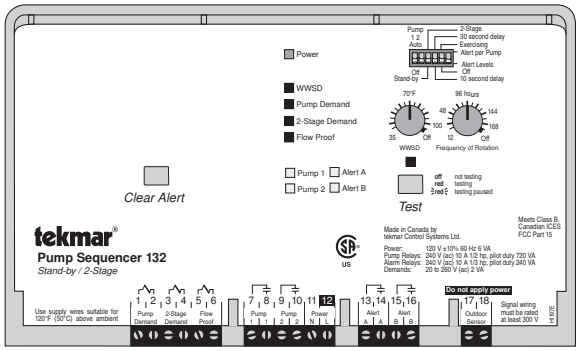
Alert per Pump / Alert Levels

The *Alert per Pump / Alert Levels* DIP switch is used to select which mode of alert operation is required. Refer to section C.

Technical Data

Pump Sequencer 132 *Stand-by / 2-Stage*

Literature	—	D 132, A 132's, D 001.
Control	—	Microprocessor control; This is **not a safety (limit) control**.
Packaged weight	—	2.9 lb. (1320 g), Enclosure A, blue PVC plastic
Dimensions	—	6-5/8" H x 7-9/16" W x 2-13/16" D (170 x 193 x 72 mm)
Approvals	—	CSA US, meets ICES & FCC regulations for EMI/RFI.
Ambient conditions	—	Indoor use only, 32 to 120°F (0 to 50°C), < 90% RH non-condensing.
Power supply	—	120 V (ac) ±10% 60 Hz 6 VA
Pump Relays	—	240 V (ac) 10 A 1/2 hp, pilot duty 720 VA
Alert Relays	—	240 V (ac) 10 A 1/3 hp, pilot duty 240 VA
Demands	—	20 to 260 V (ac) 2 VA
Sensors	—	NTC thermistor, 10 kΩ @ 77°F (25°C ±0.2°C) ß=3892
Optional	—	Outdoor Sensor 070



The installer must ensure that this control and its wiring are isolated and/or shielded from strong sources of electromagnetic noise. Conversely, this Class B digital apparatus complies with Part 15 of the FCC Rules and meets all requirements of the Canadian Interference-Causing Equipment Regulations. However, if this control does cause harmful interference to radio or television reception, which is determined by turning the control off and on, the user is encouraged to try to correct the interference by re-orientating or relocating the receiving antenna, relocating the receiver with respect to this control, and/or connecting the control to a different circuit from that to which the receiver is connected.

Cet appareil numérique de la classe B respecte toutes les exigences du Règlement sur le matériel brouilleur du Canada.

Caution The nonmetallic enclosure does not provide grounding between conduit connections. Use grounding type bushings and jumper wires.

Attention Un boîtier nonmétallique n´assure pas la continuité électrique des conduits. Utiliser des manchons ou des fils de accord spécialement conçus pour la mise á la terre.

Limited Warranty and Product Return Procedure

Limited Warranty ***The liability of tekmar under this warranty is limited. The Purchaser, by taking receipt of any tekmar product ("Product"), acknowledges the terms of the Limited Warranty in effect at the time of such Product sale and acknowledges that it has read and understands same.***

The tekmar Limited Warranty to the Purchaser on the Products sold hereunder is a manufacturer's pass-through warranty which the Purchaser is authorized to pass through to its customers. Under the Limited Warranty, each tekmar Product is warranted against defects in workmanship and materials if the Product is installed and used in compliance with tekmar's instructions, ordinary wear and tear excepted. The pass-through warranty period is for a period of twenty-four (24) months from the production date if the Product is not installed during that period, or twelve (12) months from the documented date of installation if installed within twenty-four (24) months from the production date.

The liability of tekmar under the Limited Warranty shall be limited to, at tekmar's sole discretion: the cost of parts and labor provided by tekmar to repair defects in materials and/or workmanship of the defective product; or to the exchange of the defective product for a warranty replacement product; or to the granting of credit limited to the original cost of the defective product, and such repair, exchange or credit shall be the sole remedy available from tekmar, and, without limiting the foregoing in any way, tekmar is not responsible, in contract, tort or strict product liability, for any other losses, costs, expenses, inconveniences, or damages, whether direct, indirect, special, secondary, incidental or consequential, arising from ownership or use of the product, or from defects in workmanship or materials, including any liability for fundamental breach of contract.

The pass-through Limited Warranty applies only to those defective Products returned to tekmar during the warranty period. This Limited Warranty does not cover the cost of the parts or labor to remove or transport the defective Product, or to reinstall the repaired or replacement Product, all such costs and expenses being subject to Purchaser's agreement and warranty with its customers.

Any representations or warranties about the Products made by Purchaser to its customers which are different from or in excess of the tekmar Limited Warranty are the Purchaser's sole responsibility and obligation. Purchaser shall indemnify and hold tekmar harmless from and against any and all claims, liabilities and damages of any kind or nature which arise out of or are related to any such representations or warranties by Purchaser to its customers.

The pass-through Limited Warranty does not apply if the returned Product has been damaged by negligence by persons other than tekmar, accident, fire, Act of God, abuse or misuse; or has been damaged by modifications, alterations or attachments made subsequent to purchase which have not been authorized by tekmar; or if the Product was not installed in compliance with tekmar's instructions and/or the local codes and ordinances; or if due to defective installation of the Product; or if the Product was not used in compliance with tekmar's instructions.

THIS WARRANTY IS IN LIEU OF ALL OTHER WARRANTIES, EXPRESS OR IMPLIED, WHICH THE GOVERNING LAW ALLOWS PARTIES TO CONTRACTUALLY EXCLUDE, INCLUDING, WITHOUT LIMITATION, IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, DURABILITY OR DESCRIPTION OF THE PRODUCT, ITS NON-INFRINGEMENT OF ANY RELEVANT PATENTS OR TRADEMARKS, AND ITS COMPLIANCE WITH OR NON-VIOLATION OF ANY APPLICABLE ENVIRONMENTAL, HEALTH OR SAFETY LEGISLATION; THE TERM OF ANY OTHER WARRANTY NOT HEREBY CONTRACTUALLY EXCLUDED IS LIMITED SUCH THAT IT SHALL NOT EXTEND BEYOND TWENTY-FOUR (24) MONTHS FROM THE PRODUCTION DATE, TO THE EXTENT THAT SUCH LIMITATION IS ALLOWED BY THE GOVERNING LAW.

Product Warranty Return Procedure All Products that are believed to have defects in workmanship or materials must be returned, together with a written description of the defect, to the tekmar Representative assigned to the territory in which such Product is located. If tekmar receives an inquiry from someone other than a tekmar Representative, including an inquiry from Purchaser (if not a tekmar Representative) or Purchaser's customers, regarding a potential warranty claim, tekmar's sole obligation shall be to provide the address and other contact information regarding the appropriate Representative.

tekmar® Control Systems

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